

BEYOND STATISTIC PRESERVATION: CAUSAL EVALUATION OF COMPOSED FEDERATED-LEARNING DEFENSES

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ABSTRACT

A preserved statistic is not preserved suppression. No test that admits a composition because one constituent already suppresses the attack can identify whether the downstream mechanism caused its success. Federated learning composes defenses never designed together, so requiring the downstream *statistic* to survive the upstream stage is the natural repair. It fails structurally rather than statistically: the gate selects on the very outcome to be explained, so the comparison identification needs is missing at any n . The repair is a design, not a better criterion: hold the downstream defense and the attack fixed and intervene on the upstream transform. Ignoring this costs not power but the sign. On one cell the outcome-gated design moves -0.273 where the within-defense design moves $+0.125$ at $n=20$, both 95% intervals excluding zero. The two designs disagree on five of seven cells, and the reversal replicates on CIFAR-100. Two dissociations say why, with the adversary’s coefficient pinned so the attack cannot weaken. Krum’s decision flips in up to 80% of rounds while it admits no adversary at all and suppression stays inside a pre-registered ± 0.15 margin. Conversely `cos_krum`’s selection is bit-identical while ASR falls 0.173. The negative replicates in a second invariance class and outside the theory’s rescaling family. We do *not* claim that admission predicts security. And no gate of this shape can certify *emergent* suppression, so recall is capped structurally, our own lowest-ASR composition the witness.

1 INTRODUCTION

A composition of two defenses suppresses an attack. **Is the downstream defense’s mechanism what suppressed it?** Everything below is an answer to that one question.

Federated learning (McMahan et al., 2017) is vulnerable to data poisoning (Chen et al., 2017; Bagdasaryan et al., 2020), and an arms race has produced defenses (Fung et al., 2020; Pillutla et al., 2022; Yin et al., 2018; Blanchard et al., 2017) with accuracy/robustness trade-offs and no dominant member (Appendix G). *Can we compose them for stronger robustness?* Composition does not even commute: FoolsGold→CoordMedian suppresses a pixel backdoor to ASR 0.099 while the reverse reaches 0.732, that order collapsing to FoolsGold alone bit-identically, since a rank aggregator emits one vector rather than per-client updates.

Two measured facts put the question there in the first place. Two defenses that each fail alone can jointly suppress a committed (non-adaptive) attack: FoolsGold→RFA reaches ASR 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845. And the link an evaluator leans on breaks in both directions: Krum’s selection flips in 0.733 of rounds while its admission is unchanged and its suppression does not follow ($|\Delta|=0.026$), while `cos_krum`’s selection is bit-identical at every rung and its ASR nonetheless *falls* 0.173 (both drawn in Fig. A6, App. J).

We propose no new defense. Preserving what the downstream defense reads leaves open whether the composition still suppresses the attack. The natural test admits a composition because one constituent already suppresses the attack, and so cannot answer the question at any sample size: it removes the counterfactual support identification requires.

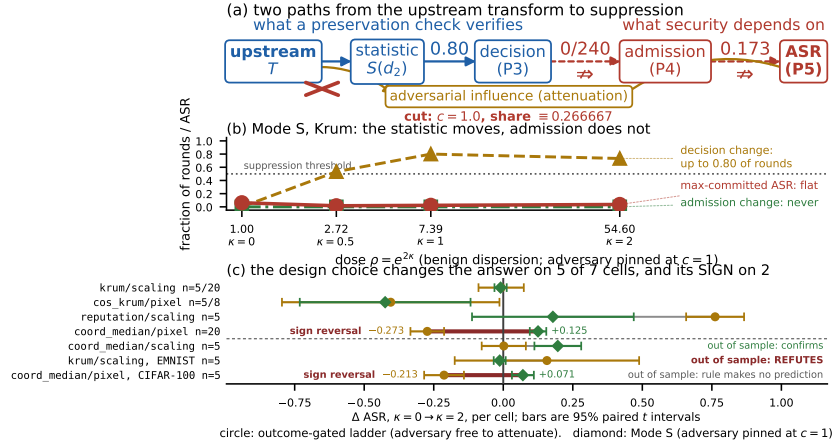


Figure 1: **The paper in one figure. Preserving the statistic a downstream defense reads does not identify preserved attack suppression.** (a) The chain a preservation check reasons along, at the levels of Def. 1: solid links hold by definition, each dashed link is marked $\nRightarrow$ for not implied, and the lower arc is the attenuation channel Mode S (§5) cuts by construction. Its three numbers are three scopes, not one experiment (App. J), and the middle one pools four CIFAR-10 arms rather than Krum alone. (b) Krum under model scaling (§5): the decision flips while admission does not, and ASR stays inside the pre-registered ± 0.15 margin (Student- t 95% bars, $n=5$). (c) Seven cells, one contrast each, estimated both ways (App. D.3): the design choice changes the answer on 5 of 7 cells, and its sign on 2. Row labels carry n per cell, outcome-gated/Mode S where the two differ, and the four cells above the dashed rule are training data for a since-withdrawn rule rather than evidence for it (Table A10).

What replaces the gated test is a design rather than a better test: hold the downstream defense and the attack fixed and intervene on the upstream transform, instead of comparing across a menu of defenses. The transform can weaken the attack itself, so an outcome-gated comparison credits the downstream defense with a fall in ASR that attenuation upstream produced. On one cell that change reverses the *sign*: -0.273 outcome-gated against $+0.125$ within-defense at $n=20$.

Why the question is sharper without a trusted coordinator. A contributor clips, a scheduler reweights by reputation, and whoever merges aggregates robustly: three independently developed mechanisms, nobody owning the stack, so the only check at each seam is that the next stage still reads what it expects. We claim no transfer beyond the controlled setting we study (p4), though the seam, rather than the scale, is what the problem is about.

Contributions. The question is scientific rather than operational: whether an evaluation can identify that a mechanism *caused* an observed suppression, not whether a deployment can detect malicious clients. Preservation is a hierarchy, and statistic-level invariance is not a causal surrogate for security-level preservation (§3). The single central contribution is therefore that evaluation-design result; the screen it implies is priced rather than validated, and we do not claim it certifies security.

- (1) **Two impossibility results (§2).** An inherited-suppression gate selects on the very outcome to be explained, so one cell of the relevant 2×2 is empty for *every* menu and no sample size repairs it (Prop. 1). The gate it describes is a documented convention whose prevalence we measured rather than assumed, at 6 of 59 coded composed-defense papers (10.2%) gating attribution on the composed outcome while running no identifying contrast, against 46 (78%) that do run one (one coder, no inter-rater statistic; App. G.1). Emergent suppression fails such a gate by definition, so any evaluator using one accepts a blind spot for emergent synergy (Prop. 2, Cor. 2).
- (2) **The design it forces, and a sign reversal (§4–§5).** Every clause of the design is forced by a term in the proposition (Cor. 1), and the screen that follows is a consequence we price rather than a third contribution (§6.3).
- (3) **The preservation hierarchy, and the two links that break (§3, §5).** Both breaks are the two measured facts above, at fixed downstream defense and attack; the demonstration is a sign in-

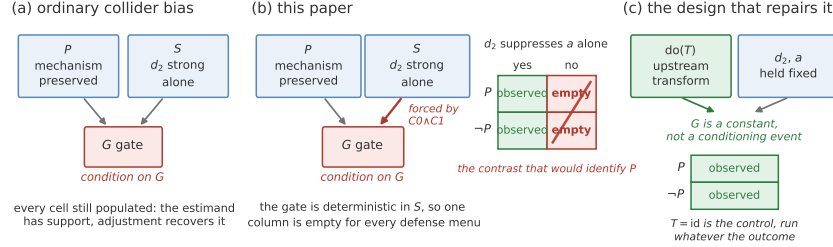


Figure 2: **Not ordinary collider bias but a structural positivity violation, which is why the remedy is a design and not an estimator.** (a) Ordinary collider bias, where every cell stays populated and adjustment recovers the estimand in principle. (b) The inherited-suppression gate $G=E$ is *deterministic* in $S=M(d_2, a)$, so the column the identifying contrast needs is empty and no sample size restores it (Prop. 1). (c) The repair: hold d_2 and a fixed, intervene on T , and run $T=id$ as the control whatever its outcome (Cor. 1; App. C.2).

version and two exact invariances, not small $n=5$ mean differences. Which level a composition preserves is settled by algebra before any run, and is reusable without our screen (Prop. 10).

The main text carries two propositions, one definition, one numbered table and two figures: the theorem, corollaries, lemmas, screening conditions, limitations (L1)–(L6) and the 59-paper audit are appendix, each cited from the sentence that needs it.

2 THE TESTABILITY BOUNDARY: WHY THE GATED TEST FAILS

Comparing many defense pairs is where identification is lost. The natural test keeps pairs whose constituents already suppress the attack and asks whether the composition preserves it. *Screen*, throughout, names a rule for choosing which compositions are worth running, never a predictor of security. Also fixed: *gate* is the condition below, *criterion* our composability rule, *protocol* §4’s clauses, *instruments* Mode S and score-only Krum. *The protocol is ours*: the gate our own criterion applies, which is why the boundary costs us our own lowest-ASR composition (§6.3).

Intuition. An admitted pair’s downstream defense is guaranteed to suppress the attack alone, so no admitted pair exists in which that defense fails alone while the composition succeeds. **That empty cell is the whole of it**, and it is empty by the gate rather than by bad luck: it is the one comparison that would credit the suppression to the mechanism rather than to the downstream defense’s own strength, and comparing the cells the gate does leave occupied varies the downstream defense, which changes the conditioning event instead of intervening on the transform.

Proposition 1 (Outcome-gated comparisons cannot identify the upstream defense’s contribution). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense.***

Nothing in this statement is about federated learning. M is any outcome metric, θ any threshold, P any condition on the two mechanisms’ definitions, and $\mathcal{D}$ any menu; App. C.2 substitutes our protocol into each, term by term. The proof is short (App. C.1).

What we estimate instead. Fix a downstream defense d_2 and a committed attack a . The **treatment** T is the upstream per-client transformation applied before d_2 aggregates, with $T=id$ the con-

trol that is bit-identical to d_2 standalone. The **outcome** Y is $\text{ASR}(T, d_2, a)$. The **mechanism-preservation variable** $P \in \{0, 1\}$ records whether T preserves the statistic d_2 reads, at whichever level is at issue. Which of the five levels is at issue is §3’s subject; the boundary below holds for any such condition. P is a deterministic function of T and d_2 ’s definition, not a measured covariate. The causal quantity is

$$\tau(T) = \mathbb{E}[Y \mid \text{do}(T), d_2, a] - \mathbb{E}[Y \mid \text{do}(T=\text{id}), d_2, a].$$

We do not estimate the causal effect of P itself. P is not manipulable apart from T , being a property of the pair, so “the effect of preservation” names no intervention. We estimate the effect of intervening on the upstream transformation while holding d_2 and a fixed, and what we report about P is whether $\tau(T) \neq 0$ occurs while P holds, and whether $\tau(T)=0$ occurs while P fails. Both occur (§5), which is what refutes P as a surrogate for Y .

Only half of this is ordinary collider bias. We claim no credit for it: that half is textbook (Elwert & Winship, 2014; Pearl, 2009). What the gate adds is a structural *positivity* violation rather than a confounding one (Fig. 2b), while reading as a reasonableness filter. The gate also earns its keep: it stops the screen certifying pairs no constituent suppresses, so wherever our conditions certify a pair, a low composition ASR is attributable to d_2 ’s standalone strength rather than to preservation, the reading we held ourselves (§6.1).

Why this is more than a standard positivity violation. The violation itself is standard; four things around it are not. The estimand is specific to composed defenses, being which *level* of a preservation hierarchy an upstream transform leaves intact, and that hierarchy has no analogue outside this setting. The gate that empties the cell is a documented convention, not an analyst’s slip, and its prevalence is measured, not asserted, at the 10.2% and 78% of §1 (App. G.1). A low rate was pre-registered as outcome (ii), so our decline to call the gate common stays, evidenced not hedged.

A screen of this shape fails a second, independent way, bounding what it can find rather than what it can be validated on.

Proposition 2 (Coverage: inherited-suppression screens cannot certify emergent synergy). ($d_1 \rightarrow d_2$)’s suppression of a is **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Prop. 1. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$, $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones and $\mathcal{S} \setminus \mathcal{E}$ for the inherited ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$, on every menu and every attack.

Our own lowest-ASR composition is the witness: §1’s FG→RFA, at max-committed ASR 0.093 ($n=30$, CI [0.062, 0.123]) against constituents that each fail alone, is emergent and a false negative of our own criterion, which is what puts the bound above the recall we report (App. C.1). It splits a menu into *inheritance*, which we screen for, and *emergence*, a search problem we do not solve. A screen reaching $\mathcal{E}$ would have to read the composition’s own suppression, the outcome it exists to avoid conditioning on: Prop. 1’s boundary met from the other side. Our blind spot is one condition, C1, so that recall is an *operating point*, not a ceiling (§6.3).

What this paper establishes, and what it does not, stated once.

Formally. Neither proposition above contains a dataset, architecture or scale, so what carries beyond this paper is the design they license and not our screen. Thm. 8 (App. C.3) and the invariance classes are proved for positive per-client rescalings into named rank and geometric aggregators, and bound no ASR.

Empirically. We identify the effect of intervening on the upstream transform at fixed downstream defense and attack, and offer no causal model of FL; on one cell that design change flips the *sign* of the estimated effect. The experiments are four aggregators of three structural kinds under two committed attacks on CIFAR-10, three arms replicated, on EMNIST-byclass ($n=3$ frozen, $n=5$ pooled), `resnet18` ($n=3$) and CIFAR-100, at $n=5$ seeds, $n=20$ on the CIFAR-10 sign-reversal cell and on Krum/scaling’s Mode S leg, $n=8$ for `cos_krum`.

Not established. We do *not* establish that admission predicts security, we claim nothing about adaptive robustness, and we claim nothing about confidentiality or availability. This is a methodology for identifying what a composition preserves, not for choosing a defense to deploy: the screen prioritizes what to evaluate, and nothing here is a deployment recommendation. Claims do not transfer upward either: scope conditions in Table A2, evidence tiers in Table A1, limits in §H, (L1)–(L6), and every claim we have withdrawn or refuted in Table A21. The criterion’s empirical validation is exploratory (L4), and our instruments read adversary identity (§I).

3 WHICH LEVEL A COMPOSITION PRESERVES

“Preserved” has five senses. The condition P the boundary quantifies over is not one condition but a graded family, and **which member of it a claim asserts decides whether the claim is checkable at all**. An upstream transform can leave a downstream defense “the same” in five senses, ordered from what the defense computes to what the attacker gets: three about the defense’s *internal* behaviour, then two about *security* behaviour, what the adversary gets.

The algebra cuts one place later, everything up to admission being *observable* and attack suppression *mechanistic*. These are five different claims, and preservation at one level does not imply preservation at the next. **Where that chain breaks is this paper’s subject**: the first two implications are logical and need no experiment, while both failures start at admission, whose preservation implies neither a preserved decision (witness Krum, §5) nor preserved suppression (witness `cos_krum`, §6.2).

To instantiate the hierarchy in FL we characterize the invariance classes, which is where the levels stop being a taxonomy and start being decidable: every per-client defense we consider acts as a positive scalar rescaling $T(u_i) = c_i u_i$, so the algebra fixes which level a given composition sits at, before any run: cosine similarity is exactly invariant, coordinate ordering only under a bounded-ratio condition, and consensus distance and residual magnitude not at all (Prop. 10, App. C.3). That class is also the theory’s whole reach: nothing we prove speaks to a transform outside it, and two composed defenses are outside it, Bulyan (El Mhamdi et al., 2018) zeroing the clients its selection drops and FLAME (Nguyen et al., 2022) adding noise to the aggregate.

Definition 1 (Preservation levels). *For an upstream transform T and a downstream defense d_2 reading statistic S :*

- (P1) **Value invariance**: $S(T(U)) = S(U)$.
- (P2) **Ordering invariance**: $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces on clients.
- (P3) **Decision invariance**: the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance**: the adversarial input d_2 admits into its aggregate is unchanged. Admission is the pre-registered term throughout for whether adversarial input survives d_2 , the support of the admitted mass rather than the mass itself. The mass is influence Λ_a , defined with the vocabulary at the end of this section and in App. K, which a preserved support leaves free to move (Table 1).
- (P5) **Attack-suppression preservation**: ASR stays below threshold, for the fixed committed attack the arm is run against.

The screening conditions the levels induce, and their names, are in App. B and C.

Five words, fixed here and used in one sense throughout, in the order the chain runs.

term	level	what it names	bears on security
statistic	(P1)	what the defense computes from the updates it reads	no
ordering	(P2)	the ranking that statistic induces over clients	no
decision	(P3)	which client or coordinate it selects	no
admission	(P4)	whether adversarial input enters its aggregate at all	yes
suppression	(P5)	whether the attack still succeeds, i.e. max-committed ASR	yes

Two further words are needed, each distinct from those five. *Influence* is Λ_a , the weight the adversary’s direction carries in the aggregate. *Attenuation* is the adversary’s own update shrunk upstream, reaching ASR without passing through the statistic at all, and it is the channel an intervention on T must close before either of the other two can be read.

Admission (P4) is the first level at which adversarial influence enters the causal graph, which is why we intervene there rather than on the statistic. It is privileged for its semantics, not for sufficiency. It is also instantiated per aggregator (App. K), so our admission result is four within-family statements and we never difference or rank across them.

4 A CAUSAL PROTOCOL: INTERVENE UPSTREAM, NOT ACROSS DEFENSES

The inadmissible comparison is replaced by a different design rather than a better criterion. Prop. 1 determines it: do not select across defenses; hold the downstream defense and the attack fixed, intervene on the upstream transformation, and measure the channels it opens separately. Writing T_1 for the upstream transformation, the estimand is $\Delta_{\text{mech}} = \text{ASR}(T_1^{\text{int}}, d_2, a) - \text{ASR}(T_1^{\text{base}}, d_2, a)$, at fixed d_2 , fixed attack a and fixed adversarial coefficient share; the third clause below is why this is a mechanism contrast and not a dose.

- (1) **Hold d_2 and the attack fixed.** *Forced by the gate $E(d_1, d_2, a)$:* vary d_2 or a and the two sides of a contrast are conditioned on different events, so they are not comparable at any sample size.
- (2) **Intervene on the upstream transform.** *Forced by the empty cell:* variation in d_2 is exactly what the gate turns into missing support, so the intervention has to live on d_1 .
- (3) **Pin the adversarial coefficient channel.** *Forced by what the proposition does not constrain:* nothing prevents T from reaching the outcome without passing through d_2 's statistic at all, by attenuating whichever client carries the poison (Lem. 2). Mode S pins it by setting every adversary's coefficient to $c=1$, which requires knowing which clients are adversarial and is therefore a laboratory instrument rather than a deployable defense (§5).
- (4) **Measure the channels, not only the outcome.** *Forced by what P asserts:* a claim about a mechanism, so a contrast reporting only ASR cannot say which channel carried the effect, and every channel the intervention opens must be reported, including the accidental ones.

Operationally, before any of that, name the downstream mechanism's invariance class and the level of Def. 1 the claim asserts (§3; Table A1's Formal row names the transforms the algebra does not reach).

Threat model: committed, not adaptive. Every clause holds the attack fixed, so the protocol establishes attack-suppression preservation under a fixed committed attack and nothing wider, which is why we do not call the object "security". One whitebox composition-aware *adaptive* stress test measures that boundary: solving the joint FoolsGold/RFA constraint knowing both stages raises $\text{FG} \rightarrow \text{RFA}$ to $\text{ASR } 0.157 \pm 0.073$ at $n=30$, $2.4\times$ its seed-matched 0.064 (paired $+0.093$, $[+0.066, +0.120]$). **Inherited robustness is not adaptive robustness**, and a protocol stated against an adaptive adversary is the sharpest thing this framework lacks (App. M).

5 STATISTIC PRESERVATION DOES NOT IDENTIFY ATTACK-SUPPRESSION PRESERVATION

Disturbing what a defense reads need not cost suppression, and **on one cell the two designs disagree in sign:** the outcome-gated design gives $\Delta\text{ASR} = -0.273$ ($[-0.334, -0.213]$) where the within-defense design gives $+0.125$ ($[+0.095, +0.155]$) at $n=20$, disjoint and both excluding zero, so an evaluator choosing between the two designs is choosing the sign of the conclusion rather than its precision. We adjudicate the implication that losing the decision means losing suppression by oracle intervention on two arms in two invariance classes rather than by a general law.

One cell is carried end to end, and it is Krum under model scaling: row 1 of Table 1, with row 6 its magnitude-controlled twin.

The instrument, as a counterfactual. The clean question asks what happens when the downstream statistic changes but the adversary cannot become weaker. **Mode S** enforces exactly that: it pins every adversary's coefficient at $c=1.0$ and spreads only the benign clients over $\rho = e^{2\kappa}$, holding the adversarial coefficient share constant at 0.266667 to within float32 read-back noise, closing Figure 1(a)'s attenuation path, so a rise in ASR here cannot be attenuation. Pinning c requires knowing which clients are adversarial, so Mode S is a measurement device and not a defense, never a quantity a deployed evaluator could compute without that ground truth.

Two readings of preservation, frozen against each other. *The statistic reading:* what matters is disturbance of d_2 's statistic, read off its *decisions*. *The admission reading:* what matters is the *adversarial input d_2 admits*. Mode S disturbs Krum's selection heavily while leaving that admission

Table 1: **The four channels, all four aggregators, under Mode S**; endpoint change $\kappa=0 \rightarrow \kappa=2$. Δ agg. is the relative displacement of the aggregate the defense *emits*, so it is what enters training and not the value of a scoring statistic. Δ adm. and $\Delta\Lambda_a$ are §3’s admission and influence, never interchangeable. Admission never moves in any measured round of any arm. The table does not order the arms, $\Delta\Lambda_a$ and Δ ASR disagreeing in rank. Row 7 varies architecture alone (ResNet18). n is seeds, not rounds; provenance, round counts and the emitter are in App. D.2.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR	n
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026	5
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178	5
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425	8
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098	5
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017	3
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023	5
Krum (ResNet18)	selector (pairwise distance)	0.824	0.556	0.000	0.000	-0.041	3

untouched, so that arm adjudicates (Table A7): as ρ climbs to 54.6 the decision changes in up to 0.80 of rounds, the adversarial mass admitted in none, and ASR shows no evidence of the predicted increase, staying inside the pre-registered practical-equivalence margin ($|\Delta|=0.026 < 0.15$).

Here practical equivalence is established, not merely consistent: TOST against that frozen ± 0.15 margin, an evaluation-design tolerance conjoined with every rung below 0.5, not a claim that 15 ASR points are negligible, gives $p=0.0014$ at $n=5$, and a pre-registered top-up to $n=20$ confirms it at $\Delta = -0.010$, 95% CI $[-0.032, +0.012]$ (App. L scores every arm this way, us included). *The margin is not load-bearing* for any reading in Table A23: each would survive at ± 0.05 , the binding arm needing $m^*=0.034$, where m^* is the smallest margin an arm’s interval fits inside; resnet18’s $n=3$ replication is the one arm that needs it, at $m^*=0.183$ (App. D.2).

The statistic reading is refuted: statistic and decision disturbance need not change suppression once the adversarial coefficient share is held fixed. *The zero is a floor here*, Krum admitting no adversary at any rung, so unchanged admission is not by itself decisive, and we claim no per-channel decomposition of Δ ASR (§D.2.1).

The table contains its own positive control, without which “we moved something and nothing happened” could not be told from too blunt an instrument: suppression moves in exactly the two arms whose *influence* channel moves, reputation and coord. median. Krum changes the decision at $\Delta\Lambda_a=0.000$ and suppression does not follow, so a decision change is not *sufficient*; cos_krum’s selection is bit-identical at every rung, also at $\Delta\Lambda_a=0.000$, yet ASR *falls*, so it is not *necessary* either. Neither reading rests on Krum’s admission floor; both are taken across all four kinds (App. D.2).

What Mode S does not hold fixed, and the control that closes it. Mode S identifies the intervention’s effect at a fixed adversarial coefficient share, not statistic disturbance in isolation, so what it yields is a *dissociation*, not proof that disturbance is inert. It pins the adversarial coefficients alone, so the dose still moves the aggregated update’s magnitude and geometry, a third channel reading Δ agg. = 0.892 here. **Score-only Krum**, frozen before any score-only ASR existed, closes it by aggregating the *untransformed* selected update at the same Δ dec. of 0.733: $\Delta = -0.023$ against -0.026 uncontrolled on the same cell, so the claim is instrumented against both confounds, attenuation and magnitude (App. D.2, §S9).

A pre-registered replication that refutes the positive half. coord_median under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the smallest decision change (0.482) and the largest admission-level change ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise, and the thresholds were frozen before its runs existed. We find $\Delta = +0.098$ at $n=5$, which **refutes the admission ordering**: the arm with the largest admission-level change rose less than one with a smaller change, and the pooled decision reading does not predict it either.

What replicates is the negative on a coordinate-wise order statistic: that decision change costs no suppression by the frozen margin, at accuracy 0.698. Equivalence here is to this cell’s own identity

rung and not to a low absolute ASR, and the rise is positive in 5/5 seeds (post hoc): small and consistent, not zero (App. D.2).

The design choice, not the defense, sets the inferred direction. The sign disagreement this section opens with is measured on this identical cell (matched aggregate displacement, decision change and clean accuracy, Table A3): the -0.273 is the outcome-gated ladder of §6.1, whose adversary may attenuate, and the $+0.125$ is the Mode S intervention above, which pins that adversary at $c=1$; both intervals are paired. The $n=5$ above is the same estimand at its frozen threshold’s own n : the two differ in n alone. Collider bias is usually taught as attenuation; **here the confounded design does not lose power, it points the other way**: a criterion calibrated against that contrast would be calibrated against the confound.

The two designs are not the same intervention: Λ_a moves in *both*, 0.081 against 0.033 ($2.4\times$), so neither is a clean statistic-only intervention and no p -value attaches to their difference. What the pair does establish, *within the rescaling class and defense family measured here* and by counterexample rather than by law, is that **statistic preservation is not sufficient to identify attack-suppression preservation**, which is not the claim that a preserved statistic is uninformative (Prop. 11, App. C.3; Table A2).

Seven cells, not one (App. D.3). To separate a property of the design from one of this aggregator and attack, we ran the contrast on seven cells. They disagree on five, across two attacks, three datasets and two CNN configurations, and **the reversal replicates**: the same cell on CIFAR-100 ($n=5$) reads -0.213 against $+0.071$, both intervals excluding zero. `coord.median` under *model scaling* disagrees too, so it is not specific to the pixel backdoor. A mechanism frozen for when they disagree is refuted out of sample and withdrawn, so §4 says to intervene rather than consult a rule.

6 WHAT THE BOUNDARY LICENSES, AND WHAT WE FOUND

The regime is CIFAR-10, a 4-layer CNN, $N=10$ clients, $f=0.2$ adversarial, with the attacks and seed counts of page 4; ASR means *max-committed ASR*, and the protocol is fixed in the supplement (§S6).

Four questions, four kinds of answer: *algebra* before any run, one *causal* intervention at fixed downstream defense and attack, *replication* on further datasets, and what *screening* costs in recall, the last three being Table A1’s tier words and the first a proof, of very unequal weight and never pooled. The core evidence is a seed-matched intervention on one cell, not a mean comparison across cells, and any cross-arm ordering or exact admission-to-suppression magnitude sits in the weakest band with nothing resting on it.

6.1 THE CONFOUNDED LADDER, AND WHERE ITS SIGN DISAGREES

The inadmissible design still told us something. Our earlier suites tried to validate ordering preservation by choosing a different downstream defense, the move Prop. 1 rules out. Varying only the *dose* at fixed d_2 and attack ($\kappa=0$ bit-identical to d_2 standalone) refutes that suite’s pooled prediction that disturbance predicts each arm’s rise, and in three of four arms ASR *falls* as the dose rises (one-sided $p=0.091$ against a frozen $p < 0.05$; Table A6). **The fault is ours and is now measured**: fixing $\text{mean}(c)=1$ leaves relative weights free, so a large spread mostly *dilutes* whichever client carries the poison: attenuation, not statistic disturbance (§S7, §S10).

6.2 THE DISSOCIATION HOLDS ACROSS FOUR AGGREGATORS AND TWO DATASETS

Under the instrument, our one confirming arm was mostly attenuation. What remains of reputation’s rise tracks the small *admission* change $\Delta\Lambda_a$ rather than the large decision change (App. D.2).

The Krum dissociation replicates on a second dataset and on a second architecture family: pre-registered reruns on EMNIST-byclass/simple_cnn and CIFAR-10/resnet18 change the decision in 0.643 and 0.556 of rounds, the adversary never admitted, and both $n=3$ point estimates fall inside the frozen ± 0.15 margin, -0.017 $[-0.033, -0.001]$ and -0.041 $[-0.183, +0.101]$, beside `cifar_cnn`’s -0.026 at $n=5$. Neither 95% interval is a plain null: EMNIST’s excludes zero, a

real effect smaller than the margin, and `resnet18`'s is wider than the margin itself, so equivalence there is indeterminate at $n=3$ rather than established (App. L, App. D.2).

The dissociation also replicates outside the rescaling family the theory covers: masking a random 0.8 fraction of each *benign* client's coordinates at fixed L_2 norm leaves adversarial updates exactly untouched, so nothing we prove predicts it: the decision changes in 0.467 of rounds at admitted adversarial mass 0.000 and suppression does not follow, and neither of the pre-registered refuting branches fired (App. D.2.1). One mask is not the class of all non-rescalings.

Two arms support the account rather than carry it. `cos_krum`'s selection is bit-identical at every rung yet ASR falls 0.173 at the first accuracy-gate-passing rung, from an identity 0.584 at which it never suppresses (one-sided $p=0.044$, $n=8$, 5/8 seeds), a pre-registered counterexample within the tested regime; and pinning payload share at fixed dispersion separates the theorem's two regimes, confirmed for `coord_median` and refuted for reputation (App. D.2, E, Table A9).

Two pre-registered regimes ask whether the disagreement needs a non-adaptive attacker or ten clients (App. D.4). Against an adversary that solves a joint evasion constraint for `coord_median`, the two designs still part: D , the seed-paired difference of the Δ s, is $-0.324 [-0.543, -0.105]$, the instrument rising where the outcome-gated ladder falls at every seed. It adapts to d_2 , not d_1 , an instrument no deployment runs, and beats the plain backdoor by 0.035 at one seed. At $N=100$, $K=20$ the disagreement reproduces and the sign reversal does not: D is $-0.126 [-0.217, -0.035]$, but both designs now read the dose as protective, so the reversal is a finding at this participation regime and the disagreement travels. Neither Δ is powered at $n=3$.

6.3 WHAT THE SCREEN CANNOT FIND, AND WHAT TO RUN INSTEAD

The screen fails exactly where Prop. 2 says it must: what it discards is $FG \rightarrow RFA$, the lowest-ASR composition in the menu, and **statistic-preservation screening is the rule that keeps it.** Against all nine variants an evaluator could run, the criterion sits at the high-precision, low-recall end (App. E.4, whose footnotes qualify both the rates and the C2 column). To find a menu's strongest composition, do not screen on inherited suppression at all. Two prospective tests price the menu and the criterion; the latter's four registered cells came out as predicted, and **that label does not survive more seeds**, where C1 fails and the criterion certifies nothing (§E.1, App. E.5).

7 CONCLUSION

Preserving a defense statistic does not suffice to identify preserved attack suppression, and the reason is structural: an outcome-gated cross-defense comparison conditions on the outcome it seeks to explain, which no additional data repairs (Prop. 1), and a screen of that shape also misses emergent suppression (Prop. 2). §5's dissociations witness both directions. *What transports is one rule, and it is not our screen:* to attribute an outcome to an upstream mechanism in a composed pipeline, hold the downstream mechanism and the threat fixed and intervene on the upstream component. Holding the threat fixed is what that rule costs: it is a diagnostic under a committed attack and not a guarantee against an adversary that adapts (§6).

What changes for an evaluator, without using our criterion at all: *do not* read a preserved statistic or an unchanged decision as evidence that suppression survived (Table 1); *do not* select the pairs you compare on the outcome you are about to attribute, the design that reversed our own sign; *do* run §4's four clauses and report where the chain broke. Both instruments read adversary identity, leaving the open question (App. I): **can admission (P4) be estimated without knowing which clients are malicious?**

ETHICS STATEMENT

This paper studies poisoning attacks in order to evaluate defenses against them, so its subject matter is dual-use. Three things bound the risk. Every attack we run is already published (model-scaling (Bagdasaryan et al., 2020), pixel-pattern backdoors (Chen et al., 2017), DBA (Xie et al., 2020)) and we introduce no new attack or new attack capability. All experiments are simulations on public benchmark datasets (CIFAR-10, CIFAR-100, EMNIST-byclass, UCI Spambase) with no

human subjects, no personal data and no deployed system involved. Our headline result is defensive and cautionary rather than offensive: it tells practitioners that a preserved downstream statistic is *not* evidence that a composed defense stack still suppresses an attack, which removes a false assurance rather than supplying an attacker with one. The instruments we build (Mode S, the payload-share instrument, score-only Krum) require knowing which clients are malicious and are therefore measurement devices, not deployable defenses; we state this at each use so that no reader mistakes them for a recommendation.

REPRODUCIBILITY STATEMENT

Every claim in the paper is tagged *general*, *formal* or *empirical* at the point it is made, and Table A2 states the scope conditions on each. For the formal claims, Propositions 1 and 2 are stated in full in the body and proved in the appendix, and Proposition 10 and Theorem 8 are stated and proved in the appendix and referenced from the body. For the empirical claims: each pre-registered arm has a decision rule git-committed before the corresponding runs existed, with the commit hash recorded in the text and a runner that refuses to start unless its own recorded commit matches; every table cell is emitted by a generator that reads only frozen artifacts rather than being transcribed; sample sizes, seed ranges and per-seed values are given for every arm, with per-seed detail in the appendix and the supplement; and the figures are recomputed per seed by the same scripts. Where a number was revised we report the revision and its cause rather than the corrected value alone, and where a derived condition turned out to be empty on our data we report it as empty rather than omitting it (§C.3). **No code or data package accompanies this submission**, and throughout this paper “the supplement” means the companion PDF only. What will be released on publication is specific rather than a general promise: every experiment runner and measurement script, the `results/` JSON artifacts every reported number is read from, the simulator itself, the generators that emit the figures, and the 26 pre-registration documents fixing the decision rules for every pre-registered arm, seed top-up and audit, which is the count of `experiments/pre-registration.*.md` and is checkable as one rather than taken on trust. Everything this paper’s tables and figures are computed from runs from the repository root on Python 3 with `numpy`, `scipy`, `matplotlib`, `torch` and `torchvision`. The analysis entry points behind them are **idempotent**: they read the frozen artifacts and rewrite them byte-identically, so every number we report is a deterministic function of a frozen artifact rather than of a re-run, and no table cell can drift between builds. What the paper itself provides is the pre-registration commit hashes, the frozen decision rules they fix, and per-seed values for every arm, which is what is needed to check the reported numbers against a reimplement.

AI USE STATEMENT

In this work, we used generative AI tools as a writing and analysis assistant: for prose editing and revision of the manuscript, for $\text{\LaTeX}$ restructuring, and for writing verification and analysis scripts that check the manuscript against the frozen result artifacts. We did not use generative AI tools to generate research ideas, to formulate the hypotheses, to design the experiments or pre-registrations, or to produce any experimental result, and no claim, number, figure or table in this paper was authored by an AI tool. We have reviewed all AI-assisted work: AI-assisted prose was checked against the underlying artifacts sentence by sentence, and AI-written analysis scripts were run against frozen inputs whose outputs are reproducible independently of the tool. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

REFERENCES

- Fatima Z. Abacha, Sin G. Teo, Yuanxiang Wu, Lucas C. Cordeiro, and Mustafa A. Mustafa. Fed-Surrogate: Backdoor defense in federated learning via layer criticality and surrogate replacement. *arXiv preprint arXiv:2605.11122*, 2026.
- Eugene Bagdasaryan, Andreas Veit, Yiqing Hua, Deborah Estrin, and Vitaly Shmatikov. How to backdoor federated learning. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 2020.

- Arjun Nitin Bhagoji, Supriyo Chakraborty, Prateek Mittal, and Seraphin Calo. Analyzing federated learning through an adversarial lens. In *International Conference on Machine Learning*. PMLR, 2019.
- Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Advances in Neural Information Processing Systems*, 2017.
- Xiaoyu Cao, Minghong Fang, Jia Liu, and Neil Zhenqiang Gong. FLTrust: Byzantine-robust federated learning via trust bootstrapping. In *Network and Distributed System Security Symposium (NDSS)*, 2021.
- Xinyun Chen, Chang Liu, Bo Li, Kimberly Lu, and Dawn Song. Targeted backdoor attacks on deep learning systems using data poisoning. *arXiv preprint arXiv:1712.05526*, 2017.
- Francesco Croce, Maksym Andriushchenko, Vikash Sehwal, Edoardo Debenedetti, Nicolas Flammarion, Mung Chiang, Prateek Mittal, and Matthias Hein. RobustBench: A standardized adversarial robustness benchmark. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- El Mahdi El Mhamdi, Rachid Guerraoui, and Sébastien Rouault. The hidden vulnerability of distributed learning in byzantium. In *International Conference on Machine Learning*. PMLR, 2018.
- Felix Elwert and Christopher Winship. Endogenous selection bias: The problem of conditioning on a collider variable. *Annual Review of Sociology*, 40:31–53, 2014.
- Lucas Fenaux, Zheng Wang, Jacob Yan, Nathan Chung, and Florian Kerschbaum. Hammer and anvil: Toward a theory of backdoors in federated learning. *arXiv preprint arXiv:2509.08089*, 2025.
- Clement Fung, Chris J. M. Yoon, and Ivan Beschastnikh. The limitations of federated learning in sybil settings. In *23rd International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, pp. 301–316, 2020.
- Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021.
- Sai Praneeth Karimireddy, Lie He, and Martin Jaggi. Byzantine-robust learning on heterogeneous datasets via bucketing. In *International Conference on Learning Representations (ICLR)*, 2022.
- Harsh Kasyap, Anil Kumar Pradhan, Ugur Ilker Atmaca, Graham Cormode, and Carsten Maple. PROVeFL: Private robust and verifiable aggregation in federated learning. *arXiv preprint arXiv:2607.06612*, 2026.
- Xuanqing Liu, Minhao Cheng, Huan Zhang, and Cho-Jui Hsieh. Towards robust neural networks via random self-ensemble. In *European Conference on Computer Vision (ECCV)*, 2018.
- H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 54, pp. 1273–1282. PMLR, 2017.
- Arnab Nath and Harsh Kasyap. CQSA: Byzantine-robust clustered quantum secure aggregation in federated learning. *arXiv preprint arXiv:2602.22269*, 2026.
- Thien Duc Nguyen, Phillip Rieger, Huili Chen, Hossein Yalame, Helen Möllering, Hossein Fereidooni, Samuel Marchal, Markus Miettinen, Azalia Mirhoseini, Shaza Zeitouni, Farinaz Koushanfar, Ahmad-Reza Sadeghi, and Thomas Schneider. FLAME: Taming backdoors in federated learning. In *USENIX Security Symposium*, 2022.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

- Krishna Pillutla, Sham M. Kakade, and Zaid Harchaoui. Robust aggregation for federated learning. *IEEE Transactions on Signal Processing*, 70:1142–1154, 2022.
- Phillip Rieger, Thien Duc Nguyen, Markus Miettinen, and Ahmad-Reza Sadeghi. DeepSight: Mitigating backdoor attacks in federated learning through deep model inspection. In *Network and Distributed System Security Symposium (NDSS)*, 2022.
- Virat Shejwalkar and Amir Houmansadr. Manipulating the byzantine: Optimizing model poisoning attacks and defenses for federated learning. In *Network and Distributed System Security Symposium (NDSS)*, 2021.
- Ziteng Sun, Peter Kairouz, Ananda Theertha Suresh, and H Brendan McMahan. Can you really backdoor federated learning? *arXiv preprint arXiv:1911.07963*, 2019.
- Nawrin Tabassum and Yanzhao Wu. STAR-FL: Secure federated learning with spatial-temporal analysis and robust aggregation. *arXiv preprint arXiv:2608.14861*, 2026.
- Adrita Rahman Tory, ABM Shawkat Ali, Md Abu Layek, and Khondokar Fida Hasan. Privacy, robustness, and fairness trade-offs in federated intrusion detection: Geometric indistinguishability at the aggregation interface. *arXiv preprint arXiv:2609.03420*, 2026.
- Florian Tramèr, Alexey Kurakin, Nicolas Papernot, Ian Goodfellow, Dan Boneh, and Patrick McDaniel. Ensemble adversarial training: Attacks and defenses. In *International Conference on Learning Representations (ICLR)*, 2018.
- Hongyi Wang, Kartik Sreenivasan, Shashank Rajput, Harit Vishwakarma, Saurabh Agarwal, Jy-yong Sohn, Kangwook Lee, and Dimitris Papailiopoulos. Attack of the tails: Yes, you really can backdoor federated learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- Chulin Xie, Keli Huang, Pin-Yu Chen, and Bo Li. DBA: Distributed backdoor attacks against federated learning. In *International Conference on Learning Representations*, 2020.
- Chulin Xie, Minghao Chen, Pin-Yu Chen, and Bo Li. CRFL: Certifiably robust federated learning against backdoor attacks. In *International Conference on Machine Learning (ICML)*, 2021.
- Cihang Xie, Jianyu Wang, Zhishuai Zhang, Zhou Ren, and Alan Yuille. Mitigating adversarial effects through randomization. In *International Conference on Learning Representations (ICLR)*, 2018.
- Jiahao Xu, Zikai Zhang, and Rui Hu. Achieving byzantine-resilient federated learning via layer-adaptive sparsified model aggregation. *arXiv preprint arXiv:2409.01435*, 2024.
- Dong Yin, Yudong Chen, Kannan Ramchandran, and Peter L. Bartlett. Byzantine-robust distributed learning: Towards optimal statistical rates. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.
- Zhengming Zhang, Ashwinee Panda, Linyue Song, Yaoqing Yang, Michael W. Mahoney, Prateek Mittal, Kannan Ramchandran, and Joseph E. Gonzalez. Neurotoxin: Durable backdoors in federated learning. In *International Conference on Machine Learning (ICML)*, 2022.

HOW TO READ THIS APPENDIX

The body is self-contained; the appendix is where each of its claims keeps its provenance. Four groups, in the order a sceptical reader would want them. (1) **What licenses a claim:** the evidence tiers, the levels in full, the proofs, and the identification argument term by term (App. A–C.3). (2) **What the screen buys, priced:** prospective test, metric-swap ablation, frontier, closed census (App. E.3). (3) **The intervention evidence:** the confounded ladder, the instrument that replaced it, seven cells, and the composition the screen cannot certify (App. D.2, D.3, F). (4) **What we do not claim:** limitations, the evidence-status ledger, notation, equivalence testing (App. H, I, K, L).

(5) **What is auxiliary, and the one block that only looks it.** Skippable without touching any claim in the body: the scale and architecture checks (App. N.1, N.2), which bound an external-validity statement and establish no

boundary of their own; the composition and adaptive-attack details (App. M); notation (App. K); the per-cell out-of-sample detail behind §E.6 (§S4); and the threshold-tolerance sweeps, whose result is that no published verdict moves (§S12). **The prospective suite (App. E.1) is not auxiliary, and we say so because its headline reads like a weakness:** it is the only out-of-sample prospective test here, it is where the criterion’s shortfall is priced at 13/20, and its FLTrust near-identity confound is the reason six of its arms cannot falsify anything. Cutting it would remove the evidence against us. **Nothing has been deleted to shorten this appendix,** and that is a considered choice rather than an oversight: every scope condition, limitation and withdrawal keeps exactly one home, since a disclosure relocated into a subordinate clause is one a later reader cannot find.

(6) One question asked and answered on its own: whether randomizing *which* composition is applied each round is itself costly at matched defense computation (App. O). It is a new pre-registered experiment with its own controls, not a reinstatement of the randomized-menu comparison this paper withdraws, and nothing else here rests on it.

Per top-level section, since the groups above cross section boundaries. One line each, in document order; *skippable* means no claim in the body rests on it.

- **App. A:** the three tier words every empirical claim in the paper is tagged with, and Table A1’s orthogonal strength axis (seed count, bit-identical, replicated). Read first if you want to know what any one sentence rests on.
- **App. B:** the five preservation levels stated in full. Reference; skippable if §3’s compressed version was enough.
- **App. C:** the screen C0–C3, the identification boundary, and every proof. Not skippable: Prop. 6 and Thm. 8 are stated and proved here, and the boundary is the paper’s first contribution.
- **App. E:** what the screen buys, priced. Precision 2/2, recall 40%, the prospective suite’s 13/20, and the two boundary conditions. Not skippable if you want the cost as well as the result.
- **App. D:** the intervention evidence, from the outcome-gated ladder through Mode S to the seven cells. Not skippable: the sign reversal the abstract leads with is measured, disclosed and replicated here, and this is where its seed counts are stated.
- **App. F:** the composition the screen cannot certify, which is our own flagship pair. Not skippable, since it is the evidence against us.
- **App. G:** related work, and the closed citation census with its 6/59 result. Reference.
- **App. H:** limitations L1–L6 in full, each with the claim it bounds. Not skippable.
- **App. I:** the evidence-status ledger, claim by claim, including what is withdrawn and what is merely supported. Not skippable if you are deciding what to believe.
- **App. J:** Figure 1(a)’s three numbers, which are three scopes rather than one experiment. Supporting; read it before quoting any of the three.
- **App. K:** notation. Reference, skippable.
- **App. L:** equivalence testing at the existing seed counts, that is, where the frozen ± 0.15 margin binds and where it does not. Not skippable if you doubt the margin.
- **App. M:** composition and adaptive-attack construction details. Skippable.
- **App. N:** scale, architecture and external validity. Skippable: it bounds an external-validity statement and establishes no boundary of its own.
- **App. O:** one self-contained pre-registered question, whether randomizing *which* composition is applied is itself costly. Skippable; nothing else rests on it.

A THE THREE TIERS OF EVIDENCE

Three tiers of evidence, of very unequal weight, never pooled. Every empirical claim about the preservation hierarchy or the criterion is tagged with one of the three words of Table A1 at the point it is made, so a reader can look up what licenses it instead of reconstructing it from hedges.

B PRELIMINARIES AND THE PRESERVATION LEVELS

Federated learning. N clients each hold a local dataset $\mathcal{D}_i \sim P_i$; the server learns $w^* = \arg \min_w \sum_i \frac{|\mathcal{D}_i|}{|\mathcal{D}|} F_i(w)$. In FedAvg (McMahan et al., 2017), each round the server broadcasts w^t , selected clients run local SGD, and the server aggregates $w^{t+1} = w^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$.

Poisoning and robust aggregation. An adversary controlling a fraction f of clients can flip labels (Chen et al., 2017), inject backdoor triggers (Bagdasaryan et al., 2020; Xie et al., 2020), or scale malicious updates to dominate aggregation (Bagdasaryan et al., 2020). The server may replace FedAvg with a Byzantine-robust rule: Krum (Blanchard et al., 2017), Trimmed Mean, Coordinate-wise Median, Norm Clipping (Sun et al., 2019), or RFA (geometric median) (Pillutla et al., 2022), each imposing an accuracy/robustness trade-off.

Table A1: **The three tiers of evidence, of very unequal weight and never pooled**, and the *orthogonal* strength axis. Every empirical claim in the paper carries one of the three tier words at the point it is made; the tiers are never combined into a single verdict. The lower panel is a second, independent axis: what a claim’s *design* licenses (upper panel) and how strongly the *statistics* support it are different questions, and a Causal-tier claim can still sit in the weakest band.

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§D.2)	statements about the <i>mechanism</i> , scoped to these cells
Replication	EMNIST-byclass, a second dataset and a differently configured CNN ($\Delta = -0.017$, $n=3$); <code>resnet18</code> , a second architecture family with architecture varying alone (-0.041 , $n=3$); <code>coord_median</code> , a second invariance class ($+0.098$, $n=5$); and CIFAR-100, a third dataset, where the sign disagreement of §D.3 replicates (-0.213 against $+0.071$, $n=5$)	“replicated on a further dataset or architecture, one arm each”, never “generalizes”; no composition suite leaves CIFAR-10
Screening	the outcome-gated 42-pair suite	evaluation prioritization only ; by Proposition 6 it cannot identify the mechanism’s contribution at any sample size
Statistical strength , an axis orthogonal to the tiers above. Read this before quoting any number.		
Strong	the invariance results (Prop. 10, Thm. 8) and the identification boundary (Prop. 6), which are <i>proofs</i> and carry no n ; <code>cos_krum</code> ’s exact score-only identity, which is bit-identical rather than estimated; the pre-registered Mode S intervention, frozen before any of its ASR existed; and the <i>direction</i> of the negative, which replicates on a second dataset, on a second architecture family and outside the rescaling family	
Moderate	the ASR practical-equivalence results at $n=5$, one of which was topped up to $n=20$ under a pre-registered clause and held ($\Delta = -0.010$); at $n=5$ these are TOST verdicts against a frozen margin, not tight estimates	
Weak, exploratory	any <i>cross-arm quantitative ordering</i> , which the admission reading predicted and which Table 1 refutes; and the exact (P4)→(P5) <i>magnitudes</i> , for which we claim the implication and not the size. Nothing in the paper’s argument rests on this band	

Game theory. A normal-form game is $G = (\mathcal{N}, \{\mathcal{S}_i\}, \{u_i\})$; an equilibrium is *pure* when every player’s strategy puts all its mass on one action and *mixed* otherwise. **No claim of this paper is game-theoretic**: this is needed only to read the residual scale check of Appendix N.1, and the type-space and Bayes-Nash machinery an earlier version used is not needed here.

C THE SCREEN, THE IDENTIFICATION BOUNDARY, AND THE PROOFS

The paper’s central practical question is *which* defense pairs compose well under deterministic joint application. Specifically, the criterion screens for *mechanism-preserving composition*: when one constituent defense is already individually effective (C1), can the other defense be added without breaking that suppression? This is the operationally relevant question (the criterion is not a general characterization of all composable pairs, but a sufficient condition for *mechanism preservation* among pairs where at least one component has known effectiveness), never a sufficient condition for low ASR. We answer it first, with a checkable criterion, and treat the alternative, temporal mixing, as outside the scope of this paper. **But the practical question is not what the paper establishes**: what it establishes is that preservation of the statistic the screen checks is not causally tied to the security the screen predicts, so the criterion below should be read as the application §E.3 scopes it to (an ordering over an evaluation budget) and not as a deployment recommendation.

Observe that each robust aggregation defense d operates through an *admission signal* $\sigma(u)$ applied to client update u : it accepts, clips, or down-weights u based on whether $\sigma(u)$ exceeds a threshold. Crucially, the admission signal varies by defense family:

- **Weighting defenses** (reputation, FoolsGold): $\sigma(u) = \text{distance from consensus } (\ell_2) \text{ or pairwise similarity.}$

- **Clipping defenses** (NormClip): $\sigma(u) = \|u\|_2$.
- **Rank-based defenses** (TrimmedMean, CoordMedian): $\sigma_k(u) = \text{coordinate-wise rank of } u_k$.
- **Geometric defenses** (RFA): $\sigma(u) = \text{distance from geometric median}$.

When two defenses are applied sequentially (d_1 transforms the updates, d_2 aggregates the result), d_1 's transformation may *corrupt* d_2 's signal or vice versa.

Criterion 3 (Signal-Preservation Composability: an evaluation prioritizer, never a security screen (L2)). *Let d_1 and d_2 be defense mechanisms. The following sufficient conditions guarantee **preservation of d_2 's suppression mechanism under composition, conditional on the standalone suppression C1 asserts**. They are **not** an independent guarantee of low composition ASR: Theorem 8, which proves them for one class, bounds no ASR. Stated once so the scope cannot be misread:*

$$C1 \wedge C2 \wedge C3 \Rightarrow d_2 \text{'s suppression mechanism survives composition, levels (P1)–(P3) of Def. 2.}$$

$$C1 \wedge C2 \wedge C3 \not\Rightarrow \text{low composition ASR.}$$

The conditions are of **two different kinds**, and we group them accordingly, because conflating an empirical screen with a mechanistic property is the main source of confusion about what the criterion claims.

Stage 0: applicability. Before either kind of condition is informative, the question the criterion asks has to be live.

(C0) The downstream mechanism is load-bearing. $\text{ASR}(d_1, a) \geq 0.5$ for at least one committed attack a . $C2$ and $C3$ concern preservation of d_2 's admission signal, so they say nothing when d_1 alone already suppresses everything: the composition inherits that suppression, destroying d_2 's signal costs nothing, and a correct prediction would be correct for a reason the criterion did not supply. The asymmetry is deliberate: a strong d_2 is the intended case, since asking whether upstream processing breaks an effective downstream defense is the framework's whole question, whereas a strong d_1 makes the question moot. $C0$ was **formulated post-hoc** and its provenance is in Remark 5.

The empirical screening condition. This one is measured rather than read off definitions, and it is the gate the identification boundary is about.

(C1) Standalone suppression. Some constituent suppresses the committed attack on its own: $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$, quantified per attack. With $C0$ this forces the suppressing constituent to be d_2 on the attack a^* that witnesses $C0$, and there only (Lemma 1), which is exactly what makes the gate a descendant of the quantity the screen is trying to estimate (Proposition 6). **On a committed attack that d_1 happens to suppress, the constituent doing the work can be d_1 instead**, because $C0$ is existential over attacks while $C1$ is universal over them; both of our certified pairs are of that kind, and §E reports what that costs.

Mechanism-preservation conditions. These are what Theorem 8 proves for the weighting→aggregator class, and they are determined by d_1 's and d_2 's definitions alone, no experiment required. **What we recommend is an evaluation-prioritization criterion, $C1 \wedge C2$, because that is the whole of what our evidence supports.** We do call it a screen throughout, in the ordinary sense of a cheap test run before an expensive one. **We disclaim predictive validity explicitly, here at first use:** the criterion orders candidate compositions for evaluation, and it does not predict whether a composition is secure. The pooled two-channel model over 26 cells is unconfirmed ($R^2=0.239$), and our own pre-registered replication **refuted** the cross-arm admission ordering the criterion would need in order to predict (§D.2); recall is 40% (Proposition 7), so the negative direction carries no information either. **It is never a security screen, and a PASS is an evaluation ordering rather than a deployment decision** (Table A2). $C3$ is retained as a **margin diagnostic** rather than a gate, because on our data it never binds on a pair $C1 \wedge C2$ already admits and shows no incremental predictive value (§E.3). The pre-registered labels of §E were frozen with $C3$ in the gate and are not relabelled; the demotion is a recommendation going forward, supported by a post-hoc factorial ablation, and it changes no reported number: $C1 \wedge C2$ selects exactly the same two in-distribution pairs.

(C2) Signal preservation. d_1 's per-client transformation T_1 does not destroy d_2 's admission signal: $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$ for malicious updates u_a .

(C3) Boundary preservation. d_1 's transformation does not compress the adversarial–benign separation below d_2 's discrimination threshold. Formally: if d_2 suppresses a by detecting $\tau_{d_2}(u_a) > \theta$ (triggering trimming/rejection), then $\tau_{d_2}(T_1(u_a)) > \theta$ must hold, i.e., d_1 must not map adversarial updates from above to below d_2 's rejection boundary.

When $C1$ holds but $C2$ fails (signal destruction), composition is no better than d_2 alone. When $C1$ and $C2$ hold but $C3$ fails (boundary compression), d_2 retains its admission signal direction but its threshold no longer fires: the attack passes because d_1 's rescaling places adversarial magnitudes within d_2 's acceptance region.

Writing C2 as preservation of d_2 's admission signal elides a distinction our own experiments later force, so we name the levels before relying on the condition. They are the paper's main conceptual instrument, and every claim below is stated at one of them.

Definition 2 (Preservation levels). *For an upstream transform T_1 and a downstream defense d_2 reading admission signal σ_2 :*

- (P1) **Value invariance:** $\sigma_2(T_1(U)) = \sigma_2(U)$.
- (P2) **Ordering invariance:** $\sigma_2(T_1(U)) = \varphi(\sigma_2(U))$ for some φ preserving the ordering σ_2 induces on clients. This is the level C2 is stated at (made explicit in (2)); the distinction from (P1) matters, because exact values are sufficient but never necessary.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the term we use throughout for whether adversarial input survives d_2 , and it is a level, **not one measurable quantity:** for a selector it is the indicator that an adversarial update is selected at all; for a weighted averager, the adversarial share of the weight mass; for a coordinate-wise order statistic, the adversarial share of selected coordinates, which is why Table A8's influence column is aggregator-specific by necessity. The committed pre-registrations say "admission", so that word is retained in every frozen rule name and no scored quantity is relabelled.
- (P5) **Attack-suppression preservation:** ASR stays below threshold.

(P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) hold, and hold by definition: equal values induce the same ordering, and a rule reading only that ordering cannot decide differently. **These three are logical implications under the definitions above, not empirical claims about how defenses behave,** and nothing is asserted by proving them: their content is that they stop at (P3). **Below (P3) the chain breaks, and it breaks in both directions.** Both breaks are measured. Holding decision and admission invariance does not deliver suppression: `cos_krum`'s selection is bit-identical at every dose and its admitted adversarial mass is 0.000 throughout, yet its ASR falls 0.173 by the first rung alone through the magnitude channel, at $p=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). Across the whole ladder it falls 0.425, but that endpoint sits below our own accuracy floor, so **the gate-passing rung is the witness we rest on** (§D.2). Losing decision invariance does not cost suppression: Krum's selection flips in up to 80% of rounds under an intervention that pins the adversarial mass, and its ASR stays inside the pre-registered practical-equivalence margin ($|\Delta|=0.026 < 0.15$) (§D.2). A criterion stated at (P1)–(P2) therefore constrains security in neither direction. **(P4) is where the security content sits, but we do not claim it is sufficient for (P5), and it is a level rather than one universal scalar security quantity.** It is instantiated per aggregator by the measure that aggregator's own rule exposes, Krum's and `cos_krum`'s selected-client indicator, `coord_median`'s share of selected coordinates, reputation's adversarial weight share (`experiments/build_channel_table.py`), so **(P4) names what to measure, not a single number comparable across defenses.** It is the first level that refers to adversarial influence on the aggregate rather than to the downstream defense's internal decision, and `cos_krum` is our own witness that (P4) $\nRightarrow$ (P5): admission identically 0.000 at every rung while suppression moves. **So the hierarchy answers "which level does a security claim have to be stated at?" and not "which level predicts ASR?":** no level in it is offered as a predictor.

Remark 4 (Which level C2 asserts, and which one is load-bearing). At $p = 54.6$ the uncontrolled dose ladder of §D, whose adversary is free to attenuate, changes Krum's decision in 86.7% of rounds while changing its admission of adversarial input in 13.3%, and Krum's suppression does not deteriorate at all. **Those two figures are the uncontrolled arm's and no other's:** Mode S, which pins the adversary at the same rung, reads 0.733 and 0.000 instead (Table I, `dose|krum|2.0|*` against `doseS|krum|2.0|*` in `results/admission_measurement.json`), which is why Table I can say admission never moves in any round of any arm it reports. A selection that flips from one benign client to another disturbs the statistic maximally and the mechanism not at all. **Terminology, once and for all: "mechanism" always means the downstream defense's discriminative operation (what separates adversarial from benign updates) and never its security behaviour, since that implication is precisely what this paper refutes.** We reserve "mechanism preservation" for the formal property Theorem 8 establishes, saying statistic or decision preservation where only the weaker property holds. C2 above is the (P2) condition (which is what makes it checkable from definitions alone) but the load-bearing reading is (P4). **This is the paper's central tension: our theorem operates at (P1)–(P3), while the experiments indicate security behaviour is decided at (P4);** §D.2 puts the two against each other in an experiment where they predict opposite outcomes, and Figure A6 is that argument in one picture.

Remark 5 (C0's provenance: an applicability condition formulated post-hoc). C0 sits in Criterion 3 as a stage-0 applicability condition because Lemma 1 is stated over $C0 \wedge C1$, but it **does not have the same evidential status as C1–C3**, and we record that here rather than let its promotion obscure it. It is **vacuous on the 7-defense menu of §E**, where the strongest single defense reaches only 0.519, and we formulated it only after the prospective suite of §E.1 introduced a defense for which it binds. It is a lesson from prospective evaluation,

applied there retrospectively; it is not part of the criterion validated on the 42 pairs, and none of the accuracy figures in §E depend on it.

A scope condition on the criterion itself, not a limitation of our runs. C1 and every prediction built on it are defined against a *committed* adversary, one whose attack is fixed before the composition is chosen. An adversary who observes the composition can re-target whichever channel the screen certifies, so **what the screen bounds is inherited robustness under a fixed attack, never robustness against an adaptive one**. Our single whitebox composition-aware stress test is consistent with that reading (0.064 $\rightarrow$ 0.157 seed-matched at $n=30$ on the pixel arm the attack is built from, 2.4 $\times$; §F).

C.1 THE TESTABILITY BOUNDARY

C0 and C1 together constrain what any validation of this criterion can look like. We did not anticipate this; it turns out to explain the central weakness of our own prospective suite (§E.1, L4), and (once stated in general form) to *derive* the class of within-defense designs in which C2’s contribution is identifiable (§D). Its content is narrower than “a general identification boundary” and worth stating precisely: **under inherited-effect gating**, cross-defense contrasts cannot identify the mechanism’s contribution, because the gate conditions on a descendant of both the mechanism and the outcome. What we offer as the transferable contribution is not that negative but **the admissible design it derives**, set out below and run in §D.2.

Lemma 1 (Testability boundary). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Proof. C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not attained at d_1 , hence it is attained at d_2 . $\square$

Nothing in the argument is specific to federated learning, to backdoors, or to our defense menu. Lemma 1 is **the enabling observation for the FL instance**; the substantive claim is the general one, and we state it as such.

Proposition 6 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating; Proposition 1 of the body, restated with proof). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense**.*

Proof. The load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are therefore unpopulated whatever $\mathcal{D}$ is, and P is never observed against a downstream defense that fails on its own. Any estimator of P ’s contribution from such contrasts is unidentified. $\square$

Corollary 1 (The design that restores identification). *Fix d_2 and a . Let $\mathcal{T}$ be a set of upstream transforms with $\text{id} \in \mathcal{T}$, let the experimenter assign $T \in \mathcal{T}$ by a rule that is a function of the design alone, and let $Y := M(T \rightarrow d_2, a)$ be run and recorded for every assigned T , the control $T=\text{id}$ included. Then for every $t \in \mathcal{T}$ the contrast*

$$\tau(t) := \mathbb{E}[Y \mid \text{do}(T=t)] - \mathbb{E}[Y \mid \text{do}(T=\text{id})]$$

is identified, and equals the difference of the two arms’ means. Its scope is narrow, and stating it is half the point. $\tau(t)$ is the total effect of the assigned transform at that one (d_2, a) : it separates into a statistic channel and an admission channel only when one of them is pinned by construction (§E.2), and it transfers to no other downstream defense, no other attack, and no adversary that adapts to T .

Proof. Assignment is a function of the design, so T is independent of the potential outcomes $\{Y(t)\}_{t \in \mathcal{T}}$, and every arm is run, so $\Pr(T=t) > 0$ on $\mathcal{T}$: ignorability and positivity hold by construction rather than by assumption, and $\mathbb{E}[Y \mid \text{do}(T=t)] = \mathbb{E}[Y \mid T=t]$. The cell that Proposition 6 empties is emptied by conditioning on E , a descendant of Y ; here nothing is conditioned on, and the control arm is run whatever its outcome. The scope clause is a restatement of $\text{do}(T=t)$: the intervention is on the transform, not on one channel of its effect. $\square$

Table A2: **Scope of claims: three kinds, and the one place each scope condition is stated. Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them. This table is the conservation record for every caveat in the paper; §H restates the same conditions individually as (L1)–(L6).

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution, and admissible designs are within-defense; an inherited-suppression gate cannot certify emergent synergy (Prop. 6, Prop. 7).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that any composition <i>is</i> secure, and none that mechanism preservation implies suppression. Recall is bounded, not achieved: the bound is 80% from $ \mathcal{E} / \mathcal{S} =1/5$, and 40% is what we measure (L2).
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators, with explicit margins (Thm. 8, Lem. 2, Cor. 4, Prop. 10, Prop. 9).	Proved on that class only: a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first admits no such T and is covered by nothing here . Bounds no ASR. C0 was formulated post-hoc and applied retrospectively to arms frozen before it (Rem. 5, L4). Regime A’s condition is not operationally checkable, r^* not being a stable per-attack constant (L5). Cor. 4’s bound goes vacuous as $m \rightarrow 1$, which heterogeneity realizes.
Empirical	§D.2’s interventions and §E–§E.1’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $K \in \{5, 10\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha \in \{0.1, 0.5\}$, star topology , three attack families; one arm on EMNIST-byclass/simple_cnn, one on CIFAR-100/cifar_cnn and one on CIFAR-10/resnet18 varying architecture alone.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos_krum</code> , $n=3$ frozen and $n=5$ pooled post-hoc on EMNIST-byclass, $n=3$ on the <code>resnet18</code> arm, $n=30$ for the FG→RFA characterization), all fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered practical-equivalence margin. ASR is thresholded at 0.5, the level at which the trigger fires on the majority of triggered inputs; a cell below the 0.35 clean-accuracy floor is <i>uninterpretable</i> , not suppressed. Dataset transfer is tested on <i>two</i> arms and no more: the flagship Mode-S arm on EMNIST-byclass/simple_cnn, and the sign-reversal cell of Table A10 on CIFAR-100, where the reversal replicates, the confounded leg falling and the controlled leg rising (§D.3). Both are “replicated on a further dataset”, never “generalizes” (L1). The EMNIST arm is a <i>channel</i> replication and carries no screen: no single defense measured on EMNIST-byclass suppresses the committed pixel backdoor alone (minimum 0.609; the menu measured there excludes <code>foolsgold</code> , <code>reputation</code> and the cosine mirrors), so C1’s per-attack precondition fails on that arm and the strategy comparison is out of scope over the measured menu rather than merely untested (L1). “ Refuted ” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved. The $\gamma \approx 0.66$ boundary is architecture-conditional (L1); the strongest development-set figures are <i>fit</i> , not prediction (L2).
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c (§D.2, supplement §S10).	They identify a causal channel and are not screens: no deployed defense knows which clients are adversarial, and estimating (P4) without that knowledge is the open problem of §I. Admission is <i>a direction to measure</i> , not a validated predictor: the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and its cross-arm ordering was refuted by our own replication (L6). C1 and everything built on it assume a committed adversary, so what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

Table A3: **The confounded ladder against the Mode-S instrument, on the CIFAR-10 arm that has both;** §D.3 adds a CIFAR-100 arm that has both and reproduces the sign disagreement. Same downstream defense, same attack, same seeds, same nominal dose, and the two designs share their $\kappa=0$ cell *numerically* over all 20 seeds (mean and worst per-seed gap 0.00e+00, asserted at 10^{-12}). **Each row carries its own n , because this table is mixed.** A pre-registered top-up (experiments/pre-registration-reversal-seed-topup.md) added seeds 47–61 to both legs at the two *endpoint* rungs, so the outcome rows are $n=20$; the mechanism rows are the frozen $n=5$ measurement on seeds 42–46, which is the only n they were ever recorded at. At the top rung the two designs are matched on aggregate displacement and decision change (at $n=5$) and on mean clean accuracy (at $n=20$), so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention, which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table (the adversarial share of selected coordinates, the same field in the same artifact) and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. **Which p this is, since the top-up changed it.** Each effect’s p is the one-sided *endpoint* paired t at that effect’s own $n=20$, on the same per-seed differences as the interval beneath it. It is *not* the four-rung Jonckheere–Terpstra trend test: the interior rungs were not topped up, so that test stays at $n=5$ where it was frozen, and its values ($p_{\downarrow}=0.0015$, $p_{\uparrow}=0.069$, both $n=5$) are reported in §D.3 rather than mixed into this column. This is a two-column contrast on one arm: it locates where the two designs differ, and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`, which also asserts that the published $n=5$ pair, -0.272 against $+0.098$, still reproduces on the frozen seed block; nothing here is retyped.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	coord_median	coord_median
attack	committed pixel	committed pixel
seeds	42–61	42–61
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share ($n=5$)	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c ($n=5$)	+0.000 $\rightarrow$ -0.336	0 by construction (read back $9e-08$)
aggregate displacement ($n=5$)	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change ($n=5$)	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$ ($n=5$)	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy ($n=20$)	0.761 $\rightarrow$ 0.692	0.761 $\rightarrow$ 0.703
ASR ($n=20$)	0.478 $\rightarrow$ 0.205	0.478 $\rightarrow$ 0.603
ASR effect ($n=20$)	-0.273 ($p_{\downarrow} = 6.4\times 10^{-9}$)	+0.125 ($p_{\uparrow} = 1.9\times 10^{-8}$)
paired 95% interval	[-0.334, -0.213]	[+0.095, +0.155]

This is the elementary converse of the proposition and we claim no depth for it. Its content is that the repair is **available and cheap, and is not what cross-defense comparison does**, including in our own first two suites: §5 runs it, and Table A3 is the CIFAR-10 cell where both designs exist, where they agree numerically at $\kappa=0$ and disagree in *sign* at the top rung; §D.3 reports a second such cell, on CIFAR-100, where the disagreement replicates.

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*, which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Proposition 7 (Inherited-suppression screens cannot certify emergent synergy). *($d_1 \rightarrow d_2$)’s suppression of a is emergent if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 6. Hence, writing S for the compositions with $M < \theta$, $\mathcal{E} \subseteq S$ for the emergent ones and $S \setminus \mathcal{E}$ for the inherited ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|S|$ on every menu and every attack.*

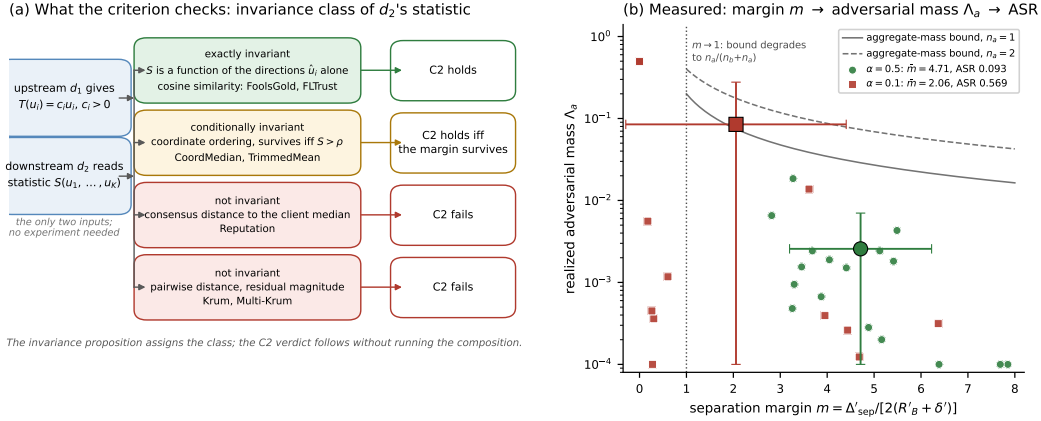


Figure A1: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 10), and the class determines the verdict: no composition needs to be run. This is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives. Each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 2 and carry no payload); large markers are mean $\pm$ s.d. Corollary 4’s bound holds in every round, and the $\alpha=0.1$ collapse is visible as a collapse in the theorem’s own quantity: m falls from 4.71 ± 1.51 ($n=17$ rounds) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a/(n_b+n_a)$, and realized Λ_a rises from 0.003 to 0.085 (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$ (that $\alpha=0.1$ cell is below our accuracy floor and so uninterpretable as ASR; the margin collapse plotted here is not). Values below 10^{-4} are drawn at 10^{-4} .

Proof. Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

Corollary 2 (The cap reaches 0, so it is not a reassurance). *If $\mathcal{E} = \mathcal{S}$, every screen carrying that eligibility gate has recall exactly 0 on that menu, at every sample size: it certifies no low- M composition at all. And $1 - |\mathcal{E}|/|\mathcal{S}|$ is a quantity the screen’s own user cannot report, since $|\mathcal{E}|$ is measurable only by running the compositions the gate rejects.*

Proof. Every member of $\mathcal{S}$ is then emergent, hence rejected by Proposition 7, so the certified set meets $\mathcal{S}$ in $\emptyset$ and recall is 0; recall is also non-negative, so the bound is attained. For the second claim, deciding membership in $\mathcal{E}$ requires $M(d_1 \rightarrow d_2, a)$ for a pair with $\min_d M(d, a) \geq \theta$, and that is a pair the gate rejects. $\square$

Our own menu witnesses $\mathcal{E} \neq \emptyset$: FoolsGold→RFA reaches 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845 (§F). One emergent pair among five low-ASR pairs puts the cap here at 80% against a measured 40%, so emergence explains part of the gap; had this menu’s low-ASR set been emergent throughout, the same criterion would have certified nothing. That is why we report recall as an operating point rather than as a residual (§E.4).

Mechanism–effectiveness confounding: Proposition 6 in our instance. The boundary has an uncomfortable consequence for validation, and it is worth spelling out concretely. In *every* in-scope pair the criterion certifies, d_2 already suppresses the attack that d_1 fails. A certified composition’s low ASR can therefore always be attributed to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3): the two are confounded **by the conditions themselves**, not by an unlucky choice of defense menu. We name this **mechanism–effectiveness confounding** and offer it as a methodological observation about validating composition criteria of this shape, applicable beyond our own experiments.

In particular it rules out the experiment we previously named as decisive: a d_2 whose admission signal is invariant under positive rescaling but which is *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. The achievable substitute is **per-attack complementarity**: a d_2 that suppresses the attack d_1 fails while remaining weak on a *different* committed attack, so its max-committed ASR stays high and it cannot suppress the composition single-handedly. Reputation is such a defense (0.017 on model-scaling, 0.842 on the pixel backdoor), and §E.2 builds the test on that observation.

Argument sketch. (C1 sufficient) If at least one of d_1, d_2 individually suppresses a , composition inherits that suppression provided $C2 \wedge C3$ preserve the signal chain. **That covers two routes and only one of them is this paper’s subject.** If the suppressing constituent is d_2 , the inheritance is exactly what $C2 \wedge C3$ are for. If it is d_1 , the suppression is inherited because d_1 sits upstream and nothing downstream can undo it, so $C2 \wedge C3$ are not required on that attack and the preservation argument contributes nothing to it. Both routes count toward a max-committed label, which is why the label can be right on an attack the mechanism has no purchase on (§E). C1 is sufficient, not necessary: two individually-weak defenses may suppress synergistically through complementary mechanisms (e.g., partial downweighting + geometric rejection), but predicting such synergy requires the quantitative margins below. The criterion is therefore *conservative* in the direction this sketch argues for: it may reject working compositions. **It is not a guarantee in the other direction, and nothing here establishes one.** That no clean $C1 \wedge C2 \wedge C3$ PASS is a false positive is a *measured* property of 42 pairs (§E.3), not a consequence of the argument above; the criterion has no established predictive validity ((L2), §H), and against an adversary that solves both constituents’ constraints at once a certified pair’s ASR rises $2.4 \times$ (§F).

(C2 role) Suppose d_2 individually suppresses a via signal $\sigma_2(u_a) > \theta$. If d_1 ’s transformation T_1 destroys this signal ($\sigma_2(T_1(u_a)) \not\approx \sigma_2(u_a)$), d_2 cannot detect the attack. Example: NormClip as d_1 maps every client norm above τ to τ . A σ_2 keyed on the update norm is then client-independent, and C2 fails outright. Reputation’s consensus distance $\sigma_2(u) = \|u - \hat{\mu}\|$ does not collapse that way: at $\|u_i\| = \tau$ for every i , $\|u_i - \hat{\mu}\|^2 = \tau^2 + \|\hat{\mu}\|^2 - 2\langle u_i, \hat{\mu} \rangle$, so client dependence survives through the inner product. Clipping removes the radial part of the separation and no more. For a purely radial attack that is the whole of it, and model-scaling is one, since it multiplies the update by a constant that clipping undoes exactly; where the adversarial direction also differs, the angular part survives and C2 fails by margin compression rather than by annihilation. This is measured, not hypothetical: NormClip→reputation leaves model-scaling ASR at 0.807 ± 0.048 and the pixel backdoor at 0.842 ± 0.047 over 5 seeds, against reputation alone at 0.017 (per-seed values in §M.1). Quantitatively: C2 holds when relative signal distortion $\Delta_{\text{signal}} := |\sigma_2(T_1(u_a)) - \sigma_2(u_a)| / \sigma_2(u_a) < \delta_2$ for a defense-specific tolerance δ_2 .

(C3 role) Even when d_1 preserves the *direction* of d_2 ’s signal (C2 holds), it may compress magnitudes below d_2 ’s threshold. Example: FoolsGold as d_1 downweights adversarial clients by $\alpha \ll 1$. TrimmedMean’s coordinate-wise rank signal still distinguishes adversarial coordinates (C2 holds), but after rescaling by α , per-coordinate values $\alpha \cdot u_{a,k}$ fall within the trimming range: the coordinate-wise extreme that would trigger trimming is no longer extreme. Quantitatively: C3 holds when the margin ratio $\Delta_{\text{margin}} := (\tau_{d_2}(T_1(u_a)) - \theta) / (\tau_{d_2}(u_a) - \theta) > 0$, i.e., adversarial scores remain above d_2 ’s rejection boundary after d_1 ’s transform.

(Sufficiency) If C1 holds, C2 holds ($\Delta_{\text{signal}} < \delta_2$), and C3 holds ($\Delta_{\text{margin}} > 0$), then d_2 detects and suppresses u_a even after d_1 ’s transformation. For the weighting→aggregator class, Theorem 8 instantiates these margins explicitly: $\rho < r^*$ for the rank aggregators, and the transformed-point separation condition $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ for the geometric one, which Corollary 3 restates in a form that can be checked without running the composition. $\square$

C.2 FROM THE FEDERATED PROTOCOL TO THE ABSTRACT STATEMENT, TERM BY TERM

Proposition 6 is stated over an abstract screen precisely so that nothing in it rests on federated learning, and Lemma 1 is its instance under our criterion. That correspondence is what carries the result into this paper and out of it again, so we display it rather than assert it. Figure 2 in the main text is the same object drawn as a graph, including the design that repairs it; Table A4 is the same object written as a substitution.

Two features of the substitution carry the whole argument, and both are legible in the table rather than argued for. **First, E and the quantity the screen predicts are the same measurement.** C1 is an ASR comparison at 0.5 and the PASS label is an ASR prediction at 0.5, so the gate is not an unlucky covariate that a larger menu would decorrelate: it is a function of the outcome. **Second, P is decided without running anything.** C2 and C3 are read off d_1 ’s and d_2 ’s definitions, which is what Prop. 10 makes algebraic, so no version of the design recovers identification by measuring P more carefully. The two together are what force the within-defense design of §D.2: with d_2, a and the seeds fixed, E is fixed too, and the intervention on T moves P alone.

What the mapping does not license. It shows our criterion is *an* instance, not that every composition screen is one. A screen whose gate reads something other than the outcome it predicts, a per-client norm statistic say, has a different E and falls outside Proposition 6 entirely; the proposition is a statement about outcome-gated screens, ours included. Nor does the mapping transfer any *empirical* claim: the middle column names measured quantities, and their scope is Table A2’s, not the proposition’s.

Table A4: **The abstract statement and our criterion are the same object under one substitution.** Left: the symbols of Proposition 6, which mentions no dataset, architecture, attack family or defense. Middle: what each becomes under Criterion 3. Nothing in the middle column is new here; the table is a reading aid for a correspondence the two statements already have.

Proposition 6	Our federated instance	Stated at
outcome metric $M(d, a)$	ASR(d, a), the committed attack’s success rate under defense d	Criterion 3
threshold θ	0.5, the level at which the trigger fires on the majority of triggered inputs	Table A2
load-bearing condition $M(d_1, a) \geq \theta$	(C0), $\text{ASR}(d_1, a) \geq 0.5$ for some committed a	Rem. 5
eligibility gate E : $\min_d M(d, a) < \theta$	(C1), $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$, quantified per attack	Criterion 3
mechanism condition $P(d_1, d_2)$	(C2) $\wedge$ (C3): $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$, and the margin survives T_1	Prop. 10, Thm. 8
menu $\mathcal{D}$	the pairs $(d_1 \rightarrow d_2)$ a suite draws from a defense library	§E
certification $M(d_1 \rightarrow d_2, a) < \theta$	the criterion’s PASS label on a pair	§E.1
the empty cell of the 2×2	$\neg P$ against a d_2 that fails alone: no admitted pair has one	Lem. 1
holding d_2, a, E fixed	Mode S and Mode A: same d_2 , attack and seeds, intervene on T	§D.2

C.3 A RESTRICTED THEOREM: $C2 \wedge C3$ FOR WEIGHTING $\rightarrow$ RANK AND WEIGHTING $\rightarrow$ GEOMETRIC COMPOSITIONS

What follows is a boundary theorem, and deliberately not a security theorem. It bounds what an *invariance* argument about composition can establish: preservation of the downstream defense’s discriminative mechanism, levels (P1)–(P3) of Definition 2, and it says nothing whatever about suppression, which lives at (P4)–(P5). We state that in the same breath as the theorem because the whole point of the paper is the gap between the two: the theorem is the best case for statistic-level reasoning, proved with explicit margins, and §D.2 then shows that even exact invariance neither implies nor preserves suppression. Read it as a *formal* claim in the sense of Table A2: proved, on one class, bounding no ASR.

The criterion above is an operational heuristic. For the class of compositions that matter most in practice, a per-client *weighting* defense d_1 (reputation, FoolsGold) followed by a *rank-based* or *geometric* aggregator d_2 (TrimmedMean, CoordMedian, RFA), which contains the flagship FoolsGold $\rightarrow$ RFA, C2 and C3 can be *proven* rather than asserted, with an explicit tolerance on how aggressively d_1 may reweight. The key structural fact is that d_2 ’s admission score is scale-equivariant, so it commutes with d_1 ’s reweighting up to a bounded ratio; the composition fails only when d_1 compresses that ratio past a margin set by d_2 ’s separation.

Theorem 8 (Bounded reweighting preserves the downstream discriminative mechanism). *Consider a round with n_b benign and n_a adversarial updates under honest majority $n_b > n_a$. Let d_1 be a per-client reweighting $T_1(u_i) = w_i u_i$ with $w_i \geq 0$, and define the ratio over strictly positive weights, $\rho := \max_i w_i / \min_{i: w_i > 0} w_i$. Suppose d_2 suppresses attack a through a client admission score $\sigma_2(\cdot; U)$ that is scale-equivariant, $\sigma_2(c u_i; c U) = c^k \sigma_2(u_i; U)$ for all $c > 0$ and some degree $k \geq 0$. Assume $w_a > 0$ for adversarial clients; the $w_a = 0$ case is Lemma 2. The theorem establishes preservation of d_2 ’s suppression mechanism; it does not bound ASR. Then:*

- (Rank aggregators; $C2 \wedge C3$ hold up to a margin.)** If $d_2 \in \{\text{TrimmedMean}, \text{CoordMedian}\}$, so σ_2 is a coordinate-wise rank, then d_2 ’s per-coordinate suppression of a is preserved when two conditions hold. (i) **Count majority:** $n_a < K/2$ for CoordMedian, and $n_a \leq \lfloor \beta K \rfloor$ for TrimmedMean with β -trimming. (ii) **Signed ordering:** on each suppression-relevant coordinate k , the adversarial transformed value lies strictly beyond every benign one, $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail). Sufficient magnitude condition for (ii): if $\text{sign}(u_{a,k})$ is common to all adversaries and $|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds; writing $r^* := \min_k \min_a \min_b |u_{a,k}| / |u_{b,k}|$ for the coordinate-wise separation margin, which minimizes over adversaries as well so that the bound binds on the least extreme one, this is $\rho < r^*$. Magnitude alone is not sufficient without the sign assumption, and the reason is not that a large adversarial coordinate could sit between the benign values: since $\rho \geq 1$, the magnitude condition gives $|u_{a,k}| > \max_b |u_{b,k}| \geq \max(\max_b u_{b,k}, -\min_b u_{b,k})$, which forces $u_{a,k} > \max_b u_{b,k}$ or $u_{a,k} < \min_b u_{b,k}$. What magnitude leaves open is which of the two. Mixed adversarial signs split the adversaries across both tails, while (ii) is a one-tail condition and (i)’s count budget is written

against a single tail; a common sign is what puts them all in one. With $n_a=1$ the sign assumption is therefore vacuous. At $\rho = 1$ ranks are preserved exactly, recovering Proposition 9.

2. **(Geometric median / RFA; C2 $\wedge$ C3 hold up to a margin.)** The composition hands RFA the transformed points $u'_i := w_i u_i$ and RFA runs its standard unweighted geometric median on them; it is not a weighted geometric median, and w_i enters only by moving the points. All quantities must therefore be defined on the transformed set. Let μ'_B be the geometric median of the benign u'_b alone, μ' that of all of U' , $R'_B := \max_b \|u'_b - \mu'_B\|$ the transformed benign radius, and $\delta' := \|\mu' - \mu'_B\|$ the displacement caused by including the adversarial points. Suppression is preserved when

$$\Delta'_{\text{sep}} := \min_{a,b} \|u'_a - u'_b\| > 2(R'_B + \delta'),$$

under which every adversarial residual $\|u'_a - \mu'\|$ strictly exceeds every benign residual $\|u'_b - \mu'_B\|$, so RFA's inverse-residual weight on a stays below the benign floor.

3. **(Norm-equalizing d_1 ; C2 fails or is compressed: the degenerate case.)** If instead d_1 maps every client norm to a common value (NormClip with τ below all client norms), the consequence depends on what σ_2 reads. A σ_2 keyed on the update norm becomes constant across clients, so it stops separating a and C2 fails outright. A σ_2 keyed on consensus distance does not become constant: at $\|u_i\|=\tau$ for all i , $\|u_i - \hat{\mu}\|^2 = \tau^2 + \|\hat{\mu}\|^2 - 2\langle u_i, \hat{\mu} \rangle$, so client dependence survives through the inner product. Equalization removes the radial part of the separation and compresses the rest to its angular part, and C2 then fails once that residual drops below d_2 's threshold, which is a margin condition of the same form as parts (1) and (2) rather than an algebraic collapse. Nor is the weight ratio infinite: $w_i = \tau/\|u_i\|$ is finite for every non-zero update, so ρ is a finite scalar and a ρ -margin formally applies. What fails is that ρ is set by the realized norms, so no margin fixed before the round bounds it; weight exactly zero is the genuinely unbounded case, and that is Lemma 2.

Parts (1) and (2) are sufficient conditions for d_2 to preserve the downstream suppression ordering, not to exclude adversarial contributions outright. Neither is a sufficient condition for low ASR, and part (2) orders residuals per client only; it constrains the aggregate adversarial coefficient solely through Corollary 4.

Proof. (1) TrimmedMean/CoordMedian act on coordinate k through the rank of $w_i u_{i,k}$ among $\{w_j u_{j,k}\}_j$; a is suppressed on k because its transformed coordinate is extreme (trimmed, or not selected by the median). Given (ii), at most n_a of the extreme order statistics are adversarial, so with (i) the selected value is benign. For the magnitude condition, $w_a > 0$ gives $\rho \geq \max_b w_b/w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a sign common to all adversaries this places $w_a u_{a,k}$ strictly beyond every benign value on that side. Scale-equivariance ($k=0$ for pure rank) makes the uniform case $\rho=1$ exact. (2) All quantities are on the transformed points $u'_i = w_i u_i$, which is what RFA receives. Every benign residual obeys $\|u'_b - \mu'\| \leq R'_B + \delta'$ by the triangle inequality. For adversarial u'_a , $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$, hence $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Demanding that this exceed the benign bound gives $\Delta'_{\text{sep}} - R'_B - \delta' > R'_B + \delta'$, i.e. the stated condition. No weighted-majority assumption is needed, and none would be appropriate: RFA weights every input point equally and the w_i act only by relocating the points. Our implementation runs Weiszfeld for 50 iterations at tolerance 10^{-6} , so the argument transfers to it. (3) Clipping to a common τ sends $\|u_i\| \mapsto \tau$ for all i . For $\sigma_2(u_i) = \|u_i\|$ (NormClip-as- d_2) the score is then a client-independent constant, separation vanishes, and that is C2-failure outright. For the consensus distance, expand $\|u_i - \hat{\mu}\|^2 = \tau^2 + \|\hat{\mu}\|^2 - 2\langle u_i, \hat{\mu} \rangle$: the first two terms are common to all clients, so every trace of magnitude separation is gone, but the inner product is not, and it carries the angular separation between u_i and the consensus direction. Equalization therefore compresses the separation onto its angular component instead of annihilating it, and C2 fails once that residual falls below d_2 's threshold θ , a margin condition of the same shape as parts (1) and (2). The norm-keyed score is the special case in which the residual is identically zero. $\square$

The condition as stated cannot be checked before running the composition, and that is fixable. Part (2) is stated on the transformed points, and $\delta' = \|\mu' - \mu'_B\|$ is by definition the displacement that including the adversarial points causes: evaluating it requires executing the very composition the criterion is meant to screen. In a screening criterion that is a real defect. It is also removable, because the displacement is controlled by the benign spread and the client counts alone.

Corollary 3 (Checkable separation margin). *In the setting of Theorem 8(2), write $S'_B := \sum_b \|u'_b - \mu'_B\|$ and let $\bar{R}'_B := S'_B/n_b$ be the mean transformed benign residual. Under honest majority $n_b > n_a$,*

$$\delta' \leq \frac{2S'_B}{n_b - n_a} = C_A \bar{R}'_B, \quad C_A := \frac{2n_b}{n_b - n_a},$$

and therefore the separation condition of Theorem 8(2) holds whenever

$$\Delta'_{\text{sep}} > 2R'_B + 2C_A \bar{R}'_B. \quad (1)$$

No quantity in (1) is computed from the composed aggregate.

Proof. μ' minimizes $F(x) := \sum_i \|x - u'_i\|$, so $F(\mu') \leq F(\mu'_B)$. Write $t := \delta'$. By the triangle inequality each benign term obeys $\|\mu' - u'_b\| \geq t - \|u'_b - \mu'_B\|$ and each adversarial term obeys $\|\mu' - u'_a\| \geq \|\mu'_B - u'_a\| - t$, so $F(\mu') \geq n_b t - S'_B + \sum_a \|\mu'_B - u'_a\| - n_a t$. Since $F(\mu'_B) = S'_B + \sum_a \|\mu'_B - u'_a\|$, the inequality $F(\mu') \leq F(\mu'_B)$ gives $(n_b - n_a)t \leq 2S'_B$, which is the displacement bound. Substituting it into $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ yields (1). $\square$

Honest majority is exactly what the geometric case needs, and it enters on the displacement, not on the weight ratio. C_A diverges as $n_a \rightarrow n_b$, so Corollary 3 is the precise point at which $n_b > n_a$ becomes indispensable, while the theorem itself needs no majority assumption on the *weights*. An earlier version of this paper summarized part (2) as a weight-ratio bound $\rho < n_b/n_a$; that summary was wrong, and the correction is not cosmetic, because the majority condition belongs on δ' instead. **Measured at Dirichlet $\alpha=0.5$:** the displacement bound holds in all 17 rounds with $w_a > 0$, and is conservative there by a factor of at least 10.5 (mean 215), because the realized displacement is set by a force balance among the residuals rather than by the worst case a bound must cover. Condition (1) is satisfied in 16 of those 17 rounds: 15/15 at $n_a=1$, where $C_A = 2.67$ and the requirement at the measured mean $\bar{R}'_B/R'_B = 0.429$ is $\Delta'_{\text{sep}} > 4.29 R'_B$ against a measured $\Delta'_{\text{sep}}/R'_B$ never below 5.88; and 1/2 at $n_a=2$, where $C_A = 6.00$ tightens the requirement past what one of the two rounds supplies. **The heterogeneity limit recorded for Corollary 4 applies here too:** at $\alpha=0.1$ condition (1) holds in only 5 of 12 positive-weight rounds. The checkable form certifies less exactly where the condition it replaces also stops binding, and it does not rescue the guarantee there.

A further reduction to ρ alone is vacuous on our runs, and we report it as vacuous. The natural next ask is the analogue of part (1)'s $\rho < r^*$: a threshold on ρ computable from the *pre-composition* round geometry, so that no pilot is needed at all. Such a threshold exists formally. Bounding $\Delta'_{\text{sep}} \geq w_{\min}[\Delta_{\text{sep}} - (\rho - 1)N_A]$ and $R'_B \leq 2w_{\min}[\rho R_B + (\rho - 1)\|\mu_B\|]$, where Δ_{sep} , R_B and μ_B are the untransformed analogues and $N_A := \max_a \|u_a\|$, makes $w_{\min}$ cancel and turns (1) into a linear inequality in ρ with a critical value ρ^* depending only on $(\Delta_{\text{sep}}, R_B, \|\mu_B\|, N_A, n_a, n_b)$. **It is empty on our data.** It requires $\Delta_{\text{sep}} > 4(1+C_A)R_B$, and the measured $\Delta_{\text{sep}}/[4(1+C_A)R_B]$ clears 1 in exactly 1 of the 17 positive-weight rounds. Its maximum is 1.015 at $\alpha=0.5$, attained in that $n_a=1$ round where $C_A = 2.67$ makes the requirement loosest, with mean 0.754 over those 15 rounds and a maximum of only 0.389 over the two $n_a=2$ rounds; at $\alpha=0.1$ the maximum is 0.853. Consequently $\rho^* \leq 1$ in 16 of 17 rounds at $\alpha=0.5$ and in all 12 at $\alpha=0.1$, against a measured ρ of at least 1.034 and 1.015 in the two settings respectively; in the one round where ρ^* does exceed 1 it reaches only 1.006, against a measured ρ of 1.196 in that round. The reduced condition is therefore met in *no* round, which is a stronger negative than a merely conservative threshold. The looseness is in the composition of the two bounds rather than in either one separately, and closing it would need a curvature assumption on the benign configuration that we are not prepared to assume. **We therefore claim no pilot-free screen for the geometric class**, and the calibration protocol below prices it accordingly.

What part (2) does not assume. Recorded explicitly, because the step invites readings it does not support: there is no honest-majority assumption on the *weights*, no requirement that adversarial updates lie outside the benign convex hull, and the count majority $n_b > n_a$ is used in part (2) only through Corollary 3, nowhere in the theorem's own proof.

Theorem 8(2) orders residuals *per client*: each adversarial point receives less inverse-residual weight than each benign point. That alone does not bound the adversary's *total* share of the aggregate: several adversaries could each be down-weighted yet collectively hold substantial mass. The separation margin controls this, but only quantitatively:

Corollary 4 (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ be the realized separation margin, so Theorem 8(2) reads $m > 1$. Every benign residual is at most $D_b := R'_B + \delta'$ and every adversarial residual at least $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$. Bounding $\sum_j 1/d_j$ below by the benign terms alone and each adversarial term above by $1/D_a$,*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a/D_a}{n_b/D_b + n_a/D_a} = \frac{n_a}{n_b(2m - 1) + n_a},$$

so μ' is a convex combination whose adversarial share is at most Λ_a : writing $\mu' = (1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for its benign and adversarial parts, the displacement from the benign-only combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.

The bound is only as good as the margin, and **the separation condition alone is not enough**: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, permitting the adversary 40% of the mass at $n_a=2, n_b=3$. A margin comfortably above 1 does the work. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds), capping the adversarial share at 5.1% for $n_a=1$ and 12.6% for $n_a=2$. The realized figure is far tighter than the bound: measuring Λ_a directly, it never exceeds 1.8% in any round, and the bound is satisfied in all 17 measured rounds with $w_a > 0$. With Lemma 2 this completes the picture, and the mechanism survives for *opposite* reasons: at $w_a=0$ the adversaries hold most of the Weiszfeld mass ($\Lambda_a = 0.71$ measured) but carry no payload; at $w_a > 0$ they carry payload but hold at most 12.6%.

This is the one place in the paper where the theory predicts a number and the measurement lands inside it, so we state the pair plainly: **bound 12.6%, realized $\leq 1.8\%$** , in every measured round with $w_a > 0$. **And it is also the one place where heterogeneity makes the guarantee measurably empty**. The bound is a function of the realized margin m alone, and at Dirichlet $\alpha=0.1$ RFA’s all-round mean margin **collapses from $m=3.24$ to $m=1.01$** , exactly the limit at which Corollary 4 degrades to $n_a/(n_b+n_a)$ and certifies nothing, with two of three certified compositions crossing the ASR threshold there on the raw reading, one of three once our own accuracy floor is applied (§E.6(E)). So the theorem’s explanatory value is genuine and measured in *both* directions: it predicts a number that holds where heterogeneity is moderate, and its own quantity records where it stops holding. **That is not a robustness check; it is one of the paper’s three central empirical findings** (§D.2, §I).

Λ_a is level (P4) in measurable form, but (P4) is a *level*, not a single metric. Before the instantiation, one clarification the rest of the paper depends on: **(P4) is a conceptual level of the hierarchy and is instantiated defense-specifically**, because different aggregator families have different notions of what it means for adversarial input to enter the aggregate at all. A *selector* admits or does not, so its (P4) is an indicator in $\{0, 1\}$; a *weighted averager* has an adversarial coefficient share in $[0, 1]$; a *coordinate-wise order statistic* has a fraction of selected coordinates that are adversarial. There is no single scalar all three compute, and this is why the influence column of Table A8 is **necessarily aggregator-specific** rather than a common measurement across rows: a fact about the level, not a shortcoming of the table. With that said: Definition 2’s levels are not equally meaningful. (P1)–(P3) are properties of a statistic and of a decision rule reading it; **(P4) is the first level whose content is the adversary’s influence on the aggregate**, and for a geometric aggregator Λ_a is exactly that level made measurable: Corollary 4 bounds it, Lemma 2 drives it to a benign-only convex combination, and §D.2’s instruments intervene on it directly. So the negative results do not leave the theory empty-handed: **the level the evidence points to is one the theory can bound, just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. That gap is the open problem we state in §I. **Three quantities are distinguished throughout and never used interchangeably**: *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate at all, binary for a selector and a coordinate fraction for an order statistic; the *adversarial coefficient share*, the fraction of the upstream coefficient vector’s total mass held by adversaries, which is what the instruments of §D.2 pin; and *aggregate adversarial influence* Λ_a , the weight the adversary’s direction actually carries in the aggregate. Unqualified, “admission” means the first. (Symbols are collected in Appendix K.)

Lemma 2 (Zero-weight annihilation). *Zero-weight clients are not removed from the composition: d_2 receives them as zero vectors and still sees all K points. Our implementation runs the ε -smoothed Weiszfeld iteration ($\varepsilon = 10^{-8}$), whose fixed point is $\mu' = \sum_i \lambda_i u'_i$ with $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$: a convex combination for every configuration, including the singular case $\mu' \in U'$ that a zero-weight client at the origin can create, so no non-degeneracy assumption is required. If $w_a = 0$ for every adversarial client, then $u'_a = 0$ and, writing $\Lambda_a := \sum_a \lambda_a$, the adversarial terms vanish identically: $\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}$ with $\bar{u}$ a convex combination of benign transformed updates. The aggregate is a contraction by $1 - \Lambda_a \in (0, 1)$ of a benign-only weighted average: the attack direction never enters μ' . The origin points’ only effect is that contraction: a reduction in effective step size, i.e. a utility cost, not a security failure. The same holds for CoordMedian, where a zeroed coordinate carries no payload but shrinks the aggregate when selected.*

Why the positive-weight restriction matters. Defining ρ over all weights would make it *infinite in every round*: FoolsGold’s normalization subtracts the minimum weight, so 2 of 5 clients receive exactly zero (measured, $N=10$, $K=5$). A bound of the form $\rho < 4$ would then never be satisfiable, and Theorem 8(2) would be vacuous on the very composition it is stated over. Restricting ρ to strictly positive weights and handling $w_a=0$ separately via Lemma 2 is what makes the analysis apply to the algorithm we actually run. **Both channels are drawn in Figure A2**, which is the composed pipeline as a DAG rather than the evaluation design of Figure 2: the statistic path C2 is stated over, and the zero-weight arc that reaches the aggregate without passing through σ_2 at all.

Operational calibration protocol. To evaluate the criterion for $(d_1 \rightarrow d_2)$: **C1** is available from standard single-defense baselines (zero cost). **C2/C3 for the theorem-backed class** (FG or reputation as d_1 ; RFA, CM, TM as d_2): for the *rank* aggregators, Theorem 8(1)’s bound $\rho < r^*$ needs only ρ and the coordinate-wise separation, both read off the C1 baseline runs, so this case really is free. **For the geometric case an earlier version of this paper claimed the same, and that was false**: Corollary 3 removes the one quantity that would have required running the composition, but the benign spread that replaces it is measured on the transformed points, so RFA carries the same short-pilot cost as the other families and we no longer exempt it. **C2/C3 for other families**: one short pilot (5–10 rounds, $\sim 10\%$ of full cost) measuring $\sigma_2(T_1(u_a))$ vs. $\sigma_2(u_a)$ suffices. The full 50-round experiment runs only for criterion-PASS candidates. $\delta_2 = 0.5$ separates all PASS from C2-FAIL cases on the development set; insensitive to choice: any $\delta_2 \in [0.3, 0.7]$ yields identical classifications on all 42 pairs.

Proposition 9 (Rank-invariance of coordinate-wise aggregators). *Let d_2 be a coordinate-wise rank aggregator (TrimmedMean or CoordMedian) with admission statistic $\sigma_k(u) = \text{rank}_k(u)$ among the received updates $\{u_1, \dots, u_m\}$. Let $T_1(u_i) = \alpha \cdot u_i$ be a uniform positive rescaling ($\alpha > 0$, same for all clients). Then:*

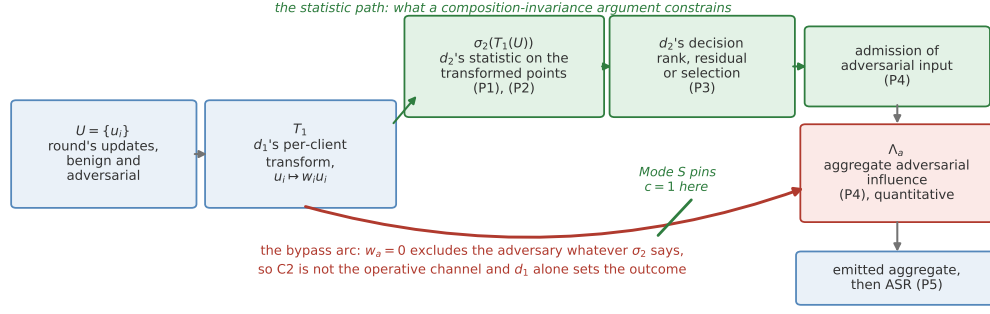


Figure A2: **The composed pipeline as a DAG, which is a different graph from Fig. 2:** that one is the *evaluation design*, this one is what a client update passes through. **The statistic path** (green) is what a composition-invariance argument constrains: d_1 's per-client transform T_1 moves the points, d_2 computes σ_2 on the transformed points, σ_2 drives d_2 's decision, and the decision determines admission, i.e. levels (P1)–(P4) of Def. 2 in order. **The bypass arc** (red) is the one the paper's argument turns on: d_1 's weights reach Λ_a (Cor. 4) without passing through σ_2 at all, so when d_1 zeroes an adversary's weight the adversary is excluded whatever σ_2 says and d_1 alone sets the outcome (Lem. 2). Mode S cuts that arc by pinning every adversary's coefficient at $c=1$, which is why it isolates C2's causal content and also why it needs adversary identity and is an instrument rather than a defense. **A statistic-level argument constrains the top path and leaves the bottom one free**, which is the whole of why (P1)–(P3) do not deliver (P5).

1. $\sigma_k(T_1(u_i)) = \sigma_k(u_i)$ for all i, k , i.e., C2 and C3 hold trivially.
2. Under heterogeneous rescaling $T_1(u_i) = \alpha_i \cdot u_i$ with $\alpha_i \neq \alpha_j$, rank preservation fails when rescaling reorders coordinates across clients: $\exists k$ s.t. $\text{rank}_k(\alpha_i u_{i,k}) \neq \text{rank}_k(u_{i,k})$.

Proof. (1) Rank is invariant to positive scalar multiplication: $u_{i,k} > u_{j,k} \iff \alpha u_{i,k} > \alpha u_{j,k}$ for $\alpha > 0$. (2) Counterexample: let $u_{1,k} = 3, u_{2,k} = 2$, with $\alpha_1 = 0.1, \alpha_2 = 1$. Then $\alpha_1 u_{1,k} = 0.3 < \alpha_2 u_{2,k} = 2$, reversing the rank. $\square$

Corollary 5. *CoordMedian (pure rank selection) is robust to C3 boundary compression under any uniform rescaling but vulnerable under heterogeneous client-specific rescaling (e.g., FoolsGold or reputation weights). This predicts that compositions $X \rightarrow \text{CoordMedian}$ will show intermediate ASR when d_1 applies heterogeneous weights: consistent with observed $\text{NC} \rightarrow \text{cm}$ (0.433) and $\text{rfa} \rightarrow \text{cm}$ (0.485), both near the 0.5 threshold.*

Proposition 9 concerns rank under *uniform* rescaling. The sharper question is which admission signals survive *heterogeneous* per-client rescaling, since that is what every per-client defense in our menu applies: NormClip ($c_i = \min(1, \tau/\|u_i\|)$), reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \epsilon)$) are all of the form $T(u_i) = c_i u_i$ with $c_i > 0$ strictly, each positive by construction. FoolsGold is of the same form with $c_i \geq 0$ but attains $c_i = 0$ in *every* round (its shift-and-clamp gives the least-trusted client weight exactly zero, 2 of 5 clients here), so it falls outside the strict hypothesis below and is governed by Lemma 2 instead: the same distinction we draw for FLTrust's ReLU zeros in §E.1. The strictly-positive case has a sharp answer, and it partitions the menu:

Proposition 10 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, strictly: the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \epsilon)$), each positive by construction. FoolsGold is deliberately not in that list: its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of 5 clients here), so a zero coefficient is not a positive rescaling and FoolsGold-as- d_1 falls outside the hypothesis, governed by Lemma 2 rather than by (a)–(c). Then:*

- (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. FoolsGold's weight vector is a function of the pairwise-cosine matrix alone, so $w(T(U)) = w(U)$ identically, and **C2 holds for every $\rightarrow \text{FoolsGold}$ composition** (the $\cdot$ standing for any upstream d_1 throughout).
- (b) **Coordinate ordering is not invariant** in general; it survives under the bounded-ratio condition of Theorem 8(1).
- (c) **Consensus distance and residual magnitude are not invariant.** Heterogeneous c_i can invert their ordering, so $\cdot \rightarrow \text{reputation}$ compositions fail C2, and $\cdot \rightarrow \text{RFA}$ needs the separation condition of Theorem 8(2).

Proof. (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so every pairwise cosine, each client’s normalized maximum similarity, and hence every FoolsGold weight is unchanged. (c) Counterexample for consensus distance: with $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3 becomes the *closest* point, reversing the ordering reputation depends on. $\square$

This corrects a natural but wrong intuition. Norm clipping does *not* blind FoolsGold: it cannot, by (a), though it does blind reputation, whose signal is consensus distance rather than similarity. We verified (a) in the shipped implementation: FoolsGold’s weight vector changes by at most 3×10^{-8} (float32 rounding) under NormClip, across 1000 synthetic trials and live FL rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against $\tau=5$). Note this also means Theorem 8(3)’s norm-equalizing degenerate case requires τ below *all* client norms; at our $\tau=5$ only 1–2 of 5 clients clip per round, so norms are not equalized and that case does not describe our runs.

C.4 THE FRAMEWORK IN GENERAL FORM

Positive rescaling is the instantiation that matters for current defenses, but the structure is more general, and stating it separates what is essential to the framework from what is an artifact of today’s defense menu. Write d_2 ’s admission mechanism as a statistic $S(U)$ of the round’s updates together with a decision rule that consumes only the *ordering* S induces over clients, which are extreme, which are far from consensus, which are down-weighted. Let T be d_1 ’s per-client transform. Then $(d_1 \rightarrow d_2)$ is **mechanism-preserving** exactly when

$$S(T(U)) = \varphi(S(U)) \quad \text{for some } \varphi \text{ preserving the ordering } d_2 \text{ acts on,} \quad (2)$$

which is C2; C3 additionally asks that φ not compress the quantitative gap below d_2 ’s operating threshold. Proposition 10 is the case $T(u_i) = c_i u_i$: for $S = \text{cosine similarity}$, $\varphi = \text{id}$, so (2) holds unconditionally; for $S = \text{coordinate value}$, φ is a per-client positive scaling, order-preserving only under the bounded-ratio condition $\rho < r^*$ of Theorem 8(1); for $S = \text{consensus distance or residual magnitude}$, no φ exists, because the reference point (median, geometric median) moves with T , so $S(T(U))$ is not a function of $S(U)$ at all. Three consequences worth stating. First, this is why the conditions are checkable without running an experiment: one reads S off d_2 ’s definition and T off d_1 ’s, and the question becomes algebraic. Second, it identifies what a genuinely new instantiation would require: an S and a non-rescaling T , e.g. a d_1 that projects updates onto a subspace or applies a coordinate-wise mask, for which (2) would need to be re-derived rather than inherited. Third, it marks the framework’s real boundary: a d_1 whose transform is not per-client (one that mixes updates across clients, or carries cross-round state) admits no T of this form, and nothing here applies to it.

D THE INTERVENTION EVIDENCE: THE CONFOUNDED LADDER, AND THE INSTRUMENT IT MOTIVATED

The controlled upstream transform. We introduce a synthetic d_1 whose only parameter is the *dispersion* of a positive per-client rescaling. For the K participants of a round, at position j in a (seed, round)-keyed pseudorandom permutation,

$$c_j \propto \exp\left(\kappa \cdot \left(\frac{2j}{K-1} - 1\right)\right), \quad \text{normalized so } \frac{1}{K} \sum_j c_j = 1. \quad (3)$$

Table A5: The four arms of the dose–response ladder, and the frozen predictions. The statistic each d_2 reads is, in order, summed pairwise Euclidean distance, distance to the client median, pairwise cosine, and coordinate ordering. *Standalone* is d_2 ’s own ASR (clean accuracy) on that attack, the C1 input, held fixed within every arm. *Measured disturbance* is the fraction of live rounds on which the rung changes that arm’s decision (for `coord.median`, the mean fraction of *coordinates* whose median-attaining client changes) measured before the freeze on the same raw updates, with no ASR computed (`results/dose_disturbance.json`). Shapes and decision rules are frozen at commit e711a95.

arm (d_2 / attack)	class (Prop. 10)	standalone ASR (acc)	measured disturbance $\kappa=0$ 0.5 1.0 2.0				predicted shape in κ
1 krum / scaling	(c) not inv.	0.061 (0.577)	0.00	0.60	0.73	0.87	monotone rise
2 reputation / scaling	(c) not inv.	0.017 (0.777)	0.00	0.87	0.87	0.93	monotone rise
3 cos.krum / pixel	(a) exact	0.300 (0.598)	0.00	0.00	0.00	0.00	FLAT (control)
4 coord.median / pixel	(b) cond. inv.	0.443 (0.767)	0.00	0.25	0.38	0.49	rise, shallower

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 actually reads, so before freezing any prediction we pushed live FL updates through the shipped transform and recorded each arm’s decision at every κ on the *same* raw updates, computing no ASR. That establishes three things before any outcome exists (Table A5): *there is a dose*: disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*: `cos_krum`’s selection changes in 0 of 120 cells at ρ up to 54.6; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). Its predictor is a *measured property of the statistic*, not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table A5, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$, with arm 3’s “flat” scored as an *equivalence* claim at a 0.15 margin fixed and justified in advance (under a quarter of the +0.629 within-defense gap for Krum (the $n=3$ value on record at the freeze; the later top-up puts it at +0.716, so the margin is if anything conservative) and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide for this arm) and arm 4’s class-(b) signature being the *shallower slope*.

One prediction we deliberately do not make. Theorem 8(1)’s ordering-survival condition $\rho < r^*$ would give arm 4 a crossing rung fixed by a measured r^* , but **no frozen prediction here uses r^*** : we measured it and it is **not a stable per-attack constant**, ranging 0.00–12.59 on model-scaling (median 0.57) and 0.61–1.63 on the pixel backdoor (median 0.92) across the 12 rounds that sampled an adversary, so an illustrative value such as $r^*=10$ is a stipulation rather than a measurement; §C.3 states the condition symbolically and stipulates no value for it. This is a limitation of our own theorem that the dose-response measurement surfaced, reported as one (L5) rather than quietly left unchecked.

D.1 OUTCOME OF THE CONFOUNDED DESIGN: ONE ARM CONFIRMED, TWO REFUTED, ONE INDETERMINATE, AND THE POOLED TEST FAILS

What did the outcome-gated ladder actually return, arm by arm? All 80 runs completed (16.5 h wall clock). Table A6 is the whole ladder; Figure A3 plots it on the nominal dose axis against the frozen shapes and Figure A4 on the independently measured disturbance axis with the real-transform anchors, scored by `analyze_dose_response.py` against the rules frozen at e711a95, every number recomputed per seed from `results/dose_response/`. The **primary** pooled test does not confirm: across the 16 cells, the rise in ASR over each arm’s own $\kappa=0$ rung correlates with the measured disturbance at Spearman $\rho_s = +0.351$, one-sided $p = 0.091$, so the frozen criterion ($\rho_s > 0$ at $p < 0.05$) is not met and **we report the primary prediction as refuted**. Of the four frozen shapes one is confirmed, two are refuted, and one is indeterminate; **in three of the four arms ASR falls as the dose rises**. This is the round’s headline result and it is a negative one. It is also the outcome the pre-registration anticipated and named in advance: “pooled correlation absent while individual arms move: the mechanism is defense-specific and measured disturbance is not a transferable predictor.”

Table A6: The dose–response ladder as measured, $n=5$ per cell, per-attack mean ASR with mean clean accuracy in parentheses. *Predicted* is the shape frozen at commit e711a95 before any run; $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability for an *increasing* trend across the four ordered rungs, which is the only direction any frozen rule tests. Verdicts apply the frozen criteria unchanged (Table A5 and the supplement §S6); an arm satisfying neither its confirmation nor its refutation criterion is reported as *indeterminate* rather than scored in our favour. $\dagger$ below the 0.35 clean-accuracy gate: a collapsed model, not suppression. Student-t 95% intervals and per-seed values are in the supplement (§S8).

arm (d_2 / attack)	predicted shape	mean ASR (clean accuracy)				JT verdict	
		$\kappa=0$	0.5	1.0	2.0	$p_{\uparrow}$	
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63	refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6}	confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) †	0.999	indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998	refuted

Arm 2 confirms: the ladder yields one prospective confirmation, while the pooled prediction is rejected. Reputation under model-scaling rises monotonically from 0.017 at the identity rung to 0.487 at $\rho = 7.39$ and 0.779 at $\rho = 54.6$ (Jonckheere–Terpstra $z = +4.64$, one-sided $p = 1.7 \times 10^{-6}$; permutation $p < 5 \times 10^{-5}$), while clean accuracy moves only $0.777 \rightarrow 0.701$, so what is lost is suppression, not the model. Same d_2 , same attack, same standalone baseline at every rung: nothing about effectiveness varies, and $55 \times$ dispersion in a *positive per-client rescaling* (a transform the criterion treats as benign for invariant statistics) carries a defense from 0.017 across the 0.5 threshold. Two qualifications keep this in proportion. It validates the *destruction*

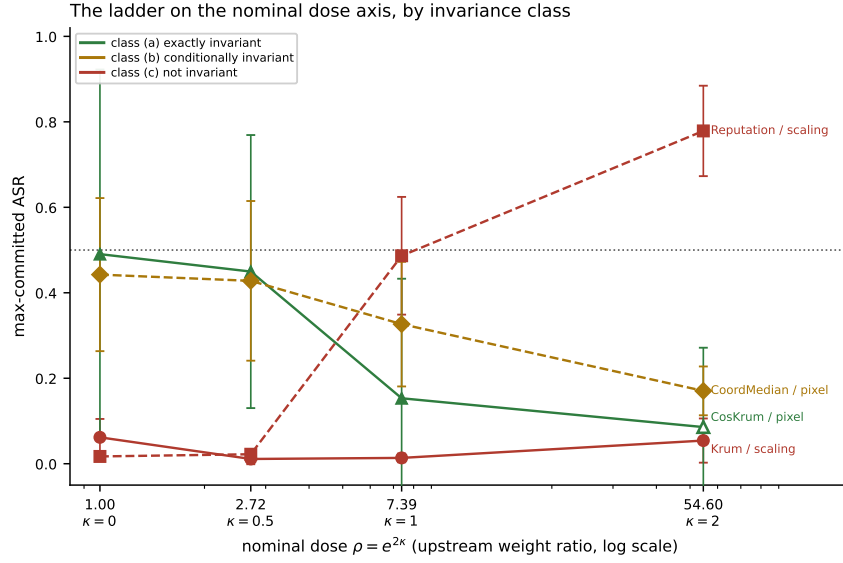


Figure A3: **The ladder and its frozen predictions.** The 16 cells on the nominal dose axis $\rho = e^{2\kappa}$ ($n=5$ per point, Student-t 95% intervals), coloured by Proposition 10 class. Of the two class-(c) arms whose statistics the framework’s fragility claim is stated over, only reputation rises; ASR *falls* with the dose in three of the four arms, against the predicted rises. Hollow markers fail the 0.35 clean-accuracy gate and are uninterpretable. Drawn by `paper/figures/dose_response_figure.py`, which recomputes every value from `results/` and imports the arm construction from the scoring script so the two cannot diverge.

half of C2’s mechanism (a non-invariant statistic, actually disturbed, loses suppression); the *preservation* half (invariance $\Rightarrow$ suppression survives) is arm 3, which came out indeterminate. And even within this arm the measured disturbance is not what orders the curve: it saturates at 0.867 by $\kappa=0.5$, where ASR is still 0.022, and the jump to 0.487 happens at the next rung at the same measured disturbance, so the *dose* moves ASR here while the flip rate does not.

Ecological validity: the real transforms do not lie on the synthetic curve. Pre-registered caveat (iv) required this check, and it fails in an informative way. RFA $\rightarrow$ Krum sits at ASR 0.779 with measured disturbance 0.667, where the synthetic ladder’s nearest rung (0.600) gives 0.011: a residual of +0.767. NormClip $\rightarrow$ Reputation sits at 0.806 with disturbance 0.333, against 0.017 at the identity rung: +0.789. Two anchors do land: RFA $\rightarrow$ Reputation, 0.895 at disturbance 1.000 against 0.779 at 0.933 (+0.116), and both `cos_krum` anchors, which coincide with its curve (+0.000 and -0.044) because that statistic is invariant under all three transforms. This is not a range mismatch: read off the same live rounds, NormClip realizes ρ of mean 3.61 (max 4.90) and RFA mean 11.93 (max 17.38), both inside the ladder’s $\{1.00, 2.72, 7.39, 54.60\}$ span. The conclusion is about our abscissa: **the rate at which a transform flips a defense’s decision is not a sufficient statistic for what that transform does to the defense**: two transforms at matched flip rate differ by up to 0.77 in ASR. **And the axis they are unmatched on is now measured, not conjectured.** Adversary-correlated dispersion $\Delta_c := \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ (§C) separates the transforms carrying those anchors from the ladder: RFA reaches $\Delta_c = -1.21$ and NormClip -0.72 , against -0.34 at the ladder’s most dispersed rung: 2–3.6 $\times$ more extremely targeted at ρ that is 3.7 and 13 $\times$ *smaller*, so **matched on the statistic, badly unmatched on the targeting, and ρ does not order Δ_c** . **This rescues no predictor**: Δ_c ’s magnitude does not order suppression either; rfa $\rightarrow$ krum still fails at 0.779 at $\Delta_c = -1.21$, while *reputation*, targeted harder still on both attacks, has rep $\rightarrow$ coord_median suppressing, and FoolsGold’s Δ_c is *positive* (+0.663). These figures are **post-hoc and exploratory**, computed with no ASR and rescoring nothing, and are tabulated in the supplement (§S10). Arm 4 has no real-transform anchor at all, since `coord_median` was not on the metric-swap menu, so its ecological validity is untested.

What we now claim less of. C2’s causal content is established prospectively for one statistic (distance to the client median) and refuted for another (summed pairwise distance) at ρ up to 54.6; the pooled claim that measured disturbance predicts loss of suppression *across* defenses is refuted, and we no longer make it. Within-defense disturbance remains the only design the testability boundary admits, and §C.1’s general statement stands untouched by this outcome; it is a statement about admissible designs, and this is the admissible design, run and reported as it landed. What changes is the empirical status of C2: one confirmed arm, not a validated screening mechanism.

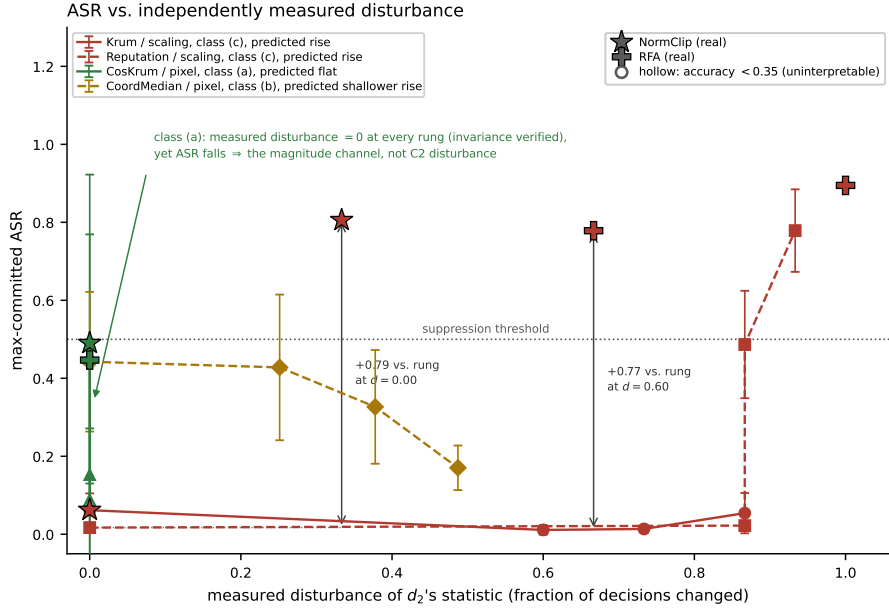


Figure A4: **Why a transform’s flip rate is not a sufficient statistic for what it does.** ASR against the *independently measured* disturbance of each arm’s own statistic ($n=5$, Student-t 95% intervals), the abscissa fixed before the freeze. Stars and crosses are the two *real* upstream transforms measured on these same (d_2 , attack) cells (NormClip, RFA), placed at their own measured disturbance rate: at matched disturbance they sit +0.77 and +0.79 ASR *above* the synthetic curve, while their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) fall *inside* the ladder’s span. The class-(a) arm’s four rungs coincide at disturbance exactly 0 (0/120 selections changed, maximum relative score drift 1.1×10^{-4}) and its ASR falls anyway: Lemma 2’s magnitude channel, not C2 disturbance.

D.2 THE TARGETED INTERVENTION: ISOLATING C2 FROM THE ATTENUATION CHANNEL

With the adversary’s share pinned, how much of each arm’s rise survives? Reading cos_krum’s entry in Table 1. Its -0.425 is the full-ladder endpoint, which lies below our accuracy floor; **the witness we rest on is instead its first, gate-passing rung** (0.173, §6.2). The rank discordance between $\Delta\Lambda_a$ and ΔASR across the two moving arms is §6.2’s refutation of our own predicted ordering. This section gives the emitter and the per-artifact provenance for every cell of that table.

Round counts and per-row provenance, relocated here from the table’s caption. *Admission* here is Table 1’s $\Delta \text{adm.}$, the *support* of the admitted adversarial mass, which that caption keeps distinct from the mass itself ($\Delta\Lambda_a$); it never moves in any measured round of any arm: 60 rounds per CIFAR-10 `cifar_cnn` arm, 56 on the EMNIST-byclass replication and 72 on the ResNet18 replication, 368 in all, including the 48 per arm where reputation and `coord_median` carry adversarial mass at baseline, so the constancy is not an artifact of arms that never admit an adversary at all. The decision/admission dissociation is accordingly not one aggregator’s property: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager, which is what the table’s *Kind* column records. Row 5 replicates row 1 on a second dataset and a differently configured CNN, with its channels measured *before* its ASR existed; row 6 is the pre-registered score-only control, row 1 with the magnitude channel closed, so its decision, admission and influence columns are row 1’s own; row 7 replicates row 1 with *architecture* varying alone, its channels also measured before its ASR existed (§D.2). **Row 7 is printed last rather than beside row 5, which is where it belongs by subject:** four sites in this paper index the score-only control as row 6, and the emitter orders the rows so that adding a seventh cannot silently falsify them. The table is *not* an out-of-sample test. **Row 5 is the frozen $n=3$ in every column, its ΔASR included,** because its three channel columns were measured at that seed count and were not re-measured: the same cell’s ΔASR at $n=5$ is -0.013 , reported with its provenance in disclosure (i) below, and substituting it here would put one $n=5$ cell in a row of $n=3$ ones.

What Mode S does to the ladder’s confirmation. Reputation was the ladder’s one confirmed arm (§D.1); Mode S shows most of that confirmation was attenuation. With the adversary’s share held fixed, reputation’s rise at $\rho=54.6$ shrinks from the ladder’s $+0.762$ to $+0.178$ (one-sided Welch $t=-5.22$, $p=0.0017$ that it is smaller), and what remains tracks a small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision change ($0.000 \rightarrow 1.000$), so H-admission is confirmed here too (Table A7). The confound is thus *explained*,

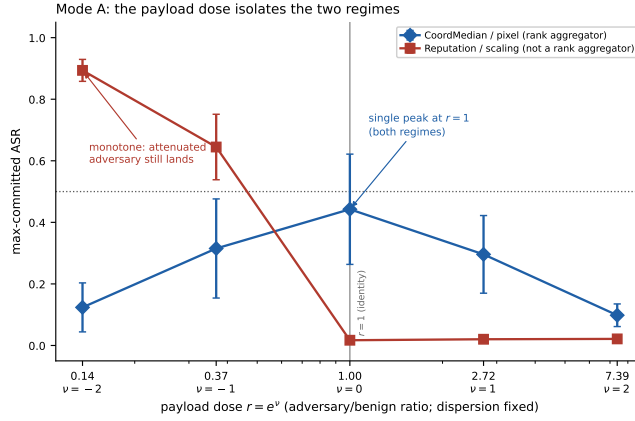


Figure A5: **Mode A, the payload dose: the two mechanism-preserving regimes, and where they hold.** With dispersion fixed and only the adversary/benign weight ratio r swept, `coordmedian` (a rank aggregator, the case Theorem 8’s regimes are stated over) is **single-peaked at $r=1$** as the theorem predicts (over-attenuated adversaries carry no payload, over-amplified ones are discarded) while reputation is monotone, off-spec and refuted for that shape. Mode A is the Δ_c -swept companion to Mode S’s Δ_c -pinned design (Proposition 12); Mode S itself is Figure 1. Bars are Student- t 95% intervals; numbers recomputed per seed by `analyze_targeted_dose.py`.

not explained away: a genuine, much smaller, admission-driven effect survives it. The class-(a) control cuts the other way and refutes us: `cos_krum`’s selection is bit-identical at every rung (decision and admission both 0.000), yet ASR is 0.584 at the identity (it never suppresses) and *falls* 0.425 across the ladder ($p_{\downarrow}=0.0008$; the $\kappa=2$ rung is below the accuracy gate). Exact invariance of the statistic neither implies suppression nor holds it fixed; the magnitude channel moves it regardless. We report this as a refutation of Proposition 10’s practical content, not a confirmation of its algebra.

Mode A: the two regimes, and where they hold. The payload dose gives Theorem 8 a single falsifiable shape: ASR single-peaked at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified one is discarded by a rank aggregator (Regime A). It is **confirmed for `coordmedian`**, the rank aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_{\downarrow}=3.0 \times 10^{-4}$ amplified, 6.4×10^{-4} attenuated) at accuracy ≥ 0.74 throughout (Figure A5). It is **refuted for reputation**, which is monotone: ASR is maximal at the most attenuated rung (0.894 at $r=0.135$) and near zero once $r \geq 1$, because an attenuated adversary reads as *consistent* to a reputation score and stops being down-weighted. Reputation is not a rank aggregator, so the two-regime shape was never licensed for it; the split is the theorem behaving exactly where it applies and not where it does not.

A pre-registered replication on a second downstream defense, and what it refutes. One arm cannot establish which channel matters, and the two Mode-S arms above move together on both candidate predictors: reputation is higher on *both* decision and admission change and also had the larger rise, so nothing in that pair separates them. `coordmedian` under the pixel backdoor is *rank-discordant* and therefore does: of the interpretable Mode-S arms it has the **smallest decision change** (0.482) and the **largest admission change** (0.033), so the decision reading predicts its rise is the *smallest* of the three and the admission reading predicts the *largest*. It is also a structurally different defense, a coordinate-wise order statistic (Proposition 10(b)) against Krum’s selector (class (c)), and it suppresses the pixel backdoor at usable accuracy, so there is suppression to lose. Thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$, indeterminate between) at commit 06d9bce before `results/dose.replication/` existed, with the identity rung imported bit-identical and 15 new runs spent (`experiments/pre_registration_dose_replication.md`).

The outcome is $\Delta = +0.098$ at $n=5$: the admission ordering is refuted. The arm with the largest admission change rose *less* than an arm with a smaller one, so **admission magnitude does not order outcomes across arms**. Neither does the decision reading, which predicted the *smallest* rise of the three when $+0.098$ is the middle one, so this arm refutes a rank prediction from *each* candidate channel. The pooled model agrees: re-fit with these cells it is still not confirmed ($R^2=0.239$; the admission coefficient’s one-sided p moves $0.055 \rightarrow 0.041$, but the adversarial-share coefficient is negative, so the pre-specified conjunction fails). **What does replicate is the negative:** a decision change of 0.482 costs no suppression by the frozen margin, at mean accuracy 0.698, so the accuracy gate does not bind and the flagship result now holds on two downstream defenses in *different* invariance classes. Two qualifications, the first pre-specified in the freeze and the second

Table A7: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`), per-attack mean ASR with the two measured channels. The decision channel moves far more than ASR throughout. **The admission row is the admitted adversarial mass Λ_a** , i.e. Table 1’s influence column $\Delta\Lambda_a$ and *not* its Δ adm. column, which records the support of that mass and is 0.000 in every measured round of every arm; this section’s hypothesis names (H-admission) are stated at the mass. [†] below the 0.35 accuracy gate. [‡] the pre-registered replication arm, *rank-discordant* by design (smallest decision change, largest admission change of any interpretable arm) and scored against thresholds frozen separately at 06d9bce before `results/dose_replication/` existed: $\Delta=+0.098$ falls below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates on a different invariance class (below). Verdicts otherwise scored against the readings frozen at 5130cec; per-seed values in the supplement (§S8).

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8 \times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 10 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ $< +0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

post-hoc: this cell’s identity ASR is 0.443, so equivalence is to the *identity rung* and not to a low absolute ASR (an “every rung <0.5 ” criterion has no headroom here and was deliberately not used); and the rise is positive in 5/5 seeds (paired $t=3.39$, one-sided $p=0.014$, *not* pre-registered), so on this class the cost of disturbing the decision is **small and consistent rather than zero**. All numbers are recomputed per seed by `analyze_dose_replication.py`, which imports its rung builder and trend test from the frozen suite’s scorer.

The design choice, not the defense, sets the inferred direction. This is the paper’s central empirical result. On this identical cell (matched aggregate displacement, decision change and clean accuracy, Table A3), **the confounded design does not lose power, it points the other way**: at $n=20$ the outcome-gated ladder moves -0.273 ($0.478 \rightarrow 0.205$) where the instrument moves $+0.125$ ($0.478 \rightarrow 0.603$), the two paired 95% intervals $[-0.334, -0.213]$ and $[+0.095, +0.155]$ disjoint and both excluding zero. Closing the attenuation channel does not sharpen a weak positive effect, it **inverts it**. **The two designs are not the same intervention**, and Table A3 states the one non-match: Λ_a moves in both (0.081 against 0.033 , both $n=5$), so neither column is a clean statistic-only intervention and no p -value attaches to the difference between them. What the contrast establishes is that the design choice sets the inferred direction, so a criterion calibrated against the confounded contrast would have been calibrated against the confound, and a cross-defense contrast would have read the fall as evidence about the statistic. That is why the identification boundary of §C.1 is not a technicality about power.

What the two arms jointly establish, stated formally. The arms above and the class-(a) control (`cos_krum`) are between them a *constructive* answer to the question C2 was written to settle. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 11 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass (indeed the same statistic and the same decision) while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Proof. Both halves are exhibited by runs reported in this section. (a) Mode S into Krum under model-scaling, $\kappa=0$ against $\kappa=2$: ρ moves from 1 to 54.6 and Krum’s decision changes in up to 0.80 of rounds, while the support of the admitted adversarial mass changes in none of the 60 measured rounds (a floor on this arm: no adversary is admitted at any rung), the adversarial coefficient share is constant at 0.266667, and suppression shows **no evidence of a practically meaningful change under the pre-registered ± 0.15 practical-equivalence margin** ($|\Delta|=0.026$). (b) `cos_krum` under the pixel backdoor: by Proposition 10(a) its statistic is invariant, and its decision and admission are measured bit-identical (0.000) at every rung, yet ASR falls 0.173 by the

first rung alone, at $\rho=2.72$, where the accuracy gate still passes ($0.599 \rightarrow 0.567$; paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). $\square$

What the proposition does not say. A preserved statistic is not thereby uninformative; it is not *sufficient*, remaining evidence about what d_2 reads and none about what it suppresses.

The corollary is why the title says *beyond* statistic preservation: no metric defined *solely* on preservation of the downstream statistic can identify attack-suppression equivalence, because (a) exhibits statistic disturbance without suppression change and (b) exhibits suppression change without statistic disturbance. **Scope, stated immediately:** this is a non-existence claim *within the positive-rescaling class on these defenses and attacks*, established by exhibited counterexample: **not** a claim that no metric of any kind can order security, and not a claim about transforms outside the class.

Proposition 12 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic's dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

Proof. By construction, with the realized values tabulated in the supplement (§S10) and computed with no ASR involved. Mode S pins every adversarial coefficient at $c=1$ and spreads only the benign clients, giving $\Delta_c \equiv 0$ at every ρ up to 54.6; Mode A holds the benign weights uniform and sweeps only the adversary-to-benign ratio r , moving Δ_c over $[-1.138, +2.469]$ at fixed statistic dispersion. $\square$

This is why the two instruments were the right ones, and it makes the supplement's §S10 warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**, so ρ cannot be a sufficient statistic for what the transform does to a defense. It does *not* follow that Δ_c is a predictor: §D.1 reports that its magnitude fails to order suppression either.

The dissociation is not Krum-specific: all four aggregators, all four channels. Table A8 collects the endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S for every downstream defense in the paper. **Recomputed, not transcribed:** every cell is emitted by `build_channel_table.py`, which only reads frozen artifacts under `results/`, `admission_measurement.json`, `femnist_admission.json` and `resnet18_admission.json` for the channel columns, the `targeted_dose/`, `dose_replication/`, `dose_femnist/` and `dose_resnet18/` summaries for the ASR column; it also asserts that the identity rung displaces nothing ($0.00e+00$ for all seven rows). The sixth row's aggregate column comes from `displacement_decomposition.json` and its ASR from `score_only/summary.json`, both read-only; the emitter refuses to merge that arm into the ASR sources it shares cell keys with, so it cannot shadow the first row's published number. The influence column is **aggregator-specific by necessity** (a selector has no weights) and its instantiation is: adversarial share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected coordinates for the order statistic, and the indicator that an adversarial update is selected at all for the two selectors. The fifth row replicates the first row's arm on a second dataset and a differently configured CNN (EMNIST-byclass with `simple_cnn`) with **its channels measured prospectively, before its ASR existed**; the result is reported below. **The seventh row replicates it with architecture varying alone** (CIFAR-10 with `resnet18`), its channels likewise measured before its ASR existed, and it is the one row that shares *both* the first row's dataset and its cell key: the ASR lookup is keyed on the triple (dataset, model, cell) precisely so that the two cannot shadow each other, and the emitter raises rather than resolving a collision by file order (§D.2). **What the Δ_{adm} column pools:** the four `cifar_cnn` CIFAR-10 rows contribute 60 measured rounds each, 240 in total, and adversarial mass is present at baseline in 100 of those 240 rounds (including the 48 per arm where reputation and `coord_median` carry it) while its support moves in none of them, which is the count Figure 1's annotation reports and why that annotation carries no single arm's name.

A second dataset and a differently configured CNN, pre-registered. The fifth row's arm is the first row's arm re-run with EMNIST-byclass (62 classes, $1 \times 28 \times 28$) and `simple_cnn` in place of CIFAR-10 and `cifar_cnn`, everything else held: Mode S, Krum as d_2 , the committed model-scaling attack, the same four frozen rungs $\kappa \in \{0, 0.5, 1, 2\}$. The decision rule lives in `experiments/pre_registration_dose_femnist.md`, git-committed before any EMNIST-byclass ASR existed, and the runner refuses to start unless its `PREREG.COMMIT` matches that hash.

One naming correction, made here rather than left to be inferred. Earlier drafts labelled this arm "FEM-NIST" in the tables while the prose said EMNIST-byclass, and the tables were the wrong ones: the `femnist` key loads `torchvision` EMNIST (`split="byclass"`) and partitions it with this paper's own label-Dirichlet split, **not** LEAF FEMNIST's partition by writer. So the arm is a second dataset and a differently configured

Table A8: **The four channels, all four aggregators, under Mode S.** Endpoint change $\kappa=0 \rightarrow \kappa=2$. **The first column, Δ agg., is the relative displacement of the aggregate the defense emits,** $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$: what enters training, and **not** the value of a scoring statistic; the measured channels are aggregate displacement, decision, admission and influence. The dissociation between the *decision* channel and the *admission* channel is not a property of one aggregator: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager. The table does **not** order the arms ($\Delta\Lambda_a$ and ΔASR disagree in rank, which is this section’s refutation stated in one row-pair) and it is *not* an out-of-sample test: all rows are Mode-S ladders reported elsewhere in this paper. **The sixth row is the score-only control:** the same arm as the first with the magnitude channel closed, so its Δ agg. is the re-selection component alone and its decision, admission and influence columns are the first row’s own numbers, score-only leaving the selection untouched. **The seventh row varies architecture alone:** the first row’s cell on `resnet18` instead of `cifar_cnn`, with dataset, N , K , f , α , attack and defense all at their frozen CIFAR-10 values, so it is the only row that differs from the first in a single factor. Its channel columns were measured *prospectively* — `results/resnet18_admission.json` records “No ASR. Freezes nothing.” and predates every ResNet18 ASR in this paper — and its two unrun interior rungs are why §D.2 reports no trend statistic for it. **n is the seed count behind the Δ ASR column and nothing else:** the channel columns are per-round measurements, whose counts are given above. It is emitted per row as $n_{\text{seeds}} = \min(n_0, n_2)$ over the row’s two rungs and it is **not uniform**, reading 3 for the second-dataset replication and 8 for `cos_krum`. The emitter also records `seeds_match` on the six rows whose two rungs come from one paired suite, and it is true on all six; the score-only row’s ASR is read from a separate directory, so it carries no such field.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR	n
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	−0.026	5
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178	5
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	−0.425	8
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098	5
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	−0.017	3
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	−0.023	5
Krum (ResNet18)	selector (pairwise distance)	0.824	0.556	0.000	0.000	−0.041	3

CNN, and it is **not** a more federated partition than the CIFAR-10 arm; it is the same partition scheme on different data. Every table now reads EMNIST-byclass. The frozen artifact paths, the pre-registration filename and the JSON dataset key retain the `femnist` spelling because renaming a committed artifact would break the provenance the rest of this appendix rests on.

The premise was measured first, with no ASR anywhere. Krum’s selection changes on 0.643–0.714 of rounds across the three disturbed rungs while the support of the adversarial mass it admits is unchanged in all 56 measured rounds (again a floor: no adversary is admitted at any rung), so the arm has a real disturbance to test. Had the decision change come out near zero the arm would have been reported ineligible rather than run: that is the `cos_krum` failure mode, and this check is what distinguishes the two.

Suppression shows no evidence of a practically meaningful change. $\Delta = -0.017$ (mean ASR 0.027 $\rightarrow$ 0.010) against the frozen ± 0.15 practical-equivalence margin, beside CIFAR-10’s -0.026 on the same cell, with mean clean accuracy *rising* 0.767 $\rightarrow$ 0.835 so the accuracy floor binds on no rung. Under the frozen rule this is a **replication on a second dataset and a differently configured CNN. That is narrower than the wording the pre-registration froze**, which reads “a second dataset and architecture”: `simple_cnn` and `cifar_cnn` are two configurations of one shallow-convnet family, so we report the narrower phrase and leave the frozen file unedited, claiming less than a rule licenses being always permitted where claiming more is not. The prohibition we keep verbatim is the one that matters: never “generalizes”, which one further dataset does not license.

Three disclosures, each of them from the freeze rather than from hindsight. (i) $n=3$ seeds per rung, not 5, fixed in advance to match the EMNIST-byclass payoff matrix’s own trial budget. **The frozen verdict is that $n=3$ one, and the same ladder also reads at $n=5$ with no new run needed:** the comparability amendment of §D.3 ran seeds 45 and 46 of this exact cell at all four rungs, so pooling them with the frozen three, frozen rows winning and the frozen directory never written, gives $\Delta = -0.013$ beside -0.017 , mean ASR 0.024 $\rightarrow$ 0.011 at accuracy 0.758 $\rightarrow$ 0.841: the same label under the same unmoved margin. Two more seeds *widen* the interval across zero, $[-0.034, +0.009]$ against $n=3$ ’s $[-0.033, -0.001]$, so they remove an apparent small fall rather than confirm one, and **one of the five carries the opposite sign** (seed 45, +0.015, its identity rung landing at ASR 0.000018), which a 3-seed mean cannot show. Both counts are reported and neither replaces the other (`analyze_dose_femnist.py --pooled`). Those two seeds were pre-registered in the comparability

document before they ran and are post-hoc in this arm’s own. (ii) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted the flat prediction with attenuation already closed by construction) and the observed trend is a *fall* ($J=16.0$, $z=-1.563$, $p_{\uparrow}=0.941$; at $n=5$, $J=57.0$, $z=-1.211$, $p_{\uparrow}=0.887$, the same direction). We report it and we do **not** score it as support for flatness; the pre-registration states in advance that at $n=3$ a null from this test is not evidence of flatness, and two more seeds do not convert one into evidence. (iii) The two branches had unequal headroom. With the identity rung at ASR 0.027, refuting us required mean ASR($\kappa=2$) > 0.177 against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was arithmetically unreachable. **Equivalence here is to the identity rung, not to a low absolute ASR.**

Harness sanity, which is an identity check and not corroboration. The $\kappa=0$ rung is identity $\circ$ Krum, i.e. Krum alone, and it lands *bit-identically* on the EMNIST-byclass Krum figure from our earlier payoff-matrix sweep (0.0269784552 at accuracy 0.7670795973). Both pipelines are deterministic given a seed and both run three trials from base seed 42, so the agreement shows they execute the same protocol rather than a similar one: the strongest form the pre-registered check can take, and for the same reason **not** independent corroboration of that figure: the same computation run twice cannot corroborate itself. **The pooled seeds of disclosure (i) now carry the same check across two runners.** Having already read those two seeds from the comparability suite, we ran this arm’s own runner over the same eight (rung, seed) cells, pre-registered as an amendment at ad5479b before those runs: it reproduces all eight to ten decimal places, largest gap 0.00e+00, so the pooled reading is not an artifact of which suite measured it (`analyze_dose_femnist.py --pooled`). The same reading applies: identical protocols, not independent confirmation.

A second architecture family, pre-registered, with the gate scored first. The seventh row’s arm is the first row’s arm with **architecture varying alone**: `resnet18` (11.2M parameters, BatchNorm replaced by GroupNorm-8, `conv1` adapted to 32×32 , `maxpool` $\rightarrow$ Identity) in place of `cifar_cnn`, and CIFAR-10, $N=10$, $K=5$, $f=0.2$, α , the committed model-scaling attack and Krum as d_2 all at their frozen values. The decision rule is `experiments/pre-registration-dose-resnet18.md`, git-committed at 89046f6 before any ResNet18 ASR existed, and the runner refuses to start unless its `PREREG_COMMIT` matches that hash. **Two endpoint rungs only**, $\kappa \in \{0, 2\}$ of the frozen four-rung ladder ($\rho = 1.00 \rightarrow 54.60$), at seeds 42–44: the interior rungs are not run, **so this arm carries no trend statistic at all** and the runner prints that restriction rather than leaving it to be inferred.

The gate was scored before the ladder, and it is what makes the arm interpretable. No ResNet18/Krum figure existed on disk, so the pre-registration named the void branch first: unless Krum alone suppresses the committed scaling attack on this architecture (mean ASR < 0.5 at mean clean accuracy ≥ 0.35) *there is no suppression for the ladder to preserve and the arm is void, not negative*. It passed: at $\kappa=0$, mean ASR 0.071 at mean clean accuracy 0.536, $n=3$. Had it failed, the runner refuses the $\kappa=2$ leg outright rather than spending it, and we would be reporting a void.

Suppression does not rise, and under the rule this arm was frozen against the negative replicates on a second architecture family. $\Delta = -0.041$: mean ASR 0.071 at $\kappa=0$ against 0.030 at $\kappa=2$, its magnitude inside the frozen ± 0.15 practical-equivalence margin, beside `cifar_cnn`’s -0.026 and EMNIST-byclass’s -0.017 on the same cell. **We do not write “shows no evidence of a practically meaningful change” about this arm, and disclosure (v) is why:** §S12 defines that phrase by the paired 95% interval lying inside the margin, and at $n=3$ this arm’s does not. **Both endpoints are $n=3$ on the same three seeds**, so the difference of rung means and the paired mean difference are the same number to four places (-0.0410), and `seeds_match` is true on the row; no leg is a mixed- n comparison. Mean clean accuracy *rises* $0.536 \rightarrow 0.557$, so the 0.35 floor binds on neither rung. Under the frozen three-way rule ($|\Delta| < 0.15$ replicates, $\Delta > +0.15$ architecture-specific, $\Delta < -0.15$ a fall) this is a **replication across architecture families**, and it is the arm that earns the phrase: `simple_cnn` and `cifar_cnn` are two configurations of one shallow-convnet family, whereas an 11.2M-parameter residual network is not.

Five disclosures about this arm, four of them limits. (i) **The channels were measured prospectively and the admission column is vacuous here**, exactly as on rows 1 and 5. `results/resnet18_admission.json` records “No ASR. Freezes nothing.” and predates the ladder; it reads the decision disturbed and the admitted adversarial mass not (Δ dec. 0.556 against Δ adm. 0.000 at the top rung), which is the premise the flagship arm rests on and the reason the cell was eligible at all. **But adversarial mass is never present at baseline in any of this arm’s 72 measured rounds**, so its Δ adm. = 0.000 is vacuous and we do not report it as evidence; `count_admission_rounds.py` prints that verdict per row rather than letting a zero speak for itself. (ii) **Clean accuracy is far below this architecture’s own reach:** 0.536–0.557 against the 0.66–0.76 the ResNet18 mini-sweep of §N.2 attains in the same regime, ResNet18 at 50 rounds being far from converged. The gate is a floor and not a claim of a converged model, and no statement here is about converged training. (iii) **One cell, one attack, and it widens nothing else.** It does not add a dataset, an N , a K , an f , a round count or an attack, it carries no composition suite, and it does not touch the sign-reversal cell of §D.3. (iv) **Why this arm fits an overnight budget when §D.3’s disclosure (iv) prices ResNet18 at 65 h:** that figure is a *comparability* cell, two designs $\times$ four rungs $\times$ five seeds = 40 runs at ≈ 5900 s. This arm is one design

× two endpoint rungs × three seeds = 6 runs, which is 13.4 h as measured. The two numbers price different objects and neither supersedes the other. (v) **The interval is wider than the margin, so the frozen ± 0.15 is load-bearing for this arm and, among the arms of the margin ladder, for no other.** The paired 95% interval is $[-0.183, +0.101]$ (per-seed ASR 0.114/0.050/0.048 at $\kappa=0$ against 0.008/0.034/0.047 at $\kappa=2$ on seeds 42–44, hence per-seed differences $-0.106, -0.017, -0.000$ at sd 0.057; the Reproducibility statement’s per-seed promise covers this arm too, and these are those values), so this arm’s $m^* = \max(|lo|, |hi|)$ is 0.183 against the comparability ladder’s binding 0.034 (§S12), and the body’s ladder claim is scoped to Table A23 for that reason rather than stated of the paper as a whole. TOST against the frozen margin *does* reject, at the 90% interval $\alpha=0.05$ reads: $p=0.040$, post hoc, the closest call in Table A22, and the smallest margin at which it still rejects is 0.137. **So this arm replicates under the rule it was frozen against and establishes equivalence at no tighter margin;** three seeds on two rungs is what that buys, and the remedy is seeds rather than wording.

The same cell at a $4\times$ longer horizon, pre-registered, which answers (ii) with a measurement. Disclosure (ii) concedes that ResNet18 at 50 rounds is far from converged, and the objection that follows is that a security negative read at 0.536–0.557 clean accuracy may be an artifact of an undertrained model rather than a property of the composition. We therefore re-ran this exact cell at 200 rounds: same $N=10$, $K=5$, $f=0.2$, the same committed model-scaling attack, Krum as d_2 , the same endpoint rungs $\kappa \in \{0, 2\}$, the same seeds 42–44 and the same imported `run_one`, with only the round count changed. The decision rule is `experiments/pre_registration_dose_resnet18_converged.md`, git-committed at `ca89f96` before the runner was written, and the runner refuses to start unless its `PREREG.COMMIT` matches that hash; its harness check first reproduced a stored 50-round per-seed row through the same imported code path to $< 10^{-9}$, a value comparison against stored rows rather than a hash of a results file. Output: `results/dose_resnet18_converged/summary.json`.

Both gates passed and the negative survives the longer horizon, but the margin does not bind, because the identity rung has less headroom than the margin. Mean clean accuracy reaches 0.652 at $\kappa=0$ and 0.688 at $\kappa=2$, clearing the pre-registered 0.65 convergence-label floor and reaching the 0.66–0.76 band (ii) names for this architecture in this regime (the $\kappa=0$ rung sitting just under its lower end), and the parent arm’s own validity gate passes as well. $\Delta = -0.022$, mean ASR 0.045 at $\kappa=0$ against 0.023 at $\kappa=2$, $n=3$ on one seed set, so $|\Delta| < 0.15$ and the frozen three-way rule returns the verdict it returned at 50 rounds. **We do not report this as a strengthened replication, and the freeze forbade doing so in advance:** an identity rung at 0.045 cannot fall by more than 0.045, so $|\Delta| < 0.15$ was attainable by arithmetic alone, and the pre-registered informativeness condition — identity ASR at or above 0.15, so that the margin binds — *failed*, exactly as that document predicted it would. The identity rung reads 0.045 at 200 rounds and 0.071 at 50 rounds, each stated at its own horizon; **no difference between the two is taken**, here or anywhere, because a longer horizon lengthens both the training and the adversaries’ poisoning exposure, so a cross-horizon Δ would confound them.

What the longer horizon does not buy, from the freeze rather than from hindsight. (a) $n=3$ again: the paired 95% interval is $[-0.182, +0.138]$ at sd 0.064, with per-seed differences $-0.096, +0.015$ and $+0.016$ (per-seed ASR 0.111/0.012/0.011 at $\kappa=0$ against 0.015/0.027/0.027 at $\kappa=2$, which the Reproducibility statement’s per-seed promise covers), so the mean is carried by seed 42, the one seed whose identity rung was appreciably backdoorable at all. This arm’s m^* is 0.182, so like its 50-round parent it fits inside no margin below the frozen 0.15, and *no* equivalence is claimed at any margin tighter than that one. The remedy for $n=3$ is more seeds, which this arm deliberately does not spend. (b) It is *not* a row of the comparability ladder, whose emitter reads only the frozen artifacts (§S12), so the ladder’s binding m^* and the body’s scoping of which single arm needs a margin above 0.15 are claims about that table and are unchanged by this arm. (c) Two endpoint rungs again, never described as a four-rung ladder, so no trend statistic is computed here either. (d) The six runs cost 51.4 h of wall time, 6.6–14.3 h each, the spread being contention with a concurrent arm on the same GPU rather than a property of the cell; the freeze budgeted 40–54 h from two measured rates that disagreed and named the slower one as the rate it planned against.

What the instrument does *not* hold fixed, named in full, and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $p=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is not a conjecture but a **number we already measure and print**: the Δ agg. column of Table A8 is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense *emits*, and it reads **0.892** on this cell. The supplement §S9 splits that 0.892, as vectors, into 0.986 of *re-selection* (a different benign client is chosen, and in 0/11 of the rounds where that happens it is an adversary) and 0.973 of *rescaling* (the chosen client arrives scaled by its own coefficient); for `cos_krum` the displacement is *entirely* rescaling, since 0/15 selections change at any rung. The claim this arm licenses is therefore sharper than a claim about the statistic alone: **a 0.892 displacement of the emitted aggregate, together with decision changes in up to 80% of rounds, leaves suppression inside the pre-registered practical-equivalence mar-**

gin, provided the adversarial coefficient share does not move. What is pinned is pinned exactly: $c_{adv}=1.0$ at every rung with maximum deviation $0.00e+00$ and the share constant to $3.5e-07$ on CIFAR-10, $3.9e-07$ on EMNIST-byclass: float32 norm read-back noise, not slack in the instrument (`measure_admission.py`). Naming (a)–(d) is what keeps the claim to the one channel the instrument closes by construction.

Closing channel (c) as well: the pre-registered score-only control. Channel (c) (the magnitude and geometry of what is actually aggregated) is the one confound Mode S leaves open, so we closed it too rather than argue about it. **Score-only Krum** scores on the transformed stack exactly as before and then aggregates the *untransformed* selected update: the dose can change *which* client contributes and nothing whatever about *what* that client contributes. Statistic and decision are disturbed identically to the uncontrolled arm (the selection is computed on the same transformed stack, so Δ dec. is the same 0.733) while Δ agg. is no longer 0.892 but the re-selection component alone, 0.986 relative to $\|u_{s_0}\|$ (the supplement §S9), the rescaling component being closed by construction. This is the reviewer-suggested control in its own terms: hold the selected update’s magnitude and direction fixed while perturbing the selection statistic independently.

Frozen before the arm existed. The decision rule is `experiments/pre_registration_score_only.md`, git-committed at 35788d9 before any score-only ASR existed, and the runner refuses to start unless its own `PREREG_COMMIT` matches that hash. *Primary:* $\Delta = \text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)$ against the suite’s existing ± 0.15 practical-equivalence margin, on `krum/committed_scaling/CIFAR-10/cifar_cnn`, the same four rungs and seeds 42–46. $\Delta > +0.15$ would have said that the **published flat result was partly an artifact of the magnitude channel** and that the paper’s central claim narrows to “with benign magnitudes free to move”: that branch is why the arm was worth three hours of compute. The $\kappa=0$ rung is imported rather than re-run, because at the identity the transform returns the stack unwrapped and score-only Krum *is* Krum alone; that is an exact claim, not an approximation, and it was verified bit-identically at one seed before the arm started.

The negative survives with the magnitude channel closed. $\Delta = -0.023$ (mean ASR $0.062 \rightarrow 0.039$) against -0.026 for the uncontrolled arm on the identical cell: **closing the magnitude channel moves the effect by 0.003**. The refuting threshold 0.212 is nowhere approached, and mean clean accuracy clears the 0.35 floor at every rung ($0.565/0.629/0.624/0.630$), so no rung is a collapsed model. **The central claim is therefore instrumented against two independent confounds** (attenuation, closed by Mode S, and magnitude, closed here) and the sharpened statement is that a 0.986 displacement of the emitted aggregate together with decision changes in 0.733 of rounds leaves suppression inside the pre-registered practical-equivalence margin.

Three disclosures, from the freeze rather than from hindsight. (i) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted flatness with attenuation already closed) and the observed trend is a *fall* ($J=58.5$, $z=-1.110$, $p_{\uparrow}=0.867$, permutation $p_{\uparrow}=0.874$). We report it and do **not** score it as support for flatness; the pre-registration states in advance that at $n=5$ a null from this test is not evidence of flatness. (ii) The two branches had unequal headroom. With the identity rung at 0.062, refuting us required mean $\text{ASR}(\kappa=2) > 0.212$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was *arithmetically unreachable*: ASR cannot fall below 0, so the largest possible fall on this cell is 0.062. (iii) **Selectors only.** A weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” is undefined for them and the control does not cover them, which means the `coord_median` arm carrying the $+0.098$ rise ($n=5$) is **not** magnitude-controlled, and neither is reputation’s $+0.178$ ($n=5$). Score-only Krum is also an instrument and not a defense: it aggregates an update its own scoring stage never saw.

The corollary the pre-registration staked on `cos_krum`, and exactly what it settles. Section 5 of the same document asserted, before the run, that score-only `cos_krum` must be *bit-identical* across rungs: its statistic is exactly invariant under positive rescaling, so the selection cannot move, and with the emitted update untransformed the whole 50-round trajectory must repeat. It does (accuracy 0.528400 and ASR 0.003111 at all four rungs, $|\Delta| = 0.00e+00$ throughout) and the $\kappa=0$ value reproduces the published cell at that seed exactly, so the two pipelines agree at 0.000000 rather than approximately. **What the check does not do is confirm the arm’s published fall.** It runs one seed, as frozen, and that seed is unrepresentative of the arm: under the uncontrolled dose its own ladder *rises* $+0.224$ where the eight-seed mean *falls* -0.425 . Closing the magnitude channel removes all 0.224 of that seed’s movement, which corroborates the *channel* (for `cos_krum` the emitted displacement is entirely rescaling, so no other route is open) but attributing the arm’s mean fall to magnitude rests on that decomposition rather than on this check, and settling it empirically would need the arm’s other seven seeds. Expanding a pre-registered seed count after seeing the data is precisely what pre-registration exists to prevent, so we report the invariance as confirmed and the arm-level attribution as **corroborated mechanically, not established at the arm mean**.

D.2.1 THE 2×2 CHANNEL FACTORIAL, AND THE DECOMPOSITION IT REFUSES TO LICENSE

Mode S opens two channels into a selector: the *statistic* channel (the dose moves Krum’s pairwise distances, so *which* client is selected changes) and the *magnitude* channel (the selected client enters scaled by its own

coefficient). Score-only closes the second. The fourth cell closes the first instead: **emit-only Krum** scores the *untransformed* stack, pinning the decision to the identity rung's, and aggregates the *transformed* selected update. The four cells are then a factorial, and if the two channels acted independently the three measured effects would satisfy $\Delta_{\text{full}} = \Delta_{\text{SO}} + \Delta_{\text{EO}}$. **They do not.** Rules were frozen in `experiments/pre-registration-emit-only.md` at b995f1b before any emit-only ASR existed, with the runner refusing to start unless its own `PREREG_COMMIT` matches that hash; the $\kappa=0$ rung is imported rather than re-run, because at the identity the transform returns the stack unwrapped and emit-only *is* Krum alone, an exact claim verified bit-identically before the arm started. All three harness checks the freeze required pass at 0.00e+00 across five published arms and the score-only control, so the refactor that added this cell perturbed no published computation.

Cell	statistic channel	magnitude channel	mean ASR	Δ ASR
$\kappa=0$ identity (imported)	closed	closed	0.062	—
Score-only	open	closed	0.032	-0.030
Emit-only	closed	open	0.107	+0.046
Full Mode S	open	open	0.023	-0.039
additive prediction $\Delta_{\text{SO}} + \Delta_{\text{EO}}$				+0.016
residual against measured Δ_{full} (frozen margin 0.05)				-0.054

The table above is the published $n=5$ design, retained as the pre-registration requires; the verdict it produced has since been withdrawn by our own top-up. Seeds 47–61 were added under rules frozen at 684b31e, and at $n=20$ $\Delta_{\text{EO}} = +0.010$ (95% CI $[-0.028, +0.047]$), so the magnitude channel is *established* inert rather than merely point-inside, while the residual is -0.019 ($[-0.061, +0.023]$), **inside** the additivity margin. **So the $n=5$ reading that the channels interact does not survive**, and what remains is *consistent with* separability and not an establishment of it, the interval reaching -0.061 and so not contained in ± 0.05 . Its half-width, 0.042, in fact came in *narrower* than the 0.057 that top-up's own pre-registration projected, so what blocks containment here is the off-centre point estimate rather than the width the projection worried about. Read the $n=5$ paragraphs below as the record of what was published and why it was fragile, not as the current verdict.

The primary rung is forced, and the additive model gets the sign wrong. At $\kappa=2$ mean clean accuracy is 0.333, below the 0.35 floor, so the primary contrast is reported at $\kappa=1$ and **that substitution is forced by the floor rather than chosen after seeing ASR**: Section 5 of the pre-registration named $\kappa=2$ as the rung at risk, and named it for this cell specifically, because emit-only aggregates the *rescaled* update and so degrades training in a way the other three cells do not. Seed-matched at $\kappa=1$, $\Delta_{\text{EO}} = +0.046$ sits inside the frozen inert margin of 0.05, so **the magnitude channel is inert on its own**, which is what we predicted. But $\Delta_{\text{SO}} + \Delta_{\text{EO}} = +0.016$ against a measured $\Delta_{\text{full}} = -0.039$: an additivity residual of -0.054 , outside the frozen 0.05 margin, and **the additive model does not merely mis-size the joint effect, it reverses its sign**. So **no per-channel decomposition of the uncontrolled arm is licensed**, and we withdraw any reading of the re-selection/rescaling split as a decomposition of suppression. That split remains sound as the thing it says it is: the supplement's algebraic decomposition (§S9) decomposes a *displacement*, and says in its own terms that it makes no claim about suppression.

Three reasons this verdict is fragile, all of them ours to report. (i) It is knife-edge on both margins at once, $|\Delta_{\text{EO}}|$ falling *inside* the inert margin by 0.004 and the residual *outside* the additivity margin by 0.004. (ii) Both pre-registered secondaries are null: the emit-only per-seed sign count is 3 up against 2 down, and Jonckheere–Terpstra across the four rungs gives $z = -0.437$, $p_{\uparrow} = 0.669$. (iii) One seed carries the result. Seed 42's emit-only run collapses to clean accuracy 0.2613, below our floor *at that seed* though the rung mean of 0.487 clears it, and its Δ is $+0.272$ against -0.007 , $+0.007$, $+0.013$, -0.058 ; excluding it would give $\Delta_{\text{EO}} = -0.011$ and residual -0.003 , reading as separable. **We do not exclude it**: Section 5 fixes $n=5$ on seeds 42–46 and applies the floor to a rung's *mean* rather than per seed, so no per-seed exclusion is licensed, and the pre-registered top-up settled it without one. **At $n=20$ the knife edge is gone and the verdict it produced is withdrawn**: $\Delta_{\text{EO}} = +0.010$, 95% CI $[-0.028, +0.047]$, so the magnitude channel is *established* inert, and the residual is -0.019 , $[-0.061, +0.023]$, inside the additivity margin, so the interaction reading does not survive. Seed 42 remains the only draw of 20 past either outlier threshold and is still excluded from nothing; realized sd falls $0.130 \rightarrow 0.080$. Separability is now *consistent with* and **still not established**, its interval reaching -0.061 and so not contained in ± 0.05 even though its half-width, 0.042, came in *narrower* than the 0.057 that top-up's pre-registration projected at this variance, and dropping the one seed that produces the unwelcome answer is precisely what pre-registration exists to prevent. Section 6.1 of the same document declared, before the run, that at $n=5$ only a large $|\Delta_{\text{EO}}|$ would be decisive, and $|\Delta_{\text{EO}}|$ is not large. **We therefore report this arm as not decisive, and act on it only in the direction that withdraws a claim**: low power is a reason to stop asserting a decomposition, never a reason to assert one. Every number above is emitted by `experiments/analyze-emit-only-factorial.py`, which reads only frozen artifacts and asserts all four primary quantities reproduce from the per-seed cells to $< 10^{-9}$ before printing. **Selectors only**: the

factorial is defined for `krum` and `cos_krum`, the harness raising for any other d_2 , so it is one selector family, one attack, one dataset, one architecture.

What this settles, and what it does not. The intervention delivers the controlled isolation of C2 the two cross-defense suites could not: within a single defense, with attenuation excluded by construction *and*, in the score-only arm, the magnitude of the aggregated update held exactly fixed as well, **substantial disturbance of the statistic d_2 reads need not produce a corresponding change in suppression once the adversarial coefficient share is held fixed** (on the adjudicating Krum arm under Mode S and, by pre-registered replication, on a downstream defense in a different invariance class, on a second dataset and a differently configured CNN, and on a second architecture family with architecture varying alone) while *admission* of adversarial input is the channel the evidence points to. The statistic-level reading the framework was first written in is refuted; the admission-level reading is confirmed as a *direction* in these interventions rather than established in general. It does *not* rescue the framework’s positive direction wholesale: the pooled two-channel model over all 26 cells is **not** confirmed ($R^2=0.239$), and the replication refutes its cross-arm ordering outright, so admission is a better single predictor than the statistic without being a quantitative one. It addresses neither recall nor discovery (FG→RFA remains a false negative), and nothing in §D.1 is rescored: its pooled refutation, its two refuted arms and its indeterminate arm stand exactly as reported.

Mode M: coordinate masking, a second transformation class. Every other arm in this paper transforms client updates by a positive per-client scalar, the family Prop. 10 classifies and Thm. 8 bounds. **Mode M** is outside it. Benign client i has a random m fraction of the coordinates of each parameter tensor zeroed, drawn from an independent Bernoulli($1-m$) seeded by `default_rng([seed, round, slot])` so it is reproducible and uncorrelated with adversary status, and the surviving coordinates are then **renormalized to the update’s original L_2 norm**. Renormalization is load-bearing and was required in the pre-registration *before any ASR existed*: an unnormalized mask shrinks benign norms and re-opens the magnitude channel that §D.2’s control exists to close. The unnormalized variant was considered and rejected for that reason, and the rejection is recorded rather than omitted. **Adversarial updates are returned unmodified, exactly**: the maximum deviation of every adversarial coefficient from 1 is 0.00 with object identity preserved, a strictly stronger pinning than Mode S’s 3.5×10^{-7} float32 read-back. Rungs are $m \in \{0, 0.2, 0.5, 0.8\}$ and $m=0$ returns the update list unwrapped, so the identity rung is bit-identical to Krum standalone by construction; we check rather than assert this, the identity rung agreeing to 10^{-9} across all three directories that carry those seeds.

The channels were measured before any ASR existed, as the EMNIST-byclass arm’s were. Krum’s decision changes in 0.000/0.133/0.400/0.467 of rounds across the ladder while the adversarial mass it admits is 0.000 at every rung, at an aggregate displacement of 1.095 against Mode S’s 0.892 at a *larger* decision change. So the arm was established prospectively as a generalization test, and the pre-registration (684b31e) then split four ways, including two that would have refuted us: a monotone rise in m would confirm the statistic reading, and masking that made Krum admit the adversary would void the equivalence claim at that rung and confirm the *admission* reading. **Neither fired.** At $n=10$ (seeds 42–51) the paired contrast $m=0 \rightarrow m=0.8$ is $\Delta = -0.016$, 95% CI $[-0.051, +0.019]$, contained in the frozen ± 0.15 margin at a half-width 23.1% of it (TOST $p=5.3 \times 10^{-6}$), every rung mean clearing the 0.5 ceiling and the 0.35 accuracy floor, and Jonckheere–Terpstra finding no increasing trend ($p_{\uparrow}=0.895$). The mask analogue of the score-only control gives -0.018 , $[-0.047, +0.010]$, agreeing with the full ladder to 0.002: since adversarial updates are unmasked in *both* conditions the two ladders differ only when a benign client is selected, so their agreement localizes the absent effect to *which client Krum picks* rather than to the content it emits.

What this arm does and does not license. Only the flat-with-admission-unchanged outcome generalizes the negative, and that is the outcome observed; had admission moved, the arm would have been reported as a confirmation of the admission reading instead. **One mask is one transformation, not the class of non-rescalings**: rotations, projections, quantization and sign compression are untested, so the supportable claim is that the negative replicates on a second, structurally different transformation class, never that it holds for arbitrary upstream processing. Mode M has nothing to impose in 24/120 measured rung-rounds, identical to Mode S’s figure, and like Mode S it is an *instrument* and not a defense, since it reads adversary identity. The `cos_krum` mask ladder was considered and dropped for compute before any ASR existed, which is why the masking column carries one outcome and three channel-only entries.

D.3 SEVEN CELLS, AND A PRE-REGISTERED MECHANISM THAT DID NOT SURVIVE THEM

Is the sign reversal a property of the design, or of this aggregator and attack? Table A3 contrasts the two designs on the one arm that motivated the instrument. That contrast is one cell, and a single cell cannot say whether the sign reversal is a property of the evaluation design or of `coord.median` under a pixel backdoor. We therefore ran the contrast on seven cells: four that already existed in frozen artifacts and are **re-scored, never re-run**, and three that did not exist and were run for this purpose (60 runs for the first two of them, plus 8 under Amendment 2 and 40 for the third under Amendment 4 below, which is the run budget that amendment froze).

Table A9: **The same aggregators under two transformation classes, every prediction frozen before its measurement.** Prop. 10 classifies aggregators under *positive per-client rescaling* only; coordinate masking is outside that family, so Thm. 8 and Prop. 10 make **no** prediction there, which is what makes the second class a test rather than a second demonstration. Rescaling predictions were frozen at 5130cec, masking predictions at 684b31e, each before the corresponding run existed; `experiments/build_prediction_matrix.py` refuses to emit if any observed cell disagrees with its frozen document. **cos_krum is the adjudicating row:** exactly invariant under rescaling, its selection bit-identical at every rung, it loses the guarantee under masking on the same attack and the same seeds, which is what it means for the classification to be a statement about a transformation *class* and not about an aggregator. **Every cell in this table is a decision-channel measurement, not an ASR:** under masking, ASR is measured for krum alone (§D.2), so the other three masking rows are channel-only and this table carries no ASR column to report them in.

aggregator	Prop. 10 class	Δ dec., rescaling		Δ dec., masking	
		predicted	obs.	predicted	obs.
cos_krum	(a) exactly invariant	unchanged	0.000	lapses	0.467
krum	(c) not invariant	changes	0.733	changes	0.467
coord_median	(b) conditionally inv.	partial	0.482	partial	0.700
reputation	(c) not invariant	changes	1.000	changes	0.533

Both designs are the same computation. The two ladders differ in exactly two tokens, the family name and whether `generic_compose` receives an `adv_mask`; data partition, participant draw, attack, local training and seeds are identical. `run_comparability_cells.py` refuses to start unless it first reproduces a published `results/dose_response/` cell bit-identically ($|\text{diff}| = 0.000\text{e}+00$), so the new cells are the same experiment as the four they are compared against, and `analyze_comparability.py` refuses to print any pooled number unless all four published contrasts reproduce.

What was frozen, and when. Before any new run, we pre-registered a mechanism (`experiments/pre-registration_comparability.md`, frozen at c986ef4; two amendments at 1026a96 and f16083b; the three hashes are kept because every other pre-registration in this paper carries its own, and dropping them for brevity would make this the one prereg citation a reader cannot check): the two designs disagree exactly where the downstream defense admits adversarial mass, gated on the admission-level change $\Delta\Lambda_a$ at the top rung. The four existing cells separate perfectly under that rule, which is precisely why **they are training data and cannot also be evidence for it**. The rule mechanically generated a prediction for each new cell, committed in advance: DISAGREE for `coord_median/scaling` ($\Delta\Lambda_a=0.0086$, disclosed at the time as the weakest margin of any cell), AGREE for `krum/EMNIST-byclass` ($\Delta\Lambda_a=0.0000$). The document names in advance what would refute it and commits to reporting refutation rather than fitting a threshold.

The out-of-sample result: one confirms, one refutes, one was never predicted (Table A10). `coord_median` under *model scaling* disagrees, as predicted, on the weakest margin in the set: the outcome-gated ladder reads $+0.002 [-0.078, +0.081]$ where the instrument reads $+0.196 [+0.112, +0.280]$. This is the cell that matters for the headline, because it is the paper’s sign-reversal aggregator under the *other* committed attack, and it says **the disagreement is not specific to the pixel backdoor**. It is a disagreement, not a second sign reversal, and we do not call it one. The EMNIST-byclass cell then **refutes the rule**: $\Delta\Lambda_a$ is exactly 0.0000, the rule predicts the designs coincide, and they do not ($+0.157$ against -0.013 , opposite signs). **Those opposite signs are not a second sign reversal, and we do not count them as one:** both intervals contain zero ($[-0.175, +0.489]$ and $[-0.034, +0.009]$), so neither estimate is separated from no effect and what this cell establishes is a failed prediction of the rule, not a reversal of the measured one. Whether the *reversal* travels is a different question from whether the *rule* does, and the third out-of-sample cell is the one that addresses it.

The third out-of-sample cell moves the sign reversal to a third dataset, and it replicates (Amendment 4, committed at 9b8a395 before the runner was pointed at CIFAR-100). The objection this cell answers is one we sharpen rather than defend: before this row the table already spanned two attacks, two datasets and two CNN configurations, but **the sign reversal itself did not**. It existed on CIFAR-10 with `cifar_cnn` and nowhere else, because the cell that added a second dataset was `krum/scaling`, a *different* cell, and that cell is explicitly not counted as a reversal. Cell 7 therefore holds the aggregator, the attack, the rung grid and the seed set fixed and changes only the dataset, to CIFAR-100 under the same architecture: a 100-way task, which moves the backdoor’s base rate and the clean-accuracy regime together. The outcome-gated ladder falls $-0.213 [-0.284, -0.141]$ where the instrument rises $+0.071 [+0.031, +0.110]$, opposite signs with both intervals excluding zero, so under the same frozen definition applied to every other row this is a **second sign reversal. It is a replication of the phenomenon, not an enlargement of the claim:** three datasets and two CNN configurations is the scope after it, the reversal is *dataset-conditional* on exactly that evidence, and nothing here licenses a statement about a dataset or an architecture outside the three measured.

Table A10: **The confounded ladder against the Mode-S instrument, on seven cells.** Δ ASR from $\kappa=0$ to $\kappa=2$, paired 95% t intervals, each design at its own full n , **which the n column now prints as confounded/controlled** so the strongest row and the weakest are distinguishable on the page. **Row 4, the sign reversal, is the one row at $n=20$ on both legs**, following a pre-registered endpoint top-up (seeds 47–61, experiments/pre_registration_reversal_seed_topup.md); at the frozen $n=5$ it read $-0.272 [-0.422, -0.122]$ against $+0.098 [+0.018, +0.178]$, the same verdict by the same definition. **AGREE** := intervals overlap and point estimates share a sign, or both lie within ± 0.05 of zero; **DISAGREE** := intervals disjoint or signs opposite; **SIGN REVERSAL** := signs differ *and* both intervals exclude zero. The four unmarked cells are the training data the rule was read off. **The rule predicted AGREE for the EMNIST cell and the designs disagree, so it is refuted**; per the frozen clause it is withdrawn, the four-cell pattern is demoted to a description, and the table is reported in full. **The CIFAR-100 cell carries no prediction and its $\Delta\Lambda_a$ entry is absent rather than zero**, because the rule that would have generated one was already dead when that cell was pre-registered (Amendment 4); it is scored on the two endpoint rungs the primary contrast reads, and because no trend secondary was pre-registered for this table, the trend across all four rungs is reported as a *post hoc* description below rather than as a test here. Every cell emitted by `analyze_comparability.py`.

cell	n	outcome-gated	Mode S	$\Delta\Lambda_a$	pred.	observed
krum/scaling	5/20	-0.008 [-0.089, +0.074]	-0.010 [-0.032, +0.012]	0.0000	AGREE	AGREE
cos.krum/pixel	5/8	-0.405 [-0.796, -0.014]	-0.425 [-0.732, -0.118]	0.0000	AGREE	AGREE
reputation/scaling	5/5	+0.762 [+0.658, +0.866]	+0.178 [-0.112, +0.469]	0.0196	DISAGREE	DISAGREE
coord.median/pixel	20/20	-0.273 [-0.334, -0.213]	+0.125 [+0.095, +0.155]	0.0332	DISAGREE	SIGN REVERSAL
coord.median/scaling*	5/5	+0.002 [-0.078, +0.081]	+0.196 [+0.112, +0.280]	0.0086	DISAGREE	DISAGREE
krum/scaling, EMNIST*	5/5	+0.157 [-0.175, +0.489]	-0.013 [-0.034, +0.009]	0.0000	AGREE	DISAGREE
coord.median/pixel, CIFAR-100*	5/5	-0.213 [-0.284, -0.141]	+0.071 [+0.031, +0.110]	—	—	SIGN REVERSAL

* out-of-sample; the other four are the cells the rule was read off.

Five disclosures about that cell, every one of them from the amendment rather than from hindsight. (i) **The admissibility gate was two-sided and frozen before the run:** the identity rung had to reach mean clean accuracy ≥ 0.35 and mean ASR inside $[0.15, 0.85]$, the upper bound because a saturated rung kills the *rise* leg exactly as EMNIST’s 0.027 identity rung killed the *fall* leg in the cell above. It passed on both sides (0.393 accuracy, 0.703 ASR, means over seeds 42–46), and the amendment named in advance the finding it would have reported instead: *no third dataset within this compute budget admits this cell*. (ii) **No prediction was made, and the $\Delta\Lambda_a$ entry is absent rather than 0.0000.** The rule was refuted by the EMNIST cell and withdrawn before this cell existed, so it generates nothing here, and **a dead mechanism is not revived by a cell that happens to agree with what it would have said.** (iii) **The cell is scored on the two endpoint rungs,** which is exactly what the frozen primary contrast reads, so the row is complete as printed. All four rungs of the frozen grid *are* run, on both designs, at seeds 42–46; **an earlier version of this disclosure said the interior rungs were not run and that the cell therefore carried no trend statistic, which was true when written and became false when the runner finished,** and we correct it here rather than silently. What the completed interior buys is a description and not a test: **the comparability pre-registration registers no trend secondary**, so the Jonckheere–Terpstra statistics that follow are *post hoc*, they are emitted with `pre_registered: false`, and no verdict in Table A10 is a function of them. Both ladders start from the same identity rung and then move apart across every rung, not only at the top: the outcome-gated design falls monotonically (0.7034, 0.6804, 0.6508, 0.4905; $J=32.0$, $z=-2.89$, $p_{\downarrow}=0.0019$) while the instrument rises monotonically (0.7034, 0.7035, 0.7220, 0.7740; $J=96.0$, $z=+1.41$), so the reversal is a divergence of the whole ladder rather than an artifact of its endpoints. **The rising leg does not clear 0.05 as a rank trend** ($p_{\uparrow}=0.079$, permutation $p_{\uparrow}=0.084$) even though its endpoint interval $[+0.031, +0.110]$ excludes zero, and **we report that weaker reading beside the stronger one** rather than the half that helps: at $n=5$ per rung a rise of this size is not separable from chance by ranks alone. Both $p_{\downarrow}$ values here are normal-approximation figures, following the supplement’s §S8, because the permutation leg is computed for the increasing direction only. The CIFAR-10 cell of the same aggregator and attack has the same shape on both legs ($z=-2.96$, $p_{\downarrow}=1.5 \times 10^{-3}$ falling against $z=+1.48$, $p_{\uparrow}=0.069$ rising), under the same caveat on the rising one; its falling leg is the arm-4 trend already reported there, and it agrees because `analyze_comparability.py` imports that suite’s trend test rather than reimplementing one. **Those four figures are at $n=5$, and stay there, even though that cell’s endpoint row is now $n=20$:** the top-up ran the two *endpoint* rungs only, so its four rungs hold $[20, 5, 5, 20]$ seeds, and Jonckheere–Terpstra’s null distribution is a function of those group sizes. A mixed- n trend is therefore not the statistic this grid was described with. The scorer computes it anyway and prints it beside the quoted one so that its exclusion is visible rather than assumed ($z=-5.54$ falling, $z=+3.07$ rising); **we do not quote it**, and no verdict reads it. The quoted values are the trend over the five seeds all four rungs share, which is the pre-top-up block, so they are the same numbers the earlier draft printed and not a re-test on new data. (iv) **CIFAR-100 was chosen on wall time and we record that as the limitation it is:** at 660 s per run it was the third dataset that fit an

overnight budget, where EMNIST-byclass (29.5 h per cell, and an identity rung that fails the gate a priori) and ResNet18 (65 h) did not. **Those are comparability-cell costs, two designs $\times$ four rungs $\times$ five seeds, and they do not price the single-design endpoint arm of §D.2**, which is six runs and did fit one night; the figures are not in conflict because they cost different objects. This is the third dataset that fit, not the third that was most informative. (v) **The two designs coincide run for run at the identity rung of this cell**, bit-identically, as they also do for `coord.median/scaling` and do *not* for the EMNIST cell, so it is a property of the cell and not of the design. Both Δ estimates therefore share their $\kappa=0$ leg and the difference *between* the two designs is carried entirely by the $\kappa=2$ rung, which is where the attenuation channel is open. We say so because two identical columns in an artifact look like duplicated data and are not.

We report the refutation as a refutation. $\Delta\Lambda_a$ does not explain when the two designs disagree. The four-cell separation that suggested it stands as a description of four cells and nothing more, and no threshold was fitted after the fact to rescue it. **The observation survives the mechanism’s death:** across seven cells the two designs disagree on five, on two different attacks, three datasets and two CNN configurations, and two of those disagreements are sign reversals. What we do not have is an account of *when* to expect it, which is the honest state of this result and a direct limit on how far the protocol of §4 can be turned into a rule of thumb.

The short form first, because it is all the argument needs: we pre-registered a mechanism before the runs that tested it, one out-of-sample cell refuted it, and it is therefore withdrawn. **One process failure is disclosed below that, because it nearly reversed this section** and because a reader entitled to the first sentence is entitled to how nearly it went the other way. The EMNIST cell was first scored on non-identical seed sets ($n=5$ against a frozen $n=3$), the disagreement carried entirely by two seeds with no counterpart in the other design. We amended the pre-registration to complete the controlled ladder to the frozen seed set rather than resolve the cell on the three shared seeds, which give AGREE: choosing them would have chosen the answer. The completed comparison is the fully paired $n=5$ in the table and it still disagrees. That same amendment asserted that cross-design pairing was the convention the freeze had used. **It was not**, and what caught the error was not our judgement but the analyzer’s own gate: under cross-design pairing the four *published* cells stop reproducing (`krum/scaling`’s controlled ladder moves -0.010 at $n=20$ to -0.026 at $n=5$), and the script refused to print. A third amendment withdraws the claim, and the full sequence is left in the pre-registration.

D.4 TWO FURTHER REGIMES: AN ADVERSARY THAT ADAPTS TO THE DOWNSTREAM DEFENSE, AND $N=100$ AT 20% PARTICIPATION

Is the design disagreement an artifact of a non-adaptive attacker, or of ten clients at full participation? Neither question is answered anywhere else on disk, and each is answered by rerunning *both* ladders on the cell the instrument was built for, `coord.median` under the committed pixel backdoor. The adaptive machinery this paper already has (§F) was pointed at FoolsGold $\rightarrow$ RFA and scored whether *the screen’s* verdict survives, so no artifact held a criterion-aware adversary and `coord.median` together; and the existing $N=100$ material stores game equilibria, which is why §N.1 labels it residual. **The pre-registration was frozen before its output directory existed** (`experiments/pre-registration/regime-dissociation.md`, at `b317af0`, Amendment 1 at `f4c3fec`), and `run-regime-dissociation.py` refuses to start unless `PREREG_COMMIT` matches the amended hash. **Each regime changes one thing:** A changes the adversary, B changes N and K . The cell, the attack, the κ grid, $f=0.2$, $\alpha=0.5$, the round count and `cifar_cnn` are held at the frozen values in both.

The pre-registered estimand is the disagreement itself and nothing else. $D = \Delta(\text{confounded}) - \Delta(\text{instrument})$, paired by seed, at a two-sided 95% t -interval. Its $\kappa=0$ leg cancels exactly, because at $\kappa=0$ both ladders return the update list unwrapped and both rungs are `coord.median` alone, so $D_s = \text{ASR}_{\text{conf}}(\kappa=2, s) - \text{ASR}_{\text{inst}}(\kappa=2, s)$. **Only D is powered at $n=3$, and the freeze says so with arithmetic rather than with a hope:** at the $n=3$ half-width $4.303 \text{ sd} / \sqrt{3} = 2.484 \text{ sd}$, the frozen variances give 0.299 for $\Delta(\text{confounded})$ against a mean of -0.272 and 0.161 for $\Delta(\text{instrument})$ against $+0.098$, so a rule requiring both intervals to exclude zero is one this arm was arithmetically guaranteed to fail; D ’s frozen mean is -0.408 at $\text{sd } 0.109$, hence a half-width of 0.272. Pre-registering the two Δ s would have been pre-registering a null. Each Δ is therefore **reported descriptively, with its half-width beside it and no significance attached**, and **no equivalence claim is made anywhere in this arm:** the phrase *no evidence of a practically meaningful change* is defined in the supplement against the paired 95% interval lying inside the margin, $n=3$ cannot earn it, and `EQUIV_MARGIN` is deliberately not imported by this runner. Table A11 reports both regimes seed by seed.

Three gates, all of them run before the arm was scored, and two of them returned exact zeros rather than small ones. (1) *The adaptive adversary must beat the plain backdoor it replaces*, or regime A tests nothing: a grid of $\epsilon \in \{1, 2, 4\}$ crossed with `decorrelate` $\in \{\text{True}, \text{False}\}$ at $\kappa=0$, seed 42 only, `argmax` mean ASR, then frozen for every rung and seed. (2) *Every rung must clear the 0.35 clean-accuracy floor*, which both regimes do (0.769/0.691/0.690 and 0.474/0.426/0.441). (3) *The shared $\kappa=0$ rung must be proved shared, not assumed:* the instrument’s own $\kappa=0$ rung is run at one seed in each regime and differenced against the confounded one, giving (0.000000, 0.000000) in accuracy and ASR at a tolerance of 10^{-6} in *both* regimes,

Table A11: **The design disagreement reproduces in both new regimes; the *sign* reversal reproduces in one.** Per-seed Δ ASR on the two endpoint rungs, paired within seed, $n=3$. D is the pre-registered primary and its interval is the only inferential quantity here; **both Δ columns are descriptive**, printed with the frozen $n=3$ half-widths 0.299 and 0.161 in mind, and no verdict reads them. Under an adversary that solves for `coord_median` the instrument still moves *up* while the outcome-gated ladder moves down, on all three seeds. At $N=100$ with $K=20$ both designs move down and only the magnitude disagrees, so **regime B reproduces the disagreement and not the reversal**, which scopes the reversal to the participation regime rather than refuting it. The frozen row is Table A3’s published cell, at the n each of its quantities was frozen at, and it is a reference point rather than a fourth arm. Emitted by `experiments/run_regime_dissociation.py --analyze-only over results/ regime_dissociation/summary.json`.

regime	seed	Δ confounded	Δ instrument	D
A: adaptive adversary $N=10, K=5$	42	-0.265	+0.122	-0.387
	43	-0.214	+0.009	-0.223
	44	-0.249	+0.113	-0.362
	mean	-0.243	+0.081	-0.324 [-0.543, -0.105]
B: $N=100, K=20$	42	-0.276	-0.183	-0.093
	43	-0.333	-0.212	-0.120
	44	-0.367	-0.201	-0.165
	mean	-0.325	-0.199	-0.126 [-0.217, -0.035]
frozen cell, as published	mean	-0.273	+0.125	-0.408

the adaptive one included. **The two frozen runners were also proved unchanged by value before either regime ran:** the three optional keyword arguments this arm adds to `run_dose_response.py` and `run_targeted_dose.py` are exercised at their defaults against stored cells of `results/dose_response/` and `results/reversal_seed_topup/`, worst absolute deviation 0.00e+00 over three runs plus a cross-ladder identity comparison. **An md5 of an artifact would not have been that proof**, which is why the check is a value comparison against per-seed rows.

The calibration grid is the reason regime A exists, and its outcome vindicates a decision taken before the data. `decorrelate` confines each adversary to a disjoint coordinate block, which is an anti-FoolsGold device; the freeze argued that against a coordinate-wise median a lone adversary inside its own block is plausibly *weaker*, and refused to import the existing suite’s `argmax` on that ground. The grid says so unambiguously: with `decorrelate` on, ASR is 0.150, 0.192 and 0.199 across the three radii, **every one of them below the plain backdoor’s** 0.395, while with it off they are 0.398, 0.430 and 0.406. Importing $\varepsilon=1$ with `decorrelation` because it won against FoolsGold→RFA would have installed an adversary weaker than no adaptation at all, failed gate (1), and licensed no sentence about adaptivity. **A hyperparameter’s `argmax` does not travel across downstream defenses**, and the general lesson is the one we would want applied to us: an adaptive-attack baseline tuned against one defense is not an adaptive-attack baseline against another.

Six disclosures, five of them limits. (i) **The winning adversary beats the plain backdoor by +0.035 at one seed** (0.4299 against 0.3946), which is a margin, not a rout. What this arm establishes is that the disagreement *survives* an adversary adapting to d_2 ; it does **not** establish that a strong adaptive attack against `coord_median` was found, and no sentence anywhere in the paper says so. (ii) **The adversary does not anticipate d_1** , and that is a property of the object rather than a gap in the attack: Mode S reads adversary identity in order to pin the adversarial coefficient, so it is an instrument and not a deployable defense, and no real attacker can adapt to a stage that does not exist in deployment. The arm therefore bounds adaptation to the pipeline a deployed attacker can see. (iii) **Regime B’s sign reversal does not reproduce, and we report that as the finding it is.** Both designs read the dose as protective at $N=100$, the instrument’s Δ landing at -0.199 and negative on all three seeds (-0.183/ -0.212/ -0.201), so what travels to 20% participation is the disagreement and not its sign. That interval happens to exclude zero, and **it is still not reported as inferential**, because the freeze forbids it at $n=3$ and a bound may not become a claim because it came out convenient. (iv) **Regime B confounds N with K** , deliberately, because the practitioner-facing regime is many clients with partial participation and separating the two would have been a third arm. Nothing here attributes the attenuation of the reversal to either factor alone. (v) **Two rungs, three seeds, one cell, one attack, one dataset, one architecture.** Neither regime is a four-rung ladder, no trend statistic is computed here and none may be quoted from it, and the arms add no dataset, attack, f or round count. (vi) **The base rung does not suppress the attack in either regime** (0.387 and 0.514), which the frozen data already showed, so no sentence from this arm may say that `coord_median` alone suppresses this backdoor.

One unplanned corroboration, reported because it is stronger than what the freeze expected. The pre-registration cautioned that its $N=100$ rung and the residual $N=100$ material “are not bit-comparable”,

and on that basis regime B ran its own $\kappa=0$ rung and never substituted the older one. It ran anyway, and the two agree exactly: freshly measured ASR 0.4261/0.5532/0.5631 at accuracies 0.4773/0.4855/0.4598 against the identical five figures per seed in `results/cifar10-100clients-rich/seed.4{2,3,4}/payoff-results.json`. **So the caution was false and it was false in the safe direction:** it caused an extra measurement rather than a substituted one, the two runners are protocol-identical on this cell, and by the standard we apply to ourselves elsewhere, a computation reproduced across two runners establishes protocol identity and not corroboration of the effect. We leave the committed pre-registration as written rather than amend it a second time after seeing results, and record the correction here.

Amendment 1, and why it was needed before any run. The freeze required the runner to assert that the adversarial tensor *changed* at the first round, as a guard against a hook that is silently never called. That assertion is false by design for the committed pixel backdoor, whose payload lives in the poisoned dataset and whose `manipulate.update` is the inherited identity, so the runner would have aborted on round zero of every non-adaptive leg. Amendment 1 replaces it with an **attack-conditional** assertion: the update must change exactly when the attack overrides `manipulate.update` or the criterion-aware layer is active, and must be *unchanged* otherwise. **This is strictly stronger than what it replaced**, because the defect the original clause existed to catch, a hook that never fires, produces “nothing changed”, which is also this attack’s correct behaviour, and that exact collision once left two composition arms in this repository computing the same thing. All four failure modes were deliberately induced and each was caught. The amendment was committed **before** `results/regime-dissociation/` existed and before any run of this arm, and the pre-amendment text remains at b317af0.

E A SUBORDINATE BOUNDARY STUDY: WHAT THE SCREEN BUYS, AND WHAT IT DOES NOT

Criterion 3 predicts for each two-defense pair whether composition achieves low max-committed ASR. All experiments in this section and §F use one deployment regime throughout (CIFAR-10, $N=10$ clients, $K=5$ per round, adversarial fraction $f=0.2$, 50 rounds, 4-layer CNN), with committed attacks `model.scaling` and `backdoor.pixel`. **Convention:** unless stated otherwise, *ASR* means *max-committed ASR* (the maximum over the two committed attacks of the per-attack mean across seeds) thresholded at 0.5: the level at which the trigger fires on the majority of triggered inputs, so that “suppresses” means the backdoor is not the model’s dominant response. **That 0.5 is an operational classification threshold, not a theoretically privileged security boundary:** nothing in the propositions depends on it, and a deployment with a lower tolerance should read the per-cell ASR rather than the label. adaptive-attack, order-sensitivity and per-seed figures are labelled explicitly where they differ. A second gate is needed because a destroyed model has a low ASR for the wrong reason: every cell in this section, §F and §D–§D.2 must also hold mean clean accuracy ≥ 0.35 (our **accuracy floor**), below which we call the cell *uninterpretable* rather than suppressed, in either direction. **That scope is wider than an earlier version of this sentence stated:** it named §D–§D.2 alone, while the paragraph it sits in scopes itself to this section and §F, so the floor was announced as a second gate and then withheld from most of the cells it governs. Widening it to what this paragraph already claims reclassifies exactly one cell, `fg→rfa` at $\alpha=0.1$ (§E.6(E)), and that reclassification *weakens* a limitation we state against ourselves, which is why both readings are printed there rather than only the gate-applied one. **Sample sizes are small and were fixed before the runs:** $n=5$ seeds per cell in the interventional suites ($n=8$ for `cos.krum`, $n=3$ frozen for the second-dataset replication, which also reads at a pooled, post-hoc $n=5$, §D.2), $n=30$ for the FoolsGold→RFA characterization, so intervals are wide, and no null result is read as flatness unless it clears a pre-registered practical-equivalence margin. **“Refuted” means one thing throughout:** a pre-registered *directional* hypothesis was rejected against the rule frozen for it, never that a mathematical claim was disproved. Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$ and each round samples K participants uniformly without replacement from all N , with no stratification, so the per-round adversary count is *hypergeometric* rather than fixed: for $N=10$, $f=0.2$, $K=5$, $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$ and $P(n_a=1) = 140/252 = 0.556$ (measured over 100 rounds: 0.19/0.54/0.27). Two consequences the data bear out: CoordMedian’s count-majority condition $n_a < K/2 = 2.5$ holds for every attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds, while TrimmedMean’s budget $\lfloor \beta K \rfloor = \lfloor 0.2 \cdot 5 \rfloor = 1$ is exceeded whenever $n_a=2$: 22.2% of rounds, ≈ 11 of 50. That is a second, count-based reason FoolsGold→TrimmedMean fails (0.508) while FoolsGold→CoordMedian succeeds (0.107) under an identical upstream transform; we retain the pre-registered C3-FAIL label, the budget violation being an additional mechanism rather than a relabelling. We validate in two waves. The **development set** (18 targeted pairs, Table A12) spans all criterion prediction categories and was used to refine the criterion (adding C3). The **held-out set** (24 pairs, Table A13) was *pre-registered*: all 24 predictions were committed to version control (commit 7e4d9b1) *before* any held-out experiment was run, eliminating post-hoc classification concerns. The criterion predicts LOW (<0.5) or HIGH (≥ 0.5) max-committed ASR. The verdict categories are:

PASS precision: 2 of the 2 selected pairs are low-ASR. That is 100% as a descriptive statistic, and the positive class is two cases, so it resolves the direction and not the rate. Under the tightened per-attack C1

Table A12: Composability criterion validation (max across committed_scaling and committed_pixel; $\pm$ is per-seed std where reported). Threshold: LOW < 0.5 , HIGH ≥ 0.5 . **Seed counts differ by row and are marked:** unmarked rows are $n=3$, * is $n=5$, $\dagger$ is $n=30$, $\S$ is split across arms and noted in the table.

Pair	Failure mode	Predicted	Max ASR	Correct?
<i>PASS: $C1 \wedge C2 \wedge C3$ all hold (predicted LOW)</i>				
foolsgold $\rightarrow$ rfa $\dagger$	—	LOW	0.093 ± 0.082	✓
foolsgold $\rightarrow$ coord_median*	—	LOW	0.107 ± 0.034	✓
reputation $\rightarrow$ coord_median $\S$	—	LOW	0.463 ± 0.136	✓
<i>C1-FAIL: neither defense individually suppresses all attacks (predicted HIGH)</i>				
reputation $\rightarrow$ trimmed_mean $\dagger$	C1 (pixel)	HIGH	0.679	✓
reputation $\rightarrow$ rfa	C1 (pixel)	HIGH	0.825	✓
reputation $\rightarrow$ foolsgold	C1 (pixel)	HIGH	0.807	✓
reputation $\rightarrow$ norm_clip	C1 (pixel)	HIGH	0.813	✓
norm_clip $\rightarrow$ foolsgold $\dagger$	C1	HIGH	0.887	✓
<i>C2-FAIL: d_1 destroys d_2's signal direction (predicted HIGH)</i>				
norm_clip $\rightarrow$ reputation	C2	HIGH	0.819	✓
rfa $\rightarrow$ reputation	C2	HIGH	0.882	✓
<i>C3-FAIL: d_1 compresses boundary below d_2's threshold (predicted HIGH)</i>				
norm_clip $\rightarrow$ trimmed_mean	C3	HIGH	0.780	✓
norm_clip $\rightarrow$ rfa	C3	HIGH	0.853	✓
rfa $\rightarrow$ trimmed_mean	C3	HIGH	0.678	✓
rfa $\rightarrow$ coord_median	C3	HIGH	0.485	×
foolsgold $\rightarrow$ trimmed_mean	C3	HIGH	0.508	✓
norm_clip $\rightarrow$ coord_median	C3	HIGH	0.433	×
<i>DEGEN: d_1 has no per-client transform (predicted HIGH)</i>				
fedavg $\rightarrow$ norm_clip*	DEGEN	HIGH	0.935	✓
fedavg $\rightarrow$ trimmed_mean*	DEGEN	HIGH	0.846	✓

Overall: 16/18 = 89%. HIGH direction: 13/15 correct (87%). Under the tightened C1 the certified PASS set is 2/2 correct; fg $\rightarrow$ rfa is reclassified as a synergy false negative.

$\dagger$ Reclassified from C2-FAIL: by Proposition 10(a) clipping preserves FoolsGold's similarities exactly, so C2 holds and the pair fails C1. Prediction and outcome unchanged.

$\S$ Pixel arm $n=30$ (seeds 47–71 pre-registered at de77a7f), scaling arm $n=5$; the pixel arm sets the max, and its interval [0.412, 0.514] still crosses 0.5 at $n=30$.

(§C) the certified in-distribution set is 2 pairs, fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.463 at $n=30$), both correct; fg $\rightarrow$ rfa (0.093) is no longer certified and is reported below as a false negative of our own criterion. This is the practitioner-relevant result: the criterion's positive predictions are reliable, at the cost of 40% recall. Overall development-set accuracy: **16/18 = 89%**. Two misses, both in the HIGH direction:

One certified pair is measured to the threshold, and a pre-registered top-up did not resolve it. Rep $\rightarrow$ CM's LOW label rests on its pixel arm, whose $n=5$ interval [0.231, 0.616] crossed 0.5 with one seed at 0.665, so we pre-registered seeds 47–71 (de77a7f: seed list, unchanged 0.5 rule and the three admissible outcomes all fixed before any run) and took that cell to $n=30$, the same power as the pair the criterion rejects. It is **still unresolved**: mean 0.463, 95% interval [0.412, 0.514], with 10 of the 30 seeds above the threshold. Six times the seeds narrowed the half-width $3.8\times (0.192 \rightarrow 0.051)$ and moved the mean *up* from 0.423, toward the threshold rather than away, so the cell is near-threshold rather than merely noisy. The LOW label holds by the frozen mean rule, the pre-registration forbids adding seeds once the outcome is seen, and we therefore stop here and report it unresolved: the certified in-distribution set is 2/2 correct on the mean, and only fg $\rightarrow$ cm is certified *and* resolved below threshold. The frozen seed reproduced bit-for-bit before the top-up ran (0.43178 at seed 42), so the 25 new seeds extend the original five rather than replacing them; the scaling arm stays at $n=5$ (0.020), where it cannot set the max. Clean accuracy is 0.755 over the 30 runs, and 0.50, 0.68, 0.76 on the three cells, all well clear of the collapse regime.

Two boundary conditions, stated alongside the primary result. Both are developed in full later, but neither should be met late. (i) **Extreme heterogeneity breaks the PASS pairs: two of three on the raw reading, one of three once our own accuracy floor is applied.** At $\alpha=0.1$, fg $\rightarrow$ rfa reaches 0.569 and rep $\rightarrow$ cm 0.548; only fg $\rightarrow$ cm survives, at 0.433. fg $\rightarrow$ rfa's cell sits at clean accuracy 0.20, below our 0.35 floor, so it is *uninterpretable* rather than a crossing and the gated count is 1/3; both counts are carried, because the floor softens a limitation we state against ourselves and rep $\rightarrow$ cm's crossing (0.548 at 0.48) survives it

either way. What makes this more than a negative datapoint is that the failure appears in the theorem’s own quantity: RFA’s separation margin $m := \Delta_{\text{sep}}/[2(R_B + \delta')]$, measured on the transformed points, collapses from an all-round mean of 3.24 to 1.01 (exactly the $m \rightarrow 1$ boundary at which Corollary 4’s bound degrades to 40% adversarial mass), so a practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy. We emphasise that this is a post-hoc measurement of a known failure, not a pre-registered prediction of it; full sweep and caveats (bimodality at $n=3$, accuracy collapse to 0.20–0.48) in §E.6(E). **(ii) Mechanism preservation is not adaptive-attack robustness.** A criterion-aware adversary solving FoolsGold’s and RFA’s constraints jointly (decorrelated coordinate blocks to evade the similarity test, ϵ -ball projection to stay inside the geometric median’s radius) raises $\text{fg} \rightarrow \text{rfa}$ from 0.064 (seed-matched at $n=30$, and on the pixel arm the attack is built from) to 0.157 within our evaluated composition-aware attack family (§F). Theorem 8 bounds what the preserved mechanism admits, not what an adversary reshaping its update to satisfy both constraints at once can do, and the $2.4\times$ increase is the size of that gap on our flagship pair.

Held-out validation (pre-registered): 22/24 = 92% (*Screening*: an evaluation ordering, and by Proposition 6 not an identification of the mechanism’s contribution). Table A13 reports the 24 pre-registered pairs. Combined across both waves: **38/42 = 90.5%**. **Read against its base rate that figure is nearly uninformative, which is why it appears here, where it is computed, and in no summarizing position.** Only 5 of the 42 pairs are low-ASR (the 12% base rate of §E.3), so the constant predictor that calls every pair HIGH already scores $37/42 = 88.1\%$; 90.5% is one pair better than that, and the tally is carried by correct negatives. The informative quantities are precision 2/2 and recall 40% (§E.3). The two held-out misses are both LOW-predicted pairs that were flagged as *speculative* in the pre-registration (neither is a clean $\text{C1} \wedge \text{C2} \wedge \text{C3}$ PASS):

Table A13: Held-out pre-registered validation (24 pairs, $n=5$ seeds; max across `committed_scaling` and `committed_pixel`). All predictions committed to version control (commit 7e4d9b1) before experiments. Threshold: $\text{LOW} < 0.5$, $\text{HIGH} \geq 0.5$. Only the two LOW predictions are shown individually; the 22 HIGH predictions (all correct) are summarized.

Pair	Category	Predicted	Max ASR	Correct?
<i>Speculative LOW predictions (not clean $\text{C1} \wedge \text{C2} \wedge \text{C3}$ PASS)</i>				
<code>fedavg</code> $\rightarrow$ <code>coord_median</code>	DEGEN	LOW	0.519	×
<code>foolsgold</code> $\rightarrow$ <code>norm_clip</code>	C1 (uncertain)	LOW	0.927	×
<i>HIGH predictions (22 pairs: 6 CM-as-d_1, 6 TM-as-d_1, 4 fedavg-as-d_1, 3 weighting-d_1, 3 RFA-as-d_1; range 0.70–0.94)</i>				22/22 ✓
Held-out: 22/24 = 92%. Combined with development set: 38/42 = 90.5%.				
Constant-HIGH baseline on the same 42 pairs: 37/42 = 88.1%.				

Rep $\rightarrow$ tm: composition dominates despite C1-FAIL. Reputation $\rightarrow$ TrimmedMean is correctly classified as C1-FAIL (pixel ASR 0.679 > 0.5), yet it is the paper’s main composition result. The resolution: the criterion predicts *max-committed* ASR, while the deployment-relevant metric is *oracle* ASR (0.316). The oracle cannot exploit the pixel gap because scaling (which reputation *does* suppress) dominates its utility. The criterion thus correctly identifies that $\text{rep} \rightarrow \text{tm}$ does not suppress every committed attack, while the game-theoretic analysis shows it is strategically dominant.

Tightening C1 costs us our own flagship, and reveals a synergy case. Under the per-attack quantifier in C1, **fg $\rightarrow$ rfa no longer certifies**: neither FoolsGold (0.732) nor RFA (0.845) suppresses the pixel backdoor individually, and FoolsGold’s 0.200 against model-scaling is hollow because it comes with 10% clean accuracy. The composition nonetheless reaches 0.093. This is genuine *synergy* (suppression neither constituent possesses) and Proposition 7 proves that *no* inherited-suppression screen can certify it, so this is a scope boundary rather than a tuning failure. We therefore report our lowest-ASR composition as a false negative of our own criterion. The certified in-distribution set is $\text{fg} \rightarrow \text{cm}$ (0.107) and $\text{rep} \rightarrow \text{cm}$ (0.463), both correct: precision 2/2, recall 2/5 = 40%.

A directional C1 would empty that certified set, and we report the price rather than pay it. The natural next tightening is to replace C1’s “some constituent suppresses a ” with the directional “ d_2 suppresses a ”, on the ground that C0 forces d_2 anyway. **C0 does not force it.** C0 is existential over attacks while C1 is universal over them, so Lemma 1 pins d_2 only on the witnessing attack a^* ; off a^* the constituent holding that arm may be d_1 , and the two definitions accordingly disagree on 4 of the 42 pairs. On all four the downstream defense is `coord_median`, whose standalone model-scaling ASR is 0.519 and so above threshold, while the upstream `foolsgold` (0.200) or `reputation` (0.017) is what holds that arm; every C1 input here is the $n=5$ single-defense baseline on seeds 42–46. **Both certified pairs are of exactly that kind**, so a directional C1 selects nothing in distribution and leaves precision *undefined* instead of 2/2. **What the substitution correctly points at is real, and it is a quantifier and not a direction:** on the model-scaling arm of these two pairs the suppression is inherited from an *upstream* defense, which nothing downstream can undo, so the signal-preservation argument contributes nothing to that arm even though the arm counts toward the max-committed

label. The criterion and Theorem 8 are therefore aligned *per attack* and not per pair. We keep C1 and disclose the asymmetry; the census is emitted by `experiments/analyze_condition_ablation.py`.

E.1 A PROSPECTIVE TEST ON A NEVER-RUN DEFENSE MENU

Does the criterion predict on a menu it has never seen? **Result: 13/20, and the shortfall is in our experimental design.** At the same 0.5 threshold: TP = 8, FP = 0, TN = 5, FN = 7 (Table A14). **The suite carried the sanity gate our out-of-sample battery lacked**, pre-registered with the labels: a predicted-FAIL pair counts only if the attack clears ASR 0.5 against FedAvg, which both committed attacks do (0.667 scaling, 0.768 pixel), so a pair cannot be scored correct because the attack failed on its own. Every predicted-LOW pair came out LOW, 8/8, and that figure is far weaker than it looks, because **FLTrust suppresses both committed attacks on its own** (max-committed ASR 0.048 at 0.575 clean accuracy), the first defense in our menu that does. All 13 pairs containing FLTrust land in [0.041, 0.078], within 0.030 of FLTrust alone, in either slot. The two slots fail for different reasons, and the distinction matters:

The two slots fail differently, and the difference is the finding. (i) *FLTrust as d_1 (7 pairs, 2/7).* d_1 alone already suppresses, so the composition inherits that suppression whatever d_2 does; all five predicted-HIGH misses have real C2/C3 failures, but destroying d_2 's mechanism is costless when d_2 was never load-bearing. This is C0, invisible on the original 42 pairs because no defense there suppresses alone (the minimum single-defense max-committed ASR is CoordMedian's 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433), so we count these 7 pairs out of scope *retrospectively*: a weaker claim than a pre-registered exclusion. (ii) *FLTrust as d_2 (6 pairs, 6/6), but the prediction is nearly an identity.* FLTrust gates on the scale-invariant $\cos(u_i, g_{\text{srv}})$ and then renormalizes every update to $\|g_{\text{srv}}\|$, discarding magnitude entirely, so a positive per-client rescaling is *exactly* cancelled: four of the six pairs are bit-identical to FLTrust alone or differ only within seed noise. The single genuine deviation is FG→FLTrust (0.064), the one d_1 producing exact *zeros* rather than positive rescalings and hence a Lemma 2 interaction. This arm is a consistency check between theory and implementation, **not** a falsifiable test: every d_1 in our menu is a positive rescaling and Proposition 10(a) permits no other outcome. (iii) *Neither defense is FLTrust (7 pairs, 5/7):* the only arm that could have surprised us. Five predicted-HIGH pairs are correct (0.870–0.943), though only 2 of the arm's 7 pairs compose two real mechanisms once the pass-through and MultiKrum degeneracies are accounted for (below); both misses are diagnosable and neither is a mechanism error: fg→krum measures 0.078 at 9.9% clean accuracy (the collapsed-model artifact, not suppression), and krum→cm measures 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519. The arm contains *zero* predicted-LOW pairs, so it bears on specificity only. Per-pair numbers for all three arms are in the supplement (§S1).

What the suite does and does not establish. Counting only in-scope pairs gives 11/13; the arm that carried real risk of falsification gives 5/7 with both misses explained. But it is not the prospective positive the framework needs, because **it contains none**: by (i) and (ii) the 8 predicted-LOW pairs are 6 near-identities plus 2 out-of-scope inheritances.

We withdraw the three matched contrasts we previously drew here, which we had read, d_1 fixed while d_2 moved from FLTrust to a Euclidean rule, as a 19–20× ASR gap turning on exactly the property Proposition 10 identifies. Two ran against MultiKrum, which at $K=5$ selects *all five* participants and is bit-identical to FedAvg (1.9×10^{-6} per aggregation; a genuine selector again at $K=10$), so they carried no downstream defense at all. Substituting Krum does not repair them, and that is the finding: all three Krum arms take their max-committed ASR from the pixel backdoor, which Krum *fails to suppress standalone* (0.583), so the gap measures C1 failure and not disturbed invariance, and nc→krum is bit-identical to Krum alone on all three seeds because under that attack no client exceeds $\tau=5$ and d_1 is the exact identity. Counting the pass-through d_1 rows too, 8 of the 20 rows reduce to a single defense rather than the 5 labelled DEGEN at freeze, and arm (iii) composes two real mechanisms in 2 pairs, not 7 (App. S2). The frozen labels and the 13/20 score stand; what we retract is our reading of them. The confound is accordingly worse than we reported: in this menu the invariant-signal d_2 is also the only d_2 that suppresses alone, so C1 and C2 co-vary. Lemma 1 explains why no other choice of defense would have helped: in scope, C0∧C1 *force* d_2 to suppress a^* alone, so the confounding is structural. Breaking it requires holding the suppression principle and standalone effectiveness fixed while varying only the invariance class, which is the design of §E.2; that section's Δ analysis is the only contrast in either paper holding C1 exactly fixed, and it is where our C2 evidence now rests.

E.2 BREAKING THE CONFOUND: A MATCHED METRIC-SWAP ABLATION

Can the confound be broken by swapping the metric instead of the defense? Lemma 1 says mechanism–effectiveness confounding cannot be escaped by picking a different defense off a menu: in scope, C0∧C1 *force* the invariant-signal d_2 to be one that suppresses a^* alone. So we break the confound by construction instead, holding the suppression *principle* fixed and varying only the invariance class of the statistic d_2 reads.

The design's premise was refuted by its own baselines. Matching failed in both arms, for two independent reasons, and because the C1 inputs are measured before the freeze we learned this before the suite ran: the

Table A14: Prospective suite: 20 pairs, labels frozen at commit 262cf35 before any run. Max-committed ASR over 2 attacks $\times$ 3 seeds; threshold 0.5. Categories are verbatim from the pre-registration, including its own uncertainty hedges.

Pair	Frozen category	Pred.	Max ASR	Acc	Hit
<i>In scope (C0 holds): d_2 load-bearing</i>					
Rep $\rightarrow$ FLTrust	certified	LOW	0.046	0.577	✓
NC $\rightarrow$ FLTrust	certified	LOW	0.046	0.576	✓
RFA $\rightarrow$ FLTrust	certified	LOW	0.047	0.574	✓
Krum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
MKrum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
FG $\rightarrow$ FLTrust	certified	LOW	0.064	0.547	✓
FG $\rightarrow$ Krum	C1-fail	HIGH	0.078	0.099	×
Krum $\rightarrow$ CM	DEGEN	HIGH	0.450	0.770	×
Krum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
MKrum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
RFA $\rightarrow$ MKrum [†]	C1-fail	HIGH	0.875	0.782	✓
Rep $\rightarrow$ Krum	C1-fail	HIGH	0.880	0.530	✓
NC $\rightarrow$ MKrum [†]	C1-fail	HIGH	0.943	0.761	✓
<i>Out of scope (C0 violated): FLTrust as d_1 suppresses alone at 0.048</i>					
FLTrust $\rightarrow$ CM	certified-uncertain	LOW	0.041	0.505	✓
FLTrust $\rightarrow$ MKrum [†]	C2-fail	HIGH	0.045	0.576	×
FLTrust $\rightarrow$ TM	C3-fail-uncertain	HIGH	0.046	0.529	×
FLTrust $\rightarrow$ NC	C2-fail	HIGH	0.047	0.576	×
FLTrust $\rightarrow$ RFA	certified	LOW	0.048	0.512	✓
FLTrust $\rightarrow$ Rep	C2-fail	HIGH	0.048	0.518	×
FLTrust $\rightarrow$ Krum	C2-fail	HIGH	0.078	0.361	×

Full suite 13/20 (TP 8, FP 0, TN 5, FN 7); in scope 11/13; non-FLTrust arm 5/7.

[†]MultiKrum at $K=5$ keeps all five participants, so it is bit-identical to FedAvg and these rows carry no downstream defense; MKrum $\rightarrow$ FLTrust and MKrum $\rightarrow$ RFA repeat their Krum rows exactly. 8 of 20 rows reduce to one defense; frozen labels unchanged (App. S2).

only reason it can be reported as a finding rather than a post-hoc excuse. `cos_reputation` does not merely weaken relative to its Euclidean original, it collapses, from Reputation’s 0.017 on model-scaling to **0.982**. `cos_krum` reaches 0.078 on model-scaling, which passes C1’s numeric threshold, but at **0.146** clean accuracy, the hollow-suppression artifact already documented for FoolsGold (0.200 at 0.10) and `fg $\rightarrow$ krum` (0.078 at 0.099), so C1 fails under the accuracy gate we apply everywhere else. The mechanism is visible in the baselines rather than asserted: undefended FedAvg under model-scaling is itself at 0.100 accuracy, so Krum’s 0.577 is *restoring* utility the attack destroyed and `cos_krum`’s 0.146 fails to. **Magnitude information is what lets this defense family handle a magnitude attack**: a direction-only mirror of a Euclidean scorer cannot see a scaling attack at all, so removing the magnitude dependence removes the suppression with it: exactly the coupling the pre-registered contingency anticipated, stated there as the possibility that “invariance and standalone effectiveness may be inherently coupled” for this family. On the magnitude-neutral pixel backdoor the mirror does survive (0.300 at 0.598), but there its control Krum fails C1 (0.583), so those arms are unmatched in the opposite direction and the pixel cells cannot carry the test either.

The consequence is that **the confound re-materialized inside an experiment built specifically to break it**, which we report as the round’s methodological finding rather than as a failed run. Lemma 1 establishes that mechanism–effectiveness confounding is structural in scope; this is what trying to escape it looks like when the mechanism being varied is the mechanism the defense relies on. Two responses were available and we took neither. We did not switch to a pixel-only protocol, which would have made `cos_krum` pass C1 and manufactured two predicted-LOW cells: choosing the attack after seeing the C1 inputs is attack-shopping regardless of the resulting labels. We did not adjust the accuracy gate to admit 0.146, which would have produced a predicted-LOW pair by weakening the very criterion that catches hollow suppression elsewhere. The pre-registration’s second non-negotiable (“no switching of committed attack, threshold, seed count, or accuracy gate”) forecloses both, and either would have bought a positive test at the cost of the property that makes the test worth running.

Outcome: 8/8 correct, and zero positive tests. Because C1 binds for all eight pairs (on model-scaling through the two baseline failures above, on the pixel backdoor by design) every frozen label was HIGH, so the suite bears on specificity only, as a consequence of inputs measured before the freeze rather than a choice made afterwards. We ran all 8 pairs $\times$ 2 attacks $\times$ 3 seeds anyway, because a suite whose labels are all one class is still falsifiable in the direction that matters: a single LOW cell would have been a false positive against a criterion claiming 100% precision. All eight came out HIGH (8/8 correct, TP = 0, FP = 0, TN = 8, FN = 0, max-

committed ASR spanning 0.508–0.935) with labels frozen in `pre_registration_metric_swap.md` at commit 356897b, timestamped before `results/metric_swap/` existed and reproduced verbatim in the runner’s PAIRS list so scoring cannot drift. We state the limitation in the same breath as the result: the suite contains **no falsifiable prospective positive test**, and 8/8 is not validation of the framework’s predictive claim: eight correct negatives on pairs our own conditions could only have called HIGH is weak evidence, and it supports a claim about specificity, not prediction.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos_krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied (0.078 passes the numeric threshold; 0.146 accuracy fails the gate), so rather than choose a reading afterwards we froze *both*, with the resolution rule fixed in advance: accuracy-gated $\Rightarrow$ HIGH, numeric-only $\Rightarrow$ LOW, and a LOW outcome at accuracy ≥ 0.35 would have recorded our two primary labels as misses. Both pairs came out **HIGH** (NormClip $\rightarrow$ `cos_krum` 0.508, RFA $\rightarrow$ `cos_krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative specification of itself on data neither of us had seen. The weaker half is flagged in the supplement (§S3): the NormClip arm sits barely above threshold and it is the RFA arm that carries the conclusion.

A post-hoc contrast that is not confounded. The cross-pair scoring above is confounded in exactly the way Lemma 1 predicts, but the frozen suite happens to contain a comparison that is not, and it is the strongest C2 evidence in either paper. Compare a composition against *the same* d_2 ’s own standalone ASR on the same attack, under two different upstream transforms. C1 is then identical by construction (same downstream defense, same attack, same measured standalone effectiveness on both sides), so the residual $\Delta = \text{ASR}(d_1 \rightarrow d_2, a) - \text{ASR}(d_2, a)$ cannot be attributed to d_2 being a stronger or weaker defense. What does vary is how much the upstream transform actually disturbs d_2 ’s statistic, and that quantity was measured on the mechanism itself (`results/cos_invariance_check.json`) rather than inferred from any ASR. The frozen label scoring is fixed at $n=3$ and must not move, so we topped the six powered cells up to $n=5$ in a separate directory, with the tested direction fixed in the top-up script before the new seeds ran (`results/metric_swap_topup/`). Because Δ is a difference, the subtrahend moves with it: each topped-up cell’s standalone is recomputed at the *same* five seeds, from the dose ladder’s $\kappa=0$ rung, which is that d_2 alone under the identical configuration (§D) and agrees with the published standalone baselines bit-for-bit on every seed they share (largest deviation 0.00e+00 in ASR and accuracy across 16 seed-level comparisons), so it is the same quantity carried to five seeds and not a second measurement of it. Scoring a five-seed composition against the three-seed baseline would hide a mixed- n comparison inside the subtrahend, where it is invisible, and on one cell below it would invert the conclusion. For Krum under model-scaling (standalone 0.062 at 0.565 accuracy over those five seeds; 0.061 at 0.577 on the frozen three): NormClip changes Krum’s selection in 0/9 rounds and the composition reaches 0.062 ($\Delta = +0.001$, per-seed 0.120/0.018/0.047/0.058/0.068); RFA changes it in 6/9 rounds and the composition reaches 0.779 ($\Delta = +0.717$, per-seed 0.974/0.116/0.982/0.904/0.916): a gap of +0.716 between two compositions sharing a downstream defense, an attack, and a baseline, and at $n=5$ it is statistically resolved (Welch $t = 4.28$, two-sided $p = 0.012$; paired $t = 4.60$, $p = 0.010$; Mann–Whitney one-sided $p = 0.008$), which it was not at $n=3$.

The pattern holds across the suite once cells without power are excluded: Δ is observable only where standalone d_2 genuinely suppresses, since a baseline already above 0.5 shows $\Delta \approx 0$ by ceiling effect whatever the transform did. Restricting to standalone ASR < 0.5 at accuracy ≥ 0.35 leaves six cells, and at $n=5$ measured disturbance still splits them perfectly: the three undisturbed cells (0/9) give $\Delta \in [-0.044, +0.001]$, the three disturbed ones ($> 0/9$) give $\Delta \in [+0.717, +0.878]$: a separation of +0.716 with no overlap, wider than the +0.629 the frozen $n=3$ scoring gives. That separation is the *same* +0.716 as the Krum contrast above and not independent evidence on top of it: both extremes are those two Krum cells, which share a standalone, so the width of the split is algebraically that contrast’s gap. The other four cells extend the two halves outward rather than the distance between them. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy on the frozen three seeds, 0.490 at 0.607 on all five, since two of the added seeds are high-ASR) and carries that suppression through under *both* upstreams. Under NormClip it carries through *exactly*: NormClip never binds on this attack (max client norm 3.04 against $\tau = 5$, 0/9 rounds), so the composition is bit-identical to the baseline at all five seeds and $\Delta = +0.000$ is an identity rather than a measurement; under RFA the composition is 0.447 against that 0.490, i.e. $\Delta = -0.044$. Kept against the frozen three-seed baseline these same rows would instead read +0.190 and +0.147, and the natural reading of *that*, the magnitude channel of Lemma 2 moving Δ where the selection is exactly invariant, would credit an effect to a transform that probably did nothing: the movement is entirely the baseline’s seed set. So this contrast has no power on the magnitude channel in either direction, because its one exactly-invariant downstream sits under an upstream that never rescales; measuring that channel requires moving the dial on purpose, which is what the ladder’s arm 3 (§D) does prospectively. The ten excluded cells are listed rather than dropped silently in the supplement (§S3).

The suite contains a second such contrast, on Reputation under model-scaling (standalone 0.017 at 0.777 accuracy), and the two are informative in opposite ways. NormClip disturbs Reputation’s weight ordering in 3/9 rounds and the composition reaches 0.806; RFA disturbs it in 9/9 and the composition reaches 0.895.

The gap is only +0.089 (both arms are far above threshold, because Reputation’s ordering is disturbed under *both* upstreams) but within-arm variance is small, all five seeds move in the predicted direction (per-seed deltas +0.149/ + 0.089/ + 0.030/ + 0.061/ + 0.115), and it is resolved at both $n=3$ and $n=5$ (Welch $t = 3.06$, two-sided $p = 0.018$; paired $t = 4.31$, $p = 0.013$). So the two contrasts trade off exactly as the disturbance measurements imply: Krum has the large effect because one of its upstreams leaves it untouched, and Reputation was resolvable even at $n=3$ because neither upstream does.

This is post-hoc, and we do not present it otherwise. It was not pre-registered, it re-uses a suite frozen for a different purpose, the two extra seeds per cell were added *after* seeing the $n=3$ result, and the six-cell split is a description of that suite rather than a test on new data. The top-up could have cut in either direction and we report what it did: it resolved the Krum gap (Welch $p = 0.16$ at $n=3$, where per-seed paired deltas +0.854/ + 0.097/ + 0.935 left one seed not separating, to $p = 0.012$ at $n=5$) and left the invariant control at $\Delta \approx 0$ (+0.000 under NormClip, −0.044 under RFA), the added seeds raising its baseline and its compositions together. What the analysis does have, which neither the FLTrust contrasts of §E.1 nor the cross-pair scoring has, is C1 held exactly fixed and a disturbance measurement taken on the statistic rather than read back from the outcome. We offer it as the cleanest available piece of C2 evidence and as the shape a properly prospective version of this test would take: fix a downstream defense that suppresses the attack, then vary the upstream along measured disturbance rather than along defense identity. Table A15 sets out, claim by claim, what this design can and cannot conclude.

E.3 AN APPLICATION: SCREENING AS EVALUATION PRIORITIZATION

Table A15: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 6). Every entry is scoped to the measured setting of §E–§D.2.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: unchanged in every one of the 60 measured rounds
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside the frozen margin (Prop. 11(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes, non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No, refuted: +0.098 ($n=5$) against a frozen +0.150
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos_krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no, open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no, structural	Empty cell by the gate, for every menu (Prop. 6)
Does any of this certify <i>deployed</i> security?	no, scope	Committed attack only; one adaptive test breaks it

What follows is a consequence of the identification boundary, not a claim the paper argues for. Proposition 6 says a criterion gated on the outcome it predicts cannot be validated by cross-defense contrast; what such a criterion can still do is *order an evaluation budget*, and that is the only use we claim for it. Proposition 7 bounds what it can find. Read the numbers below as a measurement of a prioritization heuristic’s cost, not as validation of the framework.

Does every condition earn its place? C3 does not: reported as a finding. Scoring all 42 pairs on C1, C2 and C3 *independently* rather than recording only the first failure: against a 12% base rate of low-ASR pairs, C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, and C1∧C2 selects 2 at 100%. **The surviving C2 failures are exactly the non-invariant signal class:** after the Proposition 10 reclassification every one of them is a $\cdot \rightarrow \text{reputation}$ pair ($\text{nc} \rightarrow \text{rep}$ 0.819, $\text{rfa} \rightarrow \text{rep}$ 0.882, $\text{fg} \rightarrow \text{rep}$ 0.798), consensus distance being the signal a positive rescaling does not preserve, while all three non-degenerate $\cdot \rightarrow \text{FoolsGold}$ pairs are C1 failures instead. **C3 has no demonstrated incremental predictive value in this study:** alone it selects 35 pairs at 9% precision, below the base rate, and it never binds on a pair C1∧C2 already admits. So **we demote C3 from a gate to a margin diagnostic and recommend C1∧C2 as the criterion**, retaining C3 on theoretical grounds only: Theorem 8 shows margin compression is a real failure mode, and the $\alpha=0.1$ regime below is precisely a margin collapse. **The demotion changes no number,** C1∧C2 selects the same two pairs at the same precision (recomputed by `analyze_screening_cost.py`), which is why we report it as a finding about the criterion rather than a revision of results. *The pre-registered PASS/FAIL labels and the 16/18 and*

22/24 tallies were frozen with C3 in the gate and are not relabelled; this ablation is post-hoc and labelled as such, and the recommendation is for use going forward.

If the goal is to find emergent compositions, drop C1: a recommendation we price but do not validate. Proposition 7 says an inherited-suppression gate cannot certify a composition whose suppression neither constituent possesses, so an evaluator whose object is discovery rather than prioritization should not run C1 at all, and this is a consequence of the proposition rather than a finding of ours. Two designs follow, and §E.4 prices both on our own menu. **Sample the ungated menu** when the budget allows: 42 pairs at 12% precision reaches every low-ASR pair by construction, and it is the only design with no structural blind spot. **Otherwise gate on the mechanism condition alone**, C2 without C1: it admits 14 of the 42 at 36% precision, a third of the ungated cost, and it reaches FoolsGold→RFA where every C1-bearing variant misses it. **The caveat is ours and it is not small:** C2-alone’s recall on this menu is 100%, but 11 of those 14 admissions rest on a per-pair call made in sample rather than on algebra, and all 5 low-ASR pairs are among the 11 (§E.4), so that recall is a development-set number and we offer the design as a recommendation rather than as a validated procedure. Finding emergence efficiently is a search problem, and nothing here solves it: what the propositions establish is which designs cannot solve it.

E.4 THE SCREENING FRONTIER: PRECISION, RECALL AND EVALUATION COST

What does the screen cost, and what does it cost to find out? 40% **recall is an operating point, not a verdict on the conditions.** Scoring C1, C2 and C3 independently over the 42 pairs traces a frontier, and the criterion the paper reports is its high-precision end. Emitted by `experiments/build_screening_frontier.py`, which joins the two existing generators and introduces no number of its own.

Neither screen is noise, and neither is a predictor. Both C2 alone and $C1 \wedge C2$ beat a size-matched random draw from the same 42 pairs ($P \leq 0.005$ at 10^5 resamples against the menu’s 12% base rate, seeded `rng(42)`; emitted by `experiments/blind_selection_analysis.py`). Beating a random draw is the whole of what that establishes: it separates the screens from chance and not from each other, and it says nothing about either one’s ability to predict suppression.

Variant	selected	precision	recall	mean ASR	runs	reaches FG→RFA
no screen (all pairs)	42	12%	100%	0.743	420	✓
C3 alone	35	9%	60%	0.760	350	✓
C2 alone	14	36%	100%	0.588	140	✓
C2 algebraic only [†]	3	0%	0%	0.869	30	—
$C2 \wedge C3$	8	38%	60%	0.562	80	✓
C1 alone [‡]	4	50%	40%	0.514	40	—
$C1 \wedge C3$ [‡]	4	50%	40%	0.514	40	—
$C1 \wedge C2$ [‡] (the criterion)	2	100%	40%	0.242	20	—
$C1 \wedge C2 \wedge C3$ [‡]	2	100%	40%	0.242	20	—

[†] additionally requires 60 standalone runs before the condition can be evaluated at all, so this row’s evaluation budget is $20+60 = 80$ runs, not the composition runs alone.

[‡] the same screen restricted to the pairs whose C2 verdict Prop. 10 settles algebraically.

percentages are over small denominators: 5 of the 42 pairs are low-ASR, so the C1-bearing rows’

precision is $2/2$ and their recall $2/5$. *mean ASR* averages max-committed ASR over the admitted pairs;

a lower mean is not a better screen, since the C1 end reaches it by discarding the menu’s lowest-ASR pair.

mean ASR scores every pair at the menu’s uniform $n=3$, not at the marked

seed counts of Table A12, where the two admitted pairs read 0.107 and 0.463.

This table is the only home of these rates.

The blind spot is attributable to exactly one condition, and the theory named it before the ablation was run. Every C1-bearing variant misses FoolsGold→RFA and every C1-free variant reaches it: its row is C1 *false*, C2 *true*, C3 *true*, max-committed ASR 0.045. C1 is the inherited-suppression gate, and Prop. 7 proves no gate of that form can certify an emergent composition, so the miss is a *predicted* property of one condition rather than an unexplained failure, and it is tradeable: C2 alone recovers every low-ASR pair in the menu at 36% precision. **We report the high-precision end because false positives deploy an insecure stack**, not because the conditions cannot be operated at higher recall.

The C2 column is not all one kind of evidence, and the high-recall claim depends on the weaker kind. We describe C2 as read off the two defenses’ definitions, and for two of its four branches that is literally what happens: a d_2 reading a cosine is exactly invariant under any positive per-client rescaling (Prop. 10(a)) and a d_2 reading a consensus distance, a residual magnitude or a norm is not (Prop. 10(c)). The remaining branch is *conditional*: `coord.median` and `trimmed.mean` preserve a coordinate ordering only under a

separation condition, and Thm. 8(2) gives `rfa` an analogous one, so `analyze_condition_ablation.py` marks all three `None` and `c2.holds` defers to a *per-pair call*, the stored category. Nothing in the C2 column distinguishes a verdict reached by algebra from one reached by that call, so we census it (`c2.provenance`, same generator): of the 14 pairs C2 admits, 3 **rest on algebra and 11 on the per-pair call**, the 11 being every `coord.median`, `trimmed.mean` and `rfa` row. **And all 5 low-ASR pairs are in the 11:** 0 of 5 are algebraically settled. Hence the ‡ row above. **Worse for us, all 11 are development-set pairs**, so those calls were made *in sample*, while the criterion was being written; they are not pre-registered, and the generator derives the split from which results artifact holds each pair rather than taking our word for it. **Two consequences we accept rather than repair.** First, C2 alone as we score it *is not a pure statistic-preservation screen* for those three d_2 classes, so the body’s contrast between it and C1/C2 is a contrast between two screens that both borrow from the same in-sample call, not between algebra and measurement. Second, *within this menu* the region where our own invariance result decides the question contains no low-ASR pair at all, so the algebraically settled sub-region cannot discriminate any strategy from any other *here*, and the 100% recall of C2 alone is carried entirely by in-sample judgement. **The second consequence is the one we then removed**, by enlarging the algebraic region until it closed rather than by relabelling this one: the 3 pairs of the ‡ row are all three upstream defenses against the single downstream defense `foolsgold`, which is one column of a 3×3 census over $d_1 \times d_2$, and completing that census prospectively puts a low-ASR pair inside the algebraic region and separates the two strategies more sharply there than anywhere in this table (§E.5). The *first* consequence stands and needs an invariance theorem for the conditional classes, which we do not have; what neither needs is a relabelling, and no number in this appendix or the body changes. This is (L5) in concrete form: the conditions are not always readable off a (d_1, d_2) pair.

What the screen saves, in runs rather than in pairs. `experiments/analyze_screening_cost.py`: unscreened, the 42 pairs under 2 committed attacks at 5 seeds are 420 runs; the reported criterion evaluates 16 cells, 80 runs, of which 60 are the standalone measurements C1 needs. That is 340 runs avoided, 81% of runs. We give the run-level figure rather than the pair-level 95.2%, which overstates the saving by ignoring C1’s own standalone cost. **The unit is runs, and the hours behind them are now measured rather than assumed:** the runners print per-run seconds into their logs, and the six the script reads ship with the code, so the figure is checkable; over the 228 logged CIFAR-10 pair-evaluation runs the median is 11.1 min (mean 13.1, IQR 10.7–12.5), so the 340 avoided runs are 63 GPU-hours on the one machine, or 74 at the mean. *Correcting our own caveat:* through the previous version this paragraph said no per-run timing existed anywhere in the results and the artifact assumed 11.0 min/run. The runner logs did carry it; the assumption happens to have been accurate to 0.14 min, and we report the measurement in its place because a figure that can be measured should not be assumed. EMNIST-byclass arms run 52 min/run and are excluded, so this accounting is CIFAR-10 only.

E.5 A CLOSED CENSUS OF THE ALGEBRAICALLY SETTLED REGION, PRE-REGISTERED

What happens where the algebra settles preservation exactly, with no cells unspent? **The ‡ row above is the `foolsgold` column of a 3×3 box over d_1 and d_2 , and the two cells of that box which had never been run are what this wave measured.** The region where mechanism preservation is settled on *both* sides is small and enumerable: d_1 emitting strictly positive per-client coefficients $T(u_i) = c_i u_i$, so Prop. 10(a) applies, against a d_2 reading an exactly scale-invariant statistic. Over this project’s defenses and the metric-swap mirrors that region is $\{\text{norm.clip}, \text{rfa}, \text{reputation}\} \times \{\text{foolsgold}, \text{cos.krum}, \text{cos.reputation}\}$, exactly 9 ordered pairs, of which 7 were already measured at the identical protocol and 2 were unspent. Enumeration is what turned a planned 20-pair sample into a census: 118 **ordered pairs of this project are already spent**, and once the admissibility constraints (a non-degenerate d_1 , and C0) are imposed only 6 candidates remain anywhere. **A census needs no sampling argument, and an all-negative census is as informative as a positive one**, which is why this design was worth 24 runs and 4.87 h. Pre-registered at 96f8856 with per-pair C0–C3 verdicts, a frozen scoring rule and a two-sided resolution table (`experiments/pre_registration_wave3_emergent.md`); scored by `experiments/analyze_wave3_emergent.py`, which refuses to run against an uncommitted prereg.

$d_1 \downarrow d_2 \rightarrow$	<code>foolsgold</code>	<code>cos.krum</code>	<code>cos.reputation</code>
<code>norm.clip</code>	0.887 HIGH	0.508 HIGH [†]	0.935 HIGH [†]
<code>rfa</code>	0.914 HIGH	0.566 HIGH	0.861 HIGH
<code>reputation</code>	0.807 HIGH	0.263 LOW*	0.787 HIGH*

Max-committed mean ASR over seeds 42, 43, 44; LOW iff < 0.5 .

* the two unspent cells, labels frozen at 96f8856 before the output directory existed.

† `norm.clip`’s clip is inactive on the pixel arm (max update norm $3.043 < \tau=5$), so those *pixel* cells are bit-identical to d_2 alone; both verdicts are carried by the scaling arm, where it is not.

`foolsgold` is excluded from the d_1 column by measurement, and that measurement changed the design before any prediction was frozen. It was going to supply two more C1-fail $\wedge$ C2-true pairs. But `foolsgold` normalizes with `weights - weights.min()`, forcing at least one coefficient to *exactly* zero every round,

which puts it outside Prop. 10(a)’s strict $c_i > 0$ hypothesis and under Lemma 2 instead. Measured through the shipped transform over 3 seeds $\times$ 3 rounds: 2/5 exact zeros every round, and both downstream cosine statistics moved in 9 of 9 rounds. Synthetically at 1000 trials the effect is entirely due to the zero: strictly positive coefficients change `cos_krum`’s selection 0/1000 times, one exact zero changes it 471/1000. `reputation` by contrast emits 0/5 zeros every round at a positive-part ratio up to 18839.8, and under that same extreme rescaling in the same rounds the two cosine statistics moved 0/9 while non-invariant `krum`’s selection moved 8/9 and `reputation`’s own ordering 9/9: **a contrast between statistics on identical inputs, not between conditions**. So the two `foolsgold` $\rightarrow$ `cos_*` pairs were entered as *measured-C2-fail controls* rather than as members of the census. **Had C2 been assumed from the definitions instead of measured, this wave would have reported two mislabelled pairs** (`experiments/verify_wave3_invariance.py`, `results/wave3_invariance_check.json`).

#	pair	cell	pred.	measured
1	<code>reputation</code> $\rightarrow$ <code>cos_krum</code>	certified (C1 $\wedge$ C2)	LOW	0.263 LOW ✓
2	<code>reputation</code> $\rightarrow$ <code>cos_reputation</code>	C1-fail $\wedge$ C2-true	HIGH	0.787 HIGH ✓
3	<code>foolsgold</code> $\rightarrow$ <code>cos_krum</code>	measured-C2-fail control	HIGH	HIGH [§] (raw 0.086) ✓
4	<code>foolsgold</code> $\rightarrow$ <code>cos_reputation</code>	measured-C2-fail control	HIGH	0.824 HIGH ✓

[§] scored HIGH *only* by the frozen DEGENERATE-ACC rule: mean clean accuracy 0.255 < 0.35 on the arm supplying the max, so its raw 0.086 is a collapsed model and not a suppressed one.

4/4 as registered, including the pair we flagged uncertain, and the near-identity check came back clean on every arm (0 of 3 seeds identical to d_2 alone on all 8 arms; max per-seed differences 0.073 to 1.000), so these are compositions and not d_2 in disguise. Pair 1 was the live risk and the prereg said so: its LOW rests on two *different* attacks being suppressed by two *different* constituents, `reputation` holding model-scaling (0.017 at accuracy 0.777) and `cos_krum` the pixel backdoor (0.300 at 0.598), and on the scaling arm the certification and the two nearest measured analogues (`norm_clip` $\rightarrow$ `cos_krum` 0.508, `rfa` $\rightarrow$ `cos_krum` 0.566) pointed *opposite* ways. It came out LOW on both arms (0.064 scaling, 0.263 pixel). **So the one low-ASR pair in the closed census is a pair neither constituent suppresses alone**, and it is the pair C1 $\wedge$ C2 selects.

Both strategies priced over the same closed census, which costs no further runs because C1 is defined on standalone numbers that already exist.

strategy over the 9-cell census	selected	precision	recall	runs
A: statistic preservation (C2 alone)	9	11%	100%	90
B: the criterion (C1 $\wedge$ C2)	1	100%	100%	10

In this region C2 alone is not a screen at all: it admits everything, because everything here satisfies it by construction. **That inverts the body’s trade**, where B bought precision by giving up recall; here, at the frozen seed count, B dominates A on precision, recall and cost at once, and disclosure (v) below is the qualification that phrase needs. It is also the answer to the objection of §E.4 that the algebraically settled sub-region cannot discriminate strategies: it could not, at 3 pairs; at 9 it discriminates them completely.

Five disclosures, none of which the numbers above carry on their face. (i) *B’s 1/1 is a gated reading*. C1 requires suppression at mean clean accuracy ≥ 0.35 , the gate applied throughout §E.2, and it flips exactly two census cells: `cos_krum` alone reaches ASR 0.078 on model-scaling but at accuracy 0.146, barely above CIFAR-10’s 0.10 chance, so it is a destroyed model rather than a defended one. Read at the bare 0.5 threshold, C1 additionally admits `norm_clip` $\rightarrow$ `cos_krum` and `rfa` $\rightarrow$ `cos_krum`, both HIGH, and B selects 3 for 33% precision. The gate is the paper’s pre-existing convention rather than a knob turned for this table, and `analyze_wave3_emergent.py` emits both readings. (ii) *The one positive is $n=3$ and wide*: pair 1’s pixel arm is 0.000/0.107/0.681 per seed, so one of three seeds sits above the threshold and the LOW verdict is the frozen mean rule applied to a spread that a fourth seed could move. We do not add seeds after seeing this. (iii) *The controls are weaker evidence than intended*. Every `foolsgold-as-d1` scaling arm sits at chance accuracy (0.100), and pair 3 is HIGH only through the accuracy gate, its raw ASR being 0.086, and its scaling arm tripped the frozen “ASR identically 0.0 across all 3 seeds” flag, which on a model at 0.100 accuracy means the attack had nothing left to succeed against; so pairs 3 and 4 establish that the C2 probe is not inert, which is their stated role, but they are not independent confirmations of a HIGH prediction. (iv) *7 of the 9 cells are retrospective*. Only the two `reputation` cells were blind; the other seven were read from frozen artifacts, with each transcription in the prereg checked against its source by the analyzer. So the census’s *closure* is prospective at the margin, not throughout, and pair 2’s scaling arm tripped the second of the wave’s two frozen sanity flags (clean accuracy 0.790 against the FedAvg reference 0.667). Both flags are reported, neither adjusted. (v) *The C1 input that certifies pair 1 is the lowest-seed reading of a high-variance cell*. C1 holds for pair 1 on the pixel arm because `cos_krum` alone measures 0.300 there, and that is the mean over

the three frozen seeds 42–44 (0.003/0.269/0.630). The identical cell, same protocol and identity upstream, is measured twice more in this paper at larger seed counts, and the cell’s mean climbs monotonically with the count: 0.490 at $n=5$ (Table A6, arm 3’s $\kappa=0$ rung, which adds 0.864/0.686) and 0.584 at $n=8$ (Table A7, which adds 0.578/0.845/0.795), the last being the reading §D.2 states as `cos.krum` never suppressing. Six of the eight seeds are above 0.5 and the frozen three are the two lowest plus the fourth-lowest. So B’s precision 1/1 and recall 1/1 here are contingent on the frozen seed count as well as on the accuracy gate of (i), and we state the contingency at the resolution the numbers support rather than rounding it against ourselves: 0.490 is still *below* the threshold, by 0.010, so C1 survives at $n=5$; at $n=8$ C1 fails for pair 1, B certifies nothing in this census, and its one low-ASR pair becomes another false negative of the criterion rather than its one hit. We report this instead of rescoring, because the pre-registration fixes the seed set at three and forbids adding seeds afterwards (`pre-registration-wave3-emergent.md`, §5). What the disclosure costs is the claim that B *dominates* A on this region, which holds at the frozen seeds and is not established beyond them.

E.6 OUT-OF-SAMPLE PASS VALIDATION

The development and held-out validation above tests predictions from the *same* attack families (`model_scaling`, `backdoor_pixel`) and scale ($N=10$, `cifar_cnn`) used to design the criterion. To confirm that PASS predictions generalize we pre-registered three out-of-sample tests (commit `ca627e5`, before any of them ran); all held, the accounting is in the summary below, and per-cell values for every test are in the supplement (§S4). We state plainly what this is and is not: it is evidence that the conditions keep identifying compositions which survive a change of attack family, scale and architecture; it is *not* an out-of-sample classification result, because every prediction in it is positive and the matching FAIL controls proved uninformative. Negative-direction evidence rests entirely on the in-distribution tiers.

(E) Heterogeneity: where the criterion breaks. Earlier heterogeneity sweeps in this work covered *single defenses* only; no composition had been run at any $\alpha \neq 0.5$. We ran all three PASS pairs at $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both committed attacks, 50 rounds); the $\alpha=0.5$ column is each pair’s headline value at its own seed count ($n=30$ for `fg→rfa` and `rep→cm`, $n=5$ for `fg→cm`) rather than a 3-seed rerun. Reporting max-committed ASR / clean accuracy:

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
<code>fg→rfa</code>	0.569 [§] / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
<code>fg→cm</code>	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
<code>rep→cm</code>	0.548 / 0.48	0.463 / 0.76	0.377 / 0.79	0.326 / 0.81

§ *uninterpretable* under the accuracy floor of §E:

clean accuracy $0.20 < 0.35$, so the raw 0.569 is a collapsed model and is scored as neither a crossing nor a survival. Raw reading kept above.

The criterion transfers cleanly across $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID regime where FoolsGold might plausibly founder on legitimately similar benign updates; it does not, and $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$), two of the three PASS compositions cross the threshold on the raw ASR reading** (`fg→rfa` 0.569, `rep→cm` 0.548); only `fg→cm` survives, at 0.433. `Rep→cm`’s $\alpha=0.5$ baseline is itself at the threshold (0.463 at $n=30$), so its crossing is the smaller move of the two. **Our own accuracy floor changes that count, and it changes it in our favour, which is why both readings stand here.** `fg→rfa`’s $\alpha=0.1$ cell sits at clean accuracy 0.20, below the 0.35 floor of §E, so by the paper’s own rule it is *uninterpretable* and cannot be scored as a crossing. The gate-applied reading is therefore one crossing at interpretable accuracy (`rep→cm`, 0.548 at 0.48), one uninterpretable cell, and one survivor (`fg→cm`, 0.433 at 0.40); the raw reading is 2/3. **We print the raw count as well as the gated one because applying our own rule here removes a datapoint from a limitation we report against ourselves**, and a reader should be able to see that softening rather than inherit it. Two things the gate does *not* buy: the cell is uninterpretable in both directions, so it is no evidence that the criterion holds at $\alpha=0.1$ either, and `rep→cm`’s crossing is unaffected by the floor, so extreme heterogeneity still breaks a PASS pair at accuracy we are willing to read. Two caveats cut in opposite directions: `fg→rfa` at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not; and clean accuracy collapses to 0.20–0.48, so this regime stresses utility as much as security. Crucially Theorem 8 supplies a **diagnostic quantity that retrospectively explains this boundary** rather than predicting it: re-measuring on the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%). Over the payload-carrying rounds alone (the ones Figure A1 plots) the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with the per-round minimum falling $2.82 \rightarrow 0.00$; the all-round means are lower because they average in the Lemma 2 rounds, where FoolsGold zeroes every adversary and m is recorded as 0. Either way the collapse is measured, and the all-round figure is the one that reaches the vacuum limit. This sits exactly at the $m \rightarrow 1$ boundary where Corollary 4’s bound degrades to 40% adversarial mass: extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is precisely the precondition the theorem requires. A practitioner running the

5–10 round pilot would measure $m \approx 1$ and decline to deploy, so the criterion is mechanistically diagnostic at this boundary, though we emphasise this is a post-hoc measurement of a known failure, not a pre-registered prediction of it.

The one axis not validated with composition experiments is *dataset*: every composition in this paper is CIFAR-10, and the two arms measured on another dataset are single-arm intervention ladders rather than composition suites, EMNIST-byclass/simple_cnn in §D.2 and CIFAR-100/cifar_cnn (100-class output) in §D.3. The conditions are dataset-agnostic in mechanism, but the downstream aggregator’s decision boundary may shift with the number of classes, and the CIFAR-100 ladder is direct evidence that it can: the same cell that reverses sign on CIFAR-10 reverses sign there too, at a different magnitude (L1).

F THE COMPOSITION THE SCREEN CANNOT CERTIFY: FOOLSGOLD→RFA

This section is the empirical witness for Proposition 7. FoolsGold→RFA has the *lowest observed committed* ASR of every composition evaluated here (0.093), and **our own criterion classifies it as a false negative**: neither constituent suppresses the attack alone (FoolsGold 0.732, RFA 0.845), so its suppression is *emergent* in the sense of Proposition 7 and C1 cannot hold. **We report it in detail because it is the sharpest illustration of the screen’s scope, not because it is a flagship**: any criterion that certifies inherited suppression must miss compositions of this kind, which is what caps recall *structurally*; with one emergent case among five low-ASR pairs that cap is 80%, and the 40% we measure sits below it for reasons other than emergence. The characterization below is therefore descriptive (what this composition does, measured at $n=30$) and is not evidence for the criterion. **The $n=30$ measurement corrected our own earlier figure downward in reliability, and we report the correction rather than the settled number alone**: the per-seed distribution is right-skewed (median 0.060, mean 0.093, sample skew +2.07, 8/30 seeds at or above 0.10, maximum 0.380), and seeds 42–46 fall in its left tail, so the $n=5$ value we first led with understated the pair by a factor of 1.8. Any 5-seed figure for this composition should be read against that spread. We avoid calling it security-maximal: it also sustains the *highest* ASR under our adaptive adversary (0.157 against 0.064 for the same pixel attack on the same thirty seeds; at $n=5$ those two read 0.167 and 0.045, so the ratio falls from $3.7\times$ to $2.4\times$ once the denominator leaves its left tail), and it carries a 50% accuracy cost, so it is the right choice only where utility is nearly irrelevant. We validate at $n=30$ seeds with a full adaptive attack suite ($N=10$, $K=5$, $f=0.2$, 50 rounds), in Table A16:

Results. Median ASR across all base attacks ($n=30$): 0.056 [IQR 0.044–0.090]; P95 worst-case: 0.253 (committed_scaling). **Three of the six adaptive strategies beat the base attacks and we lead with the strongest, not the median.** Five of the six are measured at $n=5$ on seeds 42–46, where the base composition reads 0.045 (pixel), 0.051 (scaling) and 0.050 (oracle); the $\epsilon=1.0$ arm the body quotes is topped up to $n=30$ on seeds 42–71, where the pixel arm’s *mean* is 0.064 (the base block above prints medians, so the same cell appears there as 0.052). Against those seed-matched bases, the criterion-aware joint-constraint adversary with decorrelation reaches mean 0.157 at $\epsilon=1.0$ and $n=30$ (median 0.159, max 0.353; paired difference against the pixel arm +0.093, 95% CI [+0.066, +0.120]), which is $2.4\times$ the 0.064 of the pixel arm it is built from and the ratio every other section of this paper quotes, and 0.130 at $\epsilon=2.0$ and $n=5$, and the projection adversary reaches 0.079; the other three sit level with the base (the same criterion-aware attack without decorrelation 0.050, Neurotoxin 0.048, consensus-shift 0.045 at rate 0.05 and 0.048 at rate 0.01). Consensus-shift is the one row here we re-ran after finding our published figure void (the defect is disclosed in Table A16’s caption): its paired differences against the pixel arm on the same five seeds are +0.0031 (95% CI [−0.0005, +0.0067]) at rate 0.01 and −0.0006 ([−0.0042, +0.0029]) at rate 0.05, so both intervals contain zero, and the attack’s second phase is by construction ordinary committed pixel at the scale=1 this arm runs, so part of the null is mechanical. Since the base arm’s own $n=30$ distribution is right-skewed and seeds 42–46 sit in its left tail, a 5-seed null is weak evidence of a null rather than a demonstration that the attack cannot work here. **So the composition does not stay under 10% against the strongest adversary we built**, and 0.157 is the figure the opening paragraph reports. **The top-up moved that ratio down and we report the move rather than the settled figure alone**: at $n=5$ the same two arms read 0.167 and 0.045, a ratio of $3.7\times$, and almost all of the fall is the denominator leaving the left tail described above rather than the attack weakening: the numerator moved only $0.167 \rightarrow 0.157$ while the denominator rose $0.045 \rightarrow 0.064$, and holding the numerator at its $n=5$ value while the denominator alone is recomputed at 30 seeds would already put the ratio at $2.6\times$. What survives is weaker and still worth stating: no adaptive *mean* reaches the base composition’s own P95 tail of 0.253 (its worst single seed is 0.380), though the $\epsilon=1.0$ maximum of 0.353 does, and the $\epsilon=1.0$ gain costs the adversary nothing in clean accuracy (0.549 against the pixel arm’s 0.492, both at $n=30$), so it is a real gain rather than a utility trade. The accuracy tradeoff (0.50 vs. 0.74 for RFA alone) and Pareto alternatives are discussed below (Table A17).

Composition-aware adversary (end-to-end). We build an adversary that *knows* the full FG→RFA pipeline is deployed (Kerckhoffs) and optimizes against the complete composition, solving both constraints simultaneously: pairwise cosine similarity below FoolsGold’s down-weighting threshold, and ℓ_2 distance to the geometric median below RFA’s rejection radius. Any gradient-based optimizer faces the same tension (coordinating a

Table A16: FG→RFA under base and adaptive attacks. Base: $n=30$, reported as median [IQR] with P95 tail; adaptive: mean $\pm$ std at the n each row names, which is $n=5$ except for the $\epsilon=1.0$ arm the body quotes, topped up to $n=30$. P95 is reported only where it was pre-registered, hence – on every adaptive row. Clean accuracy: test-set accuracy on non-triggered inputs. No-attack baseline for RFA alone: 0.737. Consensus-shift row: this is a re-run, and the row published earlier (0.045 ± 0.012) was void. That runner built its attack object from the `backdoor_pixel` strategy, which does not override `manipulate_update` and so inherits the base class identity; its shift term was therefore scaled onto a zero difference, and both rates reproduced the base pixel arm on all five seeds to twelve decimal places in both ASR and clean accuracy. Both rates were re-run on the same seeds, composition and configuration with the live `ConsensusShiftAttack` (`results/fg_rfa_consensus_shift_fixed/`, pre-registered at 784a201), and the superseded `adaptive_consensus_shift` cells are kept in `results/fg_rfa_flagship/summary.json` so the defect stays auditable.

Attack	Config	ASR	P95 / max	Acc
<i>Base composition ($n=30$; median [IQR])</i>				
	scaling	0.060 [0.048, 0.096]	0.253 / 0.380	0.497
	pixel	0.052 [0.035, 0.087]	0.127 / 0.200	0.492
	oracle	0.056 [0.044, 0.090]	0.189 / 0.236	0.494
<i>Projection adaptive ($n=5$)</i>				
	$\epsilon=2.0$ (best)	0.079 ± 0.042	– / 0.160	0.495
<i>Criterion-aware joint-constraint</i>				
	$\epsilon=1.0$, decorr ($n=30$)	0.157 ± 0.073	– / 0.353	0.549
	$\epsilon=2.0$, decorr ($n=5$)	0.130 ± 0.030	– / 0.185	0.506
	$\epsilon=2.0$, no decorr ($n=5$)	0.050 ± 0.008	– / 0.059	0.494
<i>Neurotoxin ($n=5$)</i>				
	top-20%, scale = 2	0.048 ± 0.009	– / 0.057	0.495
<i>Consensus-shift ($n=5$; re-run, see caption)</i>				
	rate = 0.05	0.045 ± 0.010	– / 0.058	0.497

backdoor raises FG-similarity, while staying near the median caps the perturbation and weakens the backdoor) and our analytical construction reaches the same boundary, the Weiszfeld iteration being contractive and the gradient at convergence determined by the final-iterate reweighting the joint constraint directly constrains.¹ The ablation is informative: without decorrelation ASR stays at baseline (0.050), RFA-evasion alone failing because FoolsGold is the binding constraint; with it the adversary gains partial signal (0.130–0.167 across the ϵ sweep, held at the uniform $n=5$ the ablation compares at; $\epsilon=1.0$ alone reads 0.157 at $n=30$), which Theorem 8 explains as bounded, each adversary’s block-restricted delta being too small to move the median outside the benign cluster.

Contrast with rep→tm. FG→RFA (median ASR 0.056) provides near-absolute suppression; rep→tm (oracle ASR 0.316) provides strategic dominance against all tested adversary policies but does not suppress committed-pixel (0.679). The two serve different roles.

Accuracy–security tradeoff. FG→RFA achieves the lowest ASR of any tested composition but imposes a substantial clean-accuracy cost (0.50 vs. 0.74 for RFA alone). This cost is intrinsic to FoolsGold’s aggressive per-client downweighting, not to composition itself. Table A17 places FG→RFA on the accuracy–security Pareto frontier alongside alternative compositions:

FG→RFA is the correct choice under a *security-first* objective (e.g., safety-critical FL where any backdoor is unacceptable). For operators with accuracy constraints, FG→CM (acc 0.68, max ASR 0.107) buys 37% more accuracy for suppression that is statistically indistinguishable on the worst-case arm the two share (Welch

¹A fully gradient-based optimizer differentiating through 50 Weiszfeld iterations would face identical binding constraints at convergence; the analytical formulation is equivalent without requiring a differentiable relaxation. **The equivalence is about the constraint set at convergence and not about the optimizer, and we do not run the unrolled attack**, so two classes of behaviour sit outside it. Path effects would make an unrolled attack *weaker* than our construction: gradients that vanish or are obfuscated through the iteration, and optimization instability or a poor local optimum. A weaker unrolled result would therefore not refute the number here. The case the argument cannot exclude runs the other way: our construction searches a family we chose and parameterized in closed form, and differentiating through the iteration may reach a point inside the same constraints that this family does not contain. So 0.157 at $n=30$ is the strongest our family attains and is not a bound over gradient-based adversaries.

Table A17: Accuracy–security Pareto frontier. Max ASR = worst-case mean across committed attacks, and Clean Acc is measured on the arm that attains it. All at $N=10$, $K=5$, $f=0.2$, cifar.cnn. The three composition rows are emitted by `experiments/analyze_headline_cis.py`, each at the seed count its own max-committed arm carries: FG→RFA and Rep→CM at $n=30$ (seeds 42–71; Rep→CM’s is the pre-registered pixel top-up, 0.463 [0.412, 0.514]), FG→CM at $n=5$ (seeds 42–46). No *composition* dominates another: FG→CM buys 0.185 accuracy for +0.014 ASR. The single-defense rows are listed for reference and are not all frontier points; Rep alone dominates RFA alone on both axes.

Defense / Composition	Clean Acc	Max ASR	Regime
Temporal mix (NC/fedavg)	0.79	0.96	No security
RFA alone	0.74	0.91	Bimodal; unreliable
Rep alone	0.78	0.84	Scaling-only
Rep→CM	0.76	0.46	Moderate
FG→CM	0.68	0.11	Strong
FG→RFA	0.50	0.09	Min ASR (evaluated)

$p = 0.52$ on committed scaling, 0.093 against 0.107). It is *not* Pareto-superior and we do not claim it is: its point estimate is the higher of the two, and on the pixel arm, which is neither pair’s worst case, the gap does reach significance in FG→RFA’s favour (0.064 against 0.099, Welch $p = 0.022$). Rep→CM (acc 0.76, max ASR 0.463) offers near-baseline accuracy with moderate security. The criterion certifies FG→CM and Rep→CM; FG→RFA reaches the lowest ASR of the three but fails the tightened per-attack C1 and is therefore a false negative of our own criterion rather than a certified recommendation. The choice among them is an operator-specific utility decision, not a criterion failure.

G RELATED WORK

The contrast in one sentence. Prior work asks whether a defense withstands an attack, and the composition literature treats composition quality as an *outcome to predict*. We ask a different question: the **identifiability of mechanistic explanations** for that outcome, under the evaluation designs commonly used to compare defenses. Concretely, whether an evaluation criterion can causally attribute a composition’s security to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum (Blanchard et al., 2017), Trimmed Mean (Yin et al., 2018), RFA (Pillutla et al., 2022) and FoolsGold (Fung et al., 2020) are each proposed and evaluated as a *single* stage. **Two names have left that list, and the audit of §G.1 is what removed them.** FLAME (Nguyen et al., 2022) combines three mechanisms in one round (model clustering, weight clipping and adaptive noise), and it gives the constituent-complementarity reason for combining them, clustering and clipping being there to “minimize the required amount of noise” a differentially private defense would otherwise need. FLTrust (Cao et al., 2021) applies a server-bootstrapped trust score *and* a magnitude renormalization to the same round, which is two mechanisms under any granularity the audit’s distinctness rule admits. **The FLTrust error was ours to catch one section earlier, and we did not:** §E.1’s near-identity finding rests on precisely that two-part structure, since FLTrust gates on the scale-invariant $\cos(u_i, g_{\text{srv}})$ and then renormalizes every update to $\|g_{\text{srv}}\|$, which is why a positive per-client rescaling is exactly cancelled. So this paper described FLTrust as two mechanisms where the finding needed two and as a single stage where the contrast needed one. Both are composed defenses and both are coded as such below. **Bulyan** (El Mhamdi et al., 2018) is the earliest composed defense among those we name and predates the framing here: it runs Krum-style selection and then a trimmed-mean-style aggregation over the selected set, so it is an ordered pair of exactly the kind we study, chosen by its authors rather than by any screen. It falls outside the audit’s frame as frozen, and precisely why matters: the frame’s conference screen requires *federated* in the title, which Bulyan’s does not carry, and the arXiv sweep runs 2019–2026 (§G.1). Its `references.bib` entry is added here, *after* the audit was coded, so the audit’s first route never saw it, and we report that rather than re-run a frozen instrument: Bulyan is named as related work and is *not* a retrospective audit hit. Every count in §G.1 is unchanged. We propose no new defense; we reason about *ordered pairs* of existing ones, which is a different object: the upstream stage’s output is the downstream stage’s input, so the question is no longer whether a mechanism works but whether it *survives* a transform applied before it. Ensemble adversarial training (Tramèr et al., 2018) *assumes* that upstream processing preserves downstream signals; we make that assumption explicit, checkable, and (as it turns out) **the wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense (Shejwalkar & Houmansadr, 2021; Bhagoji et al., 2019), and the same holds for distributed and durable backdoors (Xie et al., 2020; Wang et al., 2020; Zhang et al., 2022). Our C1 quantifies over a *committed* attack

by construction, so this literature bounds what any inherited-suppression criterion (ours included) can promise. Our single whitebox composition-aware stress test (§F) is reported in order to *measure* that gap (0.064 $\rightarrow$ 0.157 seed-matched at $n=30$ on the pixel arm, $2.4\times$), not to defend the criterion against it; a screening condition stated against an adaptive adversary is what the framework most lacks.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive (Croce et al., 2021). Our point is methodological in the same way but about **identification** rather than attack strength. Wherever a screen gates candidate pairs on standalone effectiveness and then reads the composition’s outcome as evidence about the *mechanism*, Proposition 6 shows that inference cannot work at *any* sample size: the counterfactual is excluded by the gate, not undersampled. **We do not claim this is how the FL literature generally proceeds, and that sentence is now a pre-registered measurement rather than a disclaimer.**

The rubric of §G.1 was frozen at a commit hash before any paper was retrieved, and it screens 622 retrieved records down to 59 included composed-defense papers, each coded against five criteria that every code must support with a verbatim quotation and a locator. Outcome-gated attribution, defined there as two or more mechanisms applied jointly to the same round (C-a), paired for their constituents’ standalone effectiveness or complementary threat coverage (C-b), validated by the composed system’s own adversarial outcome (C-c), carrying an actual mechanism-attribution claim (C-e), and running no identifying contrast (not C-d), holds for 6 of 59 papers, 10.2%. The rate is 5 of 59 if the audit’s mechanism-distinctness rule is dropped, and §G.1 reports both, because that rule is one of two amendments written after all 59 rows had been coded. **A low rate was pre-registered as meaning that the decline above is kept and evidenced rather than replaced**, and that is what we do with it.

The audit’s first casualty is the sentence it was built to support, and we report that ahead of anything else it found. Fenaux et al. (2025) was our one named external instance, and it codes C-d = YES on strict grounds rather than by the rubric’s benefit-of-the-doubt rule: its Table 3 holds the Type-2 removal stage fixed, varies the upstream Type-1 defense across six choices, and reads the requirement on the pairing off that variation, concluding that success “ultimately relies on selecting a Type 1 defense restrictive enough to overlap with CSFT’s effective δ -range.” That is the intervention Proposition 6 asks for, performed on the upstream stage with the downstream mechanism held fixed, so the instance is withdrawn and the audit we ran to support the claim is what withdraws it. **Nor is such a contrast unusual in the frame:** one is present in 46 of 59 papers, 78.0%, and of the 33 papers that have the outcome-gated *design*, 27 run one. The 6 are therefore papers that had the design and did not run the contrast, not papers for which the contrast was unavailable or unheard of. The secondary that would have counted against our reading, a paper with the design that makes no attributional claim at all, occurs 0 times in those 33.

What the frame does lack is narrower, and it is the measurement this paper makes: no included row varies an upstream stage and asks whether the *ordering* of two downstream arms survives it, with intervals over seeds. The three closest rows are what that gap is made of. Tabassum & Wu (2026) report an ordering that flips across datasets, “robust aggregation alone outperforms STAR-FL in terms of ASR” on FMNIST while “the full framework provides greater benefits on complex datasets such as CIFAR-10,” which is this paper’s phenomenon observed and left as a remark; the paper reports no seed count, standard deviation or confidence interval anywhere, so there is no interval on either side of the flip. Tory et al. (2026) put both factors on the bench, a graded privacy ladder upstream and four aggregation rules downstream, and decline the cross in one sentence of their own limitations: “the restriction to coordinate-wise median aggregation limits immediate generalization to other Byzantine-robust rules like Krum or FLTrust.” Xu et al. (2024) come closest to the interventional form, reporting with the filter held fixed that “the accuracy of Spar+Di is higher than that of Di,” which establishes that the upstream stage helps and not whether it preserves the downstream mechanism’s discrimination.

Two further failure modes appear in the frame that Proposition 6 does not cover, and both are worth naming separately from it. Kasyap et al. (2026) establish the invariance analytically instead of measuring it, deriving that since the “mask preserves distance ordering, the plaintext post-processing is identical to the classical Krum rule”; their empirical arm has the upstream contrast and the adversarial outcome in one figure and never crosses them, the plaintext condition existing only at zero adversaries. Nath & Kasyap (2026) do not measure a composed outcome at all, and say so: “We do not simulate FL with Byzantine-robust aggregation schemes explicitly, due to limited space. However, theoretical analysis proved that the statistical relationships are preserved.” That is a C-c failure rather than a C-d one, and it is a different defect from the one we study. **The gap is also not one of capability:** Abacha et al. (2026) run an additive component ladder whose final step is exactly an identifying contrast, “Each component contributes incrementally to MTA, with the rescue stage providing the largest single gain,” so the design that would answer the question is already ordinary practice and what is absent is the specific quantity. Composition itself is named as an open problem in the field’s survey of them (Kairouz et al., 2021).

The claim we do make is structural rather than sociological, and its clearest instance is *our own* screen: Lemma 1 shows $C0 \wedge C1$ *force* the downstream defense to suppress the attack alone, so the gate is constitutive

of an inherited-suppression screen and not an incidental filter on it. Statements of a composition-preservation intuition in that literature are informal; what is new here is the *level hierarchy* at which the intuition fails (Definition 2) and the class of designs that can test it (§C.1, §D.2). **So what makes this a machine-learning contribution and not an application of standard causal-identification theory is a combination of four things, and the three-level form of the argument is in §2 rather than repeated here:** a structural identification boundary for a shape of evaluation, the preservation hierarchy that produces the failure, an interventional instantiation in FL that measures it, and the empirical non-implications $(P4) \not\Rightarrow (P3)$ and $(P4) \not\Rightarrow (P5)$ that the collider argument alone does not predict.

Attacks and defenses. Backdoor attacks (Chen et al., 2017; Bagdasaryan et al., 2020; Wang et al., 2020; Xie et al., 2020) and robust defenses (Pillutla et al., 2022; Sun et al., 2019; Blanchard et al., 2017) form the landscape. Beyond our norm-based menu, certified robustness (Xie et al., 2021) and filtering defenses (Nguyen et al., 2022; Fung et al., 2020) build stronger single mechanisms; we compose fixed defenses deterministically rather than strengthening a single mechanism, so these are complements to our object rather than baselines for it. Randomized transformations (Xie et al., 2018; Liu et al., 2018) in centralized settings are the conceptual neighbor of a *randomized-menu* question, which this paper does not ask: every claim here concerns deterministic joint application (§C). **An earlier version of this paper compared against a randomized menu and reported that deterministic composition dominates it; that comparison is not part of this submission and no claim here rests on it.** That withdrawal stands: the artifact behind it is unpaired and compute-confounded. What replaces it is a separate, pre-registered, compute-matched test of the narrower question (App. O), whose result is reported there and is not scored against the withdrawn one.

Backdoor persistence and durability. The persistence mechanism we formalize has empirical precursors. Bagdasaryan et al. (2020) design model-scaling explicitly to embed triggers that survive gradient updates; Zhang et al. (2022) (Neurotoxin) engineer triggers to occupy “less-used” gradient directions, achieving multi-round persistence even under strong norm clipping. These works establish the phenomenon, and this paper *uses* it rather than modelling it: persistence is why a committed attack stays committed across rounds, which is what makes a single committed attack against a fixed defense pair the right setting for an identification question. **We make no formal claim about persistence.** An earlier version of this paper advanced a $(\gamma, \alpha, p_{\text{eff}})$ model of when persistence eliminates the value of defense switching, together with a heuristic boundary as $\gamma \rightarrow 1$; that material is not part of this submission, and we withdraw the claim rather than leave it asserted without the evidence that supported it. What remains is γ as a measured descriptive quantity (Appendix K) and the DBA persistence replication of §M.3, which returns an *intermediate* verdict and establishes no boundary.

Defense composition and ensembling. Composing defenses is well-studied in adversarial ML for centralized models: adversarial training combined with input purification (Crocce et al., 2021), detection-then-rejection pipelines, and ensemble adversarial training (Tramèr et al., 2018). These works implicitly require that upstream processing does not destroy downstream signals; our signal-preservation criterion formalizes this requirement explicitly with measurable margins. In FL the closest *structural* precedent is bucketing (Karimireddy et al., 2022), which restores existing robust aggregators under heterogeneous data by inserting an upstream reshuffling step ahead of them: this is the object we study, an upstream transform feeding a fixed downstream aggregator. Its target is convergence under heterogeneity rather than suppression of a committed attack, so it does not meet Proposition 6; what it shows is that the contrast our protocol demands is the natural one for this object, since establishing that an upstream stage helps at all requires comparing the same aggregator with and without it. In Byzantine-tolerant distributed systems, protocol composition (e.g., layered BFT) has a long history; the key distinction is that Byzantine agreement composes for fault tolerance among replicas, while our setting composes for poison resilience across client updates where the adversarial signal is a continuous gradient, not a binary fault. **A predictive criterion for FL composition does exist in recent work, and we withdraw the priority claim we previously made here.** Fenux et al. (2025) categorize defenses by the deviation of adversarial updates from the mean of the updates and predict that pairing a large-deviation defense with a small-deviation one closes the gap either leaves alone. Our object is prior to theirs and different in kind: not which pairings succeed, but whether a comparison across defenses can identify *why* one did. Proposition 6 therefore applies to their criterion and to ours alike, and it constrains what validating evidence for such a criterion can establish rather than whether the criterion is correct.

G.1 THE PRE-REGISTERED COMPOSITION AUDIT: RUBRIC, FRAME AND THE 59 CODED PAPERS

§G reports one rate out of this audit and withdraws one claim because of it. This subsection is the artifact behind both. The rubric is `experiments/pre_registration_literature_audit.md`, the coder’s records are one JSON file per paper under `results/literature_audit/coding/`, and every count in this subsection is emitted by `experiments/analyze_literature_audit.py` into `results/literature_audit/summary.json`. **The body of Tables A19 and A20 is a file that script writes,** `results/literature_audit/table_rows.tex`, so no code and no count in them is transcribed by hand.

Not all of the rubric predates the coding, and that is the first thing to report rather than the last. Table A18 is the chronology. Pre-registered without qualification, and unchanged from the freeze to now: the primary quantity, the five criteria’s text, the four ambiguity rules, the frame and its screening procedure, the ten-paper floor, and the three admissible outcomes. Amendment 1 re-pointed the retrieval endpoint before any paper had been retrieved, and changed no query string, no criterion and no threshold. Amendments 2 through 5 are adjudication rules for questions the frozen criteria underdetermine, and they were written with rows already in hand. **Exactly one of them moves a reported number.** Under Amendment 4a’s mechanism-distinctness rule the primary is 6/59; without it, 5/59. The single row that turns on it is 2601.06466, whose three-way threshold decision over a Mahalanobis score and a trajectory score is three components under 4a and one under the coarser reading, and both rates are reported everywhere either is.

The frame is a procedure, not a reading list. Three routes in: every `references.bib` entry that proposes or evaluates an FL poisoning or backdoor defense; five frozen query strings issued against the arXiv search interface, paginated to exhaustion, over 2019–2026; and the complete title listings of USENIX Security, NDSS, IEEE S&P and PMLR screened by the frozen regex `federated AND (backdoor OR poison OR byzantine OR robust)`. A one-hop snowball through the defense-comparison tables of Nguyen et al. (2022), Rieger et al. (2022) and Fenaux et al. (2025) closes it, and no second hop is taken. **Adding a query after seeing results is forbidden by the rubric and none was added.** The bias this realizes is stated in the same breath as the rate: coding requires quotable full text, so the frame is *arXiv-reachable* FL poisoning-defense work, ACM CCS is a coverage gap rather than a sample because its proceedings pages return HTTP 403, and **CCS is not the only gap:** OpenReview is unscreened, so ICLR and NeurIPS proceedings enter only by the `references.bib` and snowball routes, as do IEEE venues other than S&P on IEEE Xplore. A paper whose title carries none of the regex’s tokens enters only by those same two routes. **We do not close these gaps by hand:** the rubric forbids adding a retrieval route after seeing results, and a hand-added paper would void the pre-registration it is supposed to be evidence under, so the gaps are named and left open.

The funnel. 622 records retrieved, 593 after deduplication, all 593 screened on title and abstract, 59 full texts sought and 59 obtained, 59 included, 534 excluded. Of the exclusions, 362 are single-mechanism or non-pipeline on their own title and abstract, 71 are a retrieval artifact in which a bibliography line was captured as a record, 38 were adjudicated individually on other stated grounds, 24 name two mechanism families with no joint application, 18 are retrieval-stage losses with neither an abstract nor a resolvable identifier, 15 are surveys, reviews or position papers, and 6 are outside federated learning. Of the 593 screening decisions, 495 are mechanical and every one of those is an exclusion; the remaining 98 were adjudicated by hand, the 59 inclusions and 39 further exclusions, because the mechanical rule can only exclude and *no* paper enters the audit without a hand decision. Every exclusion carries its reason string in `results/literature_audit/screening.json`, and the analyzer refuses to print any rate if one does not.

The criteria, and why the primary is a conjunction with a negated term. C-a: two or more mechanisms applied jointly to the same round. C-b: the pairing is selected or motivated by the constituents’ standalone effectiveness or their complementary threat coverage, and by nothing else. C-c: the composed system’s own adversarial outcome, its ASR, backdoor accuracy or robust accuracy, is what validates it. C-d: an identifying contrast, meaning an experiment or derivation that varies the *upstream* stage with the downstream mechanism and the attack held fixed. C-e: a mechanism-attribution claim is actually made. **The primary is $C-a \wedge C-b \wedge C-c \wedge C-e \wedge \neg C-d$,** and the C-e term is what stops it from measuring “nobody runs the contrast we invented”, which would be near total and worth nothing. Any UNCLEAR criterion sends the paper’s verdict to UNCLEAR, and UNCLEAR papers leave both the numerator and the denominator, so the measured prevalence can only be smaller than a guessing coder’s. No row resolved to UNCLEAR.

The result, with the secondaries it must be read against. Primary 6/59 = 10.2%, or 5/59 without Amendment 4a. The outcome-gated *design*, $C-a \wedge C-b \wedge C-c$, holds for 33/59 = 55.9%. An identifying contrast is *present* in 46/59 = 78.0%, and among the 33 design papers, 27 run one. **So the six are papers that had the design and did not run the contrast, not papers for which it was unavailable,** and any sentence implying the contrast is rare in this literature is false against our own audit. The pre-registered secondary that would count against the sharp reading, a design paper making no attributional claim at all, is 0/33; the one C-e = NO row in the frame, 1909.05125, has no design either, and a paper with neither is evidence in neither direction. The six are 2307.00356, 2310.13403, 2502.07011, 2507.16134, 2601.01053 and 2601.06466.

What the 21 C-a = NO rows are made of, in three kinds. First, Amendment 3’s purpose test excises constituents whose own stated purpose is privacy, communication efficiency or server verifiability rather than resisting corrupted updates, which is what leaves 2607.06612 and 2609.03420 with exactly one defense-purposed mechanism per round. Second, several papers decompose a *single* mechanism into stages that exist to serve it: 2505.17226’s median filter repairs an estimate the Krum-derived rule needs, and 2311.15894’s four steps are four steps of one rule. Third, 2302.07173 evaluates its schemes as alternatives to one another, which C-a’s own exclusion list names as a menu rather than a composition. The per-row basis is in each record’s C-a locator, and the reader who disagrees with a call can see the sentence it rests on.

C-b fails on five of the 38 C-a = YES rows, for four distinct reasons, and the reasons matter more than the count. 2508.12978 composes for *guarantee preservation*, proving that adding its compression does not destroy what the aggregator already provides, which is a reason about the composition rather than about the constituents’ effectiveness. 2601.04930 and 2602.22269 give a real constituent-property reason of the residual-coverage kind, but the surface being closed is one an Amendment-3-excluded constituent creates, so the pairing reason does not connect two counted mechanisms. 2503.18284 names the deficiencies of *prior* schemes as motivation for its framework and states the integration of its own two stages as a fact and an optimization convenience. 1909.05125’s only stated reason is computational and communication efficiency. **We record these rather than fold them in:** three of the four are pairing reasons of kinds the rubric could plausibly have admitted, and it does not, so C-b is stricter than it might have been in a direction that shrinks the primary.

The C-d route field stratifies the 46 contrasts and decides nothing. Of them, 21 are hyperparameter sweeps of an upstream defense parameter with an adversarial outcome read at each setting, 21 are component contrasts in which an upstream stage is removed, substituted or cumulatively built up, and 4 are analytic. Precedence when a row satisfies more than one is sweep, then contrast, then analytic, fixed in advance of the backward pass rather than deferred to which finding a paper calls central. **Amendment 5’s backward pass was run across all 46 rows and relabelled four,** 2409.01435, 2502.00587, 2505.10297 and 2604.03862, moving the stratification from 25/17/4 to 21/21/4; no rate moved, because the field enters none. **It is a floor and not a census,** since it records only routes a record actually quotes. LASA is the proof: its appendix sweep of the upstream sparsification level, with accuracy under three attacks at every rung and the downstream filter fixed, was in the paper and merely pointed at in the record until the backward pass quoted it.

One contrast in the frame is confounded by its own design, and it is coded YES anyway. 2609.03420 varies its upstream privacy budget as a graded dose with the downstream median and the attacks fixed, and states the reason: “This selection isolates privacy-fairness interactions from aggregation-induced variance.” But the client learning rate is a declared function of that budget, so two things move together along the ladder, and the paper demonstrates that this matters by reporting collapse rather than a graded decline under a static rate. The rubric’s fourth ambiguity rule gives C-d the benefit of the doubt, and the code follows the rule rather than our preference: a rule that resolves toward finding the contrast present is a rule that shrinks our own primary.

Quotation integrity, and the substitutions this table cannot avoid. Every quotation in every record is asserted verbatim against the paper’s own extracted full text by `experiments/verify_quotes_literature_audit.py`, which exits non-zero on any failure: 59 records, 295 quotations, 334 spans, all verbatim, with four spans under 40 characters and two pairs out of source order reported rather than failed, the locator saying in each case that the spans are not one passage. **The first run of that check over the whole set failed 28 quotations in 15 records, and the repair log is committed** at `results/literature_audit/quote-repair_log.md` with the pre-repair records beside it. **No code changed in any repair,** and two repairs produced *stronger* evidence for the code already recorded, which is the uncomfortable part: the defective quotations were faithful paraphrases, so a reader checking the argument would not have caught them and only the machine check did. One consequence lands in this subsection. Verbatim spans contain a true minus, an en dash inside a citation range, typographic apostrophes, `ff` and `fi` ligatures, Greek and math-italic letters and a set-membership sign, none of which this paper’s source may carry, so quoted spans are rendered with a LaTeX equivalent, for instance $\text{CSFT}'s$ for the typographic apostrophe and δ for math-italic delta. **The JSON records, not this appendix, are the authority for what each paper says.**

Extraction hazards, named because they are what a re-adjudication will hit first. `pdftotext` interleaves the columns of a two-column layout, which spliced a word from the adjacent column into a sentence in 2503.18284 and produced adjacent-but-unrelated spans in 2602.16480 and 2607.06612; three papers, 2606.31066, 2608.08574 and 2608.21172, emit table bodies in column-major order, so a number’s row label is not the label above it in the stream; and displayed fractions and page numbers appear mid-sentence. Every affected locator says so. Separately, 2609.03064 carries a journal line dated 2024 against a 2609 arXiv identifier; we code it under the arXiv year and record the inconsistency rather than choose silently.

What this audit does not establish, recorded before the numbers existed and unchanged by them. It is one coder with no inter-rater statistic, and the mitigation is the quotation rather than a claim of reliability. Four rows were known to us before the freeze and are marked in the tables: coding them is not blind. The rate is about this frame, which is arXiv-reachable, preprint-heavy and 2026-heavy, and it is not an estimate of “the FL literature”. And the rate carries no claim that the six papers’ conclusions are wrong: the audit measures whether an evaluation design can identify the mechanism it attributes to, which is a question about what the evidence licenses and not about whether the defense works.

H LIMITATIONS, IN FULL

(L1) Limited scope and architecture-conditional boundaries. Three attack families (model_scaling, pixel, DBA), two architecture families (CifarCNN, ResNet18; plus `simple_cnn`, a third CNN configuration and not

Table A18: What was frozen when. The rubric is part pre-registration and part post-hoc adjudication record, and this is the only honest way to report that. Rows in hand is the number of papers already coded at the moment the text was written.

what	state	coded rows in hand
Base rubric: five criteria, four ambiguity rules, the floor	committed 9b8a395	0
Amendment 1: retrieval endpoint re-pointed	committed 3c61301	0
Amendment 2: blind-coding disclosure widened by one paper	written during coding	5
Amendment 3: purpose test for what counts as a defense	written during coding	5
Amendment 4: distinctness, distributive test, ordering	written after coding	59
Amendment 5: <code>c_d.route</code> routes and precedence	written after coding	59

a third family, on the replicated arm), and one dataset (CIFAR-10) for *every* composition suite without exception; the two arms that leave CIFAR-10, EMNIST-byclass and CIFAR-100, are single-arm intervention ladders and neither carries a composition. The collapse boundary is architecture-conditional (ResNet18 suppression at NC10/rep90: 0.865 vs. CifarCNN 0.097, a $9\times$ shift; per-seed values in §N.2), so results at other N , K or datasets may differ. **That gap is also evidence for this paper’s own thesis:** a swing of that size under a change of architecture alone is why we state the preservation conditions as mechanisms to be measured per setting rather than as architecture-independent empirical laws. **The Mode-S replication of §D.2 does not soften this, and the two are not in tension:** what holds across both architecture families there is the *dissociation*, a large decision disturbance costing no suppression, and not any suppression *level* — which is precisely the quantity this $9\times$ swing moves. A mechanism can transfer where the number it produces does not, and we claim only the former.

Cross-distribution transfer is validated across architecture, scale ($N \in \{10, 100\}$) and adversarial fraction ($f \in [0.05, 0.6]$; §E.6(D)), and *dataset* transfer is now tested on two arms and no more: the flagship Mode-S arm on EMNIST-byclass with `simple.cnn`, where the negative replicates at $\Delta = -0.017$ (§D.2), and the `coord.median/pixel` comparability cell on CIFAR-100, where the confounded-versus-controlled sign disagreement replicates, -0.213 against $+0.071$ (§D.3).

For the screen itself the obstacle turned out to be sharper than an untested transfer: on EMNIST-byclass C1’s precondition has no input on the committed pixel arm. C1 requires that some constituent suppress each committed attack *alone*, and under the identical composition protocol the pixel backdoor is essentially unsuppressed there: re-measured here, `coord.median` reaches 0.609 at clean accuracy 0.836 (seeds 0.717/0.493/0.618), and over the seven single-defense pixel cells measured there that 0.609 is the *minimum*, with `krum` next at 0.660 and the other five between 0.988 and 0.998 (six *distinct* aggregators, since `multi.krum` reduces to `fedavg` at $K=5$). **So no pair drawn from them can satisfy C1 there:** strategy B certifies nothing over them, every such pair is predicted HIGH, and the strategy comparison of §6.3 cannot be made on that dataset from these inputs. **The coverage is exactly those cells, and we state the gap rather than generalize past it:** `foolsgold`, `reputation` and the cosine mirrors have no EMNIST-byclass pixel cell, so what is established is C1’s failure over the measured menu and *not* its unsatisfiability over the whole one. **The failure is the pixel arm’s, and we do not escape it by dropping the other.** On the model-scaling arm `krum` does suppress there (0.027 at clean accuracy 0.767), so a scaling-only protocol would supply a C1 input and manufacture the comparison. We decline it for the reason §E.1 declines the mirror move: the committed protocol requires *each* attack, and choosing the arm that happens to have an input after seeing which does is attack-shopping regardless of the resulting labels. **On any of these readings it is a scope limit on the criterion’s precondition and not a failed replication:** no prediction was made and refuted, and the question a screen answers does not arise where nothing suppresses the attack under test. We report it rather than lower f , subset the classes or lengthen training until a positive appeared, each of which is a new design axis chosen after seeing the obstacle. The `coord.median` figure is *bit-identical* to the corresponding cell of `results/femnist/payoff_results.json` (0.6091787658... at 0.8363207047...), which by the standard we apply to ourselves in §D.2 (the same computation run twice cannot corroborate itself) establishes protocol *identity* and not corroboration; what it does buy is that the matrix’s other cells are protocol-comparable, so we did not spend 21 h re-measuring them (`experiments/measure_femnist.c1.inputs.py`, `results/femnist.c1.inputs/summary.json`). **We then screened the scaling arm for a further comparability cell and report the screen rather than a cell** (supplement §S13): of the five defenses with no scaling cell there, `fltrust` is excluded by argument (its admission is bit-identical under this rescaling) and we measured the other four, of which `foolsgold` and `cos.reputation` are the only two that would add a downstream defense and both fail the *accuracy* floor, $10\times$ rescaling collapsing the model to 0.046 and 0.100 clean accuracy, so **no defense outside Table A10’s four is admissible on that axis.** `reputation` does meet the precondition (0.3465 at 0.704), but on seeds 0.9947/0.0152/0.0296, and a replication resting on a spread that wide is one we decline to run.

Dataset transfer therefore remains untested for every composition suite, both arms that leave CIFAR-10 being single-arm ladders: the preservation conditions are dataset-agnostic in mechanism (cosine similarity, geometric

Table A19: The pre-registered composition audit, rows 1–30 of 59, ordered by arXiv identifier. ✓ is YES and × is NO; the C-d route cell carries an absent-value dash exactly when C-d is NO, since a paper that runs no contrast has no route. *primary* is the outcome-gated verdict $C-a \wedge C-b \wedge C-c \wedge C-e \wedge \neg C-d$. Every cell is emitted by `experiments/analyze_literature_audit.py` from the per-paper records, and each YES or NO in those records carries a machine-checked verbatim quotation and a locator. *short name* is that script’s handle, an acronym where the paper has one and otherwise the head of the title before any colon, kept whole up to 24 characters and cut at 21 with an ellipsis beyond that, which is why four rows here read as a stem: 2207.01982 is “Defending against the Label-flipping Attack in Federated Learning”, 2302.07173 is “An Experimental Study of Byzantine-Robust Aggregation Schemes in Federated Learning”, 2409.17754 is “Byzantine-Robust Aggregation for Securing Decentralized Federated Learning” and 2409.19302 is “Leveraging MTD to Mitigate Poisoning Attacks in Decentralized FL with Non-IID Data”. We extend them here rather than in the table, whose body is machine-written; every full title is in `results/literature_audit/screen_pool.json` under the identifier in the first column.

arXiv id	short name	yr	C-a	C-b	C-c	C-d	C-e	C-d route	primary
1909.05125	Adaptive FedAvg	2019	✓	×	✓	×	×	—	no
2012.13995 [†]	FLTrust	2021	✓	✓	✓	✓	✓	contrast	no
2101.02281 [†]	FLAME	2022	✓	✓	✓	✓	✓	contrast	no
2104.06685	BROADCAST	2021	×	×	✓	✓	✓	analytic	no
2110.02940	SHARE	2021	×	×	✓	✓	✓	sweep	no
2201.00763 [†]	DeepSight	2022	✓	✓	✓	✓	✓	contrast	no
2206.12100	zPROBE	2022	×	×	✓	✓	✓	sweep	no
2207.01982	Defending against the...	2022	×	×	✓	×	✓	—	no
2208.10161	MUDGUARD	2022	×	×	✓	✓	✓	contrast	no
2210.01437	Robust-FL	2022	×	×	✓	✓	✓	sweep	no
2210.07714	CrowdGuard	2024	✓	✓	✓	✓	✓	contrast	no
2302.07173	An Experimental Study...	2023	×	×	×	×	✓	—	no
2306.04984	G ² uardFL	2023	✓	✓	✓	✓	✓	sweep	no
2306.12608	DP-BREM	2023	×	×	✓	✓	✓	sweep	no
2307.00356	Fedward	2023	✓	✓	✓	×	✓	—	yes
2309.10607	SPFL	2023	×	×	✓	✓	✓	contrast	no
2310.13403	BRFL	2023	✓	✓	✓	×	✓	—	yes
2311.15894	refined-Krum	2023	×	×	✓	×	✓	—	no
2402.04409	FRECA	2024	×	×	✓	×	✓	—	no
2404.04139	FedZZ	2024	×	×	✓	✓	✓	sweep	no
2409.01435	LASA	2024	✓	✓	✓	✓	✓	sweep	no
2409.17754	Byzantine-Robust Aggr...	2024	✓	✓	✓	✓	✓	contrast	no
2409.19302	Leveraging MTD to Mit...	2024	✓	✓	✓	✓	✓	contrast	no
2411.01040	MASA	2024	×	×	✓	✓	✓	sweep	no
2502.00587	RKD	2025	✓	✓	✓	✓	✓	sweep	no
2502.05547	DDFed	2025	✓	✓	✓	✓	✓	sweep	no
2502.07011	DROP	2025	✓	✓	✓	×	✓	—	yes
2503.18284	FedSAC	2025	✓	×	✓	✓	✓	contrast	no
2505.01454	SafeSparse	2025	✓	✓	✓	✓	✓	sweep	no
2505.10297	FeRA	2025	✓	✓	✓	✓	✓	sweep	no

[†] known to us before the rubric was frozen; coding is not blind.

Table A20: The pre-registered composition audit, rows 31–59 of 59. Columns and conventions as in Table A19. Rows are identified by arXiv identifier and short name rather than by citation: citing all 59 would add 55 entries to the bibliography, most of them preprints no resolver carries, and the identifier is what a re-adjudication needs.

arXiv id	short name	yr	C-a	C-b	C-c	C-d	C-e	C-d route	primary
2505.12851	FLTG	2025	✓	✓	✓	✓	✓	sweep	no
2505.17226	ArKrum	2025	×	×	✓	×	✓	—	no
2507.14588	FORTA	2025	×	×	✓	✓	✓	contrast	no
2507.16134	DP2Guard	2025	✓	✓	✓	×	✓	—	yes
2508.10315	CLIP-Fed	2025	✓	✓	✓	✓	✓	contrast	no
2508.12978	Fed-DPRoC	2025	✓	×	✓	✓	✓	sweep	no
2509.08089 [†]	Hammer and Anvil	2025	✓	✓	✓	✓	✓	contrast	no
2509.08449	DSFL	2025	×	×	✓	×	✓	—	no
2509.18044	HRA	2025	✓	✓	✓	✓	✓	sweep	no
2511.09294	GuardFed	2025	✓	✓	✓	✓	✓	contrast	no
2512.11760	SpectralKrum	2025	✓	✓	✓	✓	✓	contrast	no
2601.01053	PQ-BRFL	2026	✓	✓	✓	×	✓	—	yes
2601.04930	PPFA-BA	2026	✓	×	✓	✓	✓	contrast	no
2601.06466	SecureDyn-FL	2026	✓	✓	✓	×	✓	—	yes
2602.16480	SRFed	2026	×	×	✓	✓	✓	analytic	no
2602.22269	CQSA	2026	✓	×	×	✓	✓	analytic	no
2603.04422	FedEMA-Distill	2026	×	×	✓	✓	✓	contrast	no
2603.23472	Byz-Clip21-SGD2M	2026	×	×	✓	✓	✓	sweep	no
2603.28652	FedBBA	2026	✓	✓	✓	✓	✓	sweep	no
2604.03862	SecureAFL	2026	✓	✓	✓	✓	✓	sweep	no
2605.11122	FedSurrogate	2026	✓	✓	✓	✓	✓	contrast	no
2605.22506	EnCAgg	2026	✓	✓	✓	✓	✓	contrast	no
2606.31066	Secure-CHG	2026	✓	✓	✓	✓	✓	contrast	no
2607.06612	PRoVeFL	2026	×	×	✓	✓	✓	analytic	no
2608.08574	R2CFL	2026	✓	✓	✓	✓	✓	sweep	no
2608.14861	STAR-FL	2026	✓	✓	✓	✓	✓	contrast	no
2608.21172	Thermo-FL / TERRA	2026	✓	✓	✓	✓	✓	contrast	no
2609.03064	DP-BR-FedAvg	2026	✓	✓	✓	✓	✓	sweep	no
2609.03420	GeoIndist-FL-NIDS	2026	×	×	✓	✓	✓	sweep	no

[†] known to us before the rubric was frozen; coding is not blind.

Table A21: **Every claim we have withdrawn or refuted, enumerated in one place.** Each row’s full statement, the evidence that overturned it and what survives it are at the reference in the third column; this table is an index and is not itself the home of any disclosure. **It is hand-maintained.** No results/ artifact records claim status, so unlike every number in this paper these rows have no emitter, and a later revision must update them by hand.

Claim, as we once stated it	status	stated in full at
Preserving the statistic or the decision d_2 reads preserves its attack suppression	refuted	cos_krum, §D.2
Measured disturbance transfers as a predictor, and orders the arms by admission	refuted	(L6), §H
$\Delta\Lambda_a$ explains when the two designs disagree	withdrawn	§D.3
The outcome-gated and the controlled design agree	refuted	5 of 7 cells, §D.3
An inherited-suppression screen can reach emergent compositions	refuted by construction	FG→RFA, §F
The two disturbance channels combine additively	indeterminate at $n=20$	§D.2.1
Three matched contrasts show a 19–20× gap turning on disturbed invariance	withdrawn; the frozen 13/20 score stands	§E.1
C0 belonged to the criterion before the first prospective suite scored it	withdrawn as post hoc	(L4), §H
Fenaux et al. (2025) is a named external instance of the practice we criticize	withdrawn	§G, §G.1
Randomizing which composition runs each round costs suppression	withdrawn as unpaired and compute-confounded	§G, §O
Consensus-shift is ineffective against FG→RFA, as measured	figure void, withdrawn; conclusion re-established by re-running the arm	§F

distance) while the downstream decision boundary may shift with output dimensionality. *The 2×2 channel factorial is narrower still:* it is defined only for the two selectors whose decision can be transplanted (krum, cos_krum) and the harness raises for any other d_2 , so it is one attack on one aggregator family, reported at $n=5$ as **not decisive** (§D.2.1) and acted on only where it withdraws a claim. *Two further gaps in the intervention:* neither DBA nor Neurotoxin was run through it, and the score-only control is a single arm, one aggregator, one attack. *And scale is a prediction, not a measurement:* the pre-registered PASS predictions hold at $N=100$ /ResNet18 (FG→RFA 0.069) but far from converged accuracy, so they bound nothing about converged training, and **the FAIL controls run at that scale proved uninformative**, both landing below 0.5 (§S4). *What transfers:* the qualitative PASS/FAIL prediction and the mechanism. *What does not:* the exact suppression ASR. A practitioner at $N=10^4$ should use the criterion to select candidates and then pilot at their target scale: it identifies *what to try*, and architecture-specific calibration determines *how well it works*.

(L2) The composability criterion is an evaluation-prioritization rule, not a security screen and not a discovery tool. It has no established predictive validity: it orders candidates for evaluation, and the pooled model that would let it predict is unconfirmed ($R^2=0.239$) with its cross-arm ordering refuted (§D.2). The criterion gives *sufficient* conditions ($C1 \wedge C2$, with $C3$ as a margin diagnostic): it reliably identifies mechanism-preserving compositions in the common case where one defense is known-effective, but it cannot certify synergistic compositions where no single defense suppresses the attack alone. The conservative bias is by design: false negatives are operationally cheap to recover from (try the composition anyway) while false positives would deploy an insecure stack. Accuracy is $16/18 = 89\%$ on the development set, $22/24 = 92\%$ on the pre-registered held-out set (90.5% combined 2-way) and $9/10 = 90\%$ on pre-registered 3-way triples (supplement §S5). **None of those tallies should be read as discrimination:** only 5 of the 42 two-way pairs are low-ASR, so a constant HIGH call scores $37/42 = 88.1\%$ and the combined 90.5% beats it by one pair. Precision ($2/2$) and recall (40%) are the quantities that carry information, and they are why we present the criterion as a prioritizer. All clean $C1 \wedge C2 \wedge C3$ PASS predictions are correct; the four misses are either the predicted intermediate behavior of rank-based aggregators (Proposition 9: dev-set $NC \rightarrow cm$, $rfa \rightarrow cm$; held-out fedavg→cm) or one speculative LOW call the criterion does not license (held-out $fg \rightarrow NC$, a $C1$ failure). Formalization is partial: Theorem 8 proves $C2 \wedge C3$ for the weighting→rank/geometric class with explicit margins, the remaining families are em-

pirically validated, and C3’s threshold condition depends on d_2 ’s discrimination boundary, which in practice requires calibration runs.

(L3) Per-round aggregator switching only. We test per-round defense selection. Randomized *internals* (DP noise, subsampling smoothing à la CRFL, random participation masks) modify the embedding mechanism directly and may retain value even under persistence. Three adaptive attacks are tested against composition: projection-based against rep+tm (ASR saturating at 0.639 for $\epsilon \geq 2$, 0.597 at $\epsilon=1$; the same attack against fg→rfa reaches only 0.079, Table A16), Neurotoxin-style (Zhang et al., 2022) top- k gradient projection, and a two-phase consensus-shifting attack against rep+tm that biases updates toward the target at rate r for 25 rounds before injecting full backdoor updates (App. M). At shift_rate $\in \{0.01, 0.05\}$: mean ASR 0.684 ± 0.138 ($n=10$) and 0.667 ± 0.137 ($n=8$), comparable to committed_pixel (0.679) and *below* the rep30/tm70 temporal mix’s 0.915. **Those last two are not the same metric, and the comparison is the weaker one because of it:** 0.679 and the two shift-rate means are single committed attacks against the composition, whereas 0.915 is the mix’s full-information reading over both attacks, which no single attack can be expected to match. What the row supports is that the two-phase attack does not exceed the mix, not a calibrated gap between them. At $f=0.2$ the honest 80% anchor the coordinate-wise median, limiting Phase-1 drift to $\sim r/(1-f)$ per round, so Phase 2 degenerates to standard committed_pixel: a mechanism that weakens monotonically with f , and at $r \leq 0.05$ over $T \geq 200$ rounds cumulative drift may eventually overcome it. The robustness claim is therefore specific to the tested regime ($f=0.2$, $T=50$).

(L4) The falsifiable prospective positive evidence is one cell, and C0 was added retrospectively. This is the sharpest limitation we can state about our own design, and we state it rather than let a reader find it. Until §E.5 there was none at all, and one cell is what changed. Of the prospective suite’s 8 predicted-LOW pairs (§E.1), 6 are →FLTrust near-identities: FLTrust gates on a scale-invariant cosine and renormalizes every update to the server-gradient norm, discarding exactly the magnitude information an upstream positive rescaling modifies (verified bit-identical per seed for pass-through and non-clipping d_1), so Proposition 10(a) permits no other outcome and these confirmations cannot falsify it; the other 2 have FLTrust as d_1 , where suppression is inherited and C0 puts them out of scope. C0 was itself formulated *after* seeing this suite, so its 7 exclusions are retrospective. The three matched contrasts that appeared to probe Proposition 10 most directly (a 19–20× ASR gap at fixed d_1) are **withdrawn**: two ran against MultiKrum, which at $K=5$ selects all five participants and is bit-identical to FedAvg, and the genuine selector leaves a gap that measures C1 failure rather than disturbed invariance (§S2).

The decisive experiment does not exist, and that is structural. We had named it as a downstream defense with a rescaling-invariant admission signal that is *weak in isolation*; Lemma 1 shows no such pair can be certified in scope. §E.2 runs the achievable substitute, holding the suppression *principle* fixed while varying only the invariance class, and it did not deliver a positive test either, standalone effectiveness being unholdable here because the magnitude information the mirrors discard is what the originals use to suppress a magnitude attack (8/8 frozen labels, TP = 0). So the strongest C2-specific evidence is the *post-hoc* within-defense contrast of §E.2, which holds C1 exactly fixed but was not pre-registered and is resolved on only one of its two cells at $n=3$.

Three pre-registered attempts followed, and each refuted something of ours. §D delivers exactly one falsifiable prospective positive (arm 2, reputation under model-scaling, $0.017 \rightarrow 0.779$ at $p = 1.7 \times 10^{-6}$ with clean accuracy nearly unchanged) while refuting two of its own four predictions and its pooled prediction, leaving the *destruction* half of C2’s mechanism confirmed and the preservation half’s control indeterminate (§D.1). The targeted intervention of §D.2 closes the attenuation channel by construction and isolates C2 within a single defense: substantial disturbance of the statistic does not change suppression *on the adjudicating Krum arm under Mode S*, with admission the channel the evidence points to there, and it *refutes* the preservation-half reading, the exactly-invariant cos_krum neither suppressing nor holding ASR fixed. The third was designed to be the falsifiable prospective *positive*: a replication arm chosen because the statistic and admission readings order its rise *oppositely*, frozen in git before the arm ran. It replicated the negative in a different invariance class and **refuted the positive ordering**, the arm with the largest measured admission change rising *less* (+0.098 at $n=5$) than one whose admission change was smaller.

The census of §E.5 is the one place this changed, and it changed by exactly one cell. We closed the 3×3 region where Prop. 10(a) settles C2 algebraically, registering its two unspent cells at 96f8856; the single certified pair was registered LOW with *low* confidence and an explicit *uncertain* flag, and measured LOW at 0.263. Unlike every predicted-LOW pair above it is neither a near-identity (0 of 3 seeds identical to d_2 alone, on both arms) nor an inherited suppression, since reputation holds only model-scaling and cos_krum only the pixel backdoor. **That is a falsifiable prospective positive, and it is one cell at $n=3$ with a per-seed spread of 0.000 to 0.681:** it establishes that the criterion’s precision survives a blind test in the one region our invariance result decides exactly, and nothing quantitative. A validated *positive* predictor therefore remains absent: the pooled two-channel model is unconfirmed over 26 cells ($R^2 = 0.239$) and its cross-arm ordering refuted outright, so admission is *more causally proximal* than the statistic without being a quantitative predictor.

(L5) Theorem 8’s conditions need a pilot, and (1)’s is not operationally checkable as stated. Neither $\rho < r^*$ nor $\Delta_{\text{sep}}^l > 2(R_B + \delta')$ is readable off a (d_1, d_2) definition pair: both need a pilot measurement. **So two kinds of cheapness have to be kept apart, and the paper has previously run them together.** Prop. 10’s classes are *algebraically* checkable: (a) and (c) are decided from the definitions of d_1 and d_2 alone, with no data, no runtime cost and no per-deployment work. Theorem 8’s conditions are closed-form inequalities and are *still* none of those things, because their terms have to be instantiated by measurement on the deployment they are being checked for. **Algebraically checkable does not imply deployment-time cheap**, and only the first of the two is something this paper delivers. Regime A is worse than that. Its ordering-survival condition is $\rho < r^*$ with r^* the attack’s coordinate extremeness against the benign scale. Building the dose–response ladder (§D) required us to *measure* r^* rather than stipulate it, and it is not a stable per-attack constant: on model-scaling it ranges 0.00–12.59 across the 12 measured rounds that sampled an adversary (median 0.57), on the pixel backdoor 0.61–1.63 (median 0.92). An illustrative $r^*=10$ is a stipulation, not a measurement, and §C.3 stipulates no value for r^* at all: the theorem states the condition symbolically. Two consequences we state rather than work around. First, a practitioner cannot check $\rho < r^*$ from a single round: the quantity swings by orders of magnitude round to round, and on both committed attacks its median is below 1, so Regime A’s condition fails even at the identity rung under this estimator. Second, we therefore built *no* prediction in §D on r^* , resting the ladder on directly measured disturbance of d_2 ’s statistic instead. The margin form of the theorem remains valid as a statement about ρ and r^* ; what is not established is that r^* admits a stable estimator at the round granularity a deployment would have to check it at, and a usable Regime A needs an r^* estimator with characterized variance, which we do not provide. **And where the geometric condition is measurable it stops being informative in the regime that matters most:** the margin it rests on is exactly what heterogeneity closes, RFA’s all-round mean falling from 3.24 to 1.01 at Dirichlet $\alpha=0.1$, the limit at which Corollary 4 degrades to $n_a/(n_b+n_a)$ and certifies nothing (§C.3, §E.6(E)). So under extreme heterogeneity the conditions are not merely awkward to check but empty, which is the sense in which §3 calls the bound vacuous there.

(L6) Measured disturbance is not a transferable predictor, and our dose instrument varies the wrong quantity for three of its four arms. The pre-registered ladder (§D.1) refutes the pooled claim it was built to test ($\rho_s = +0.351$, $p = 0.091$) and refutes two of four per-arm shapes, with ASR *falling* as the dose rises in three arms. Two distinct faults, both ours. First, the abscissa: at matched measured disturbance the real upstream transforms sit up to +0.79 ASR above the synthetic curve, and Krum’s within-margin arm does not reproduce the very contrast (§E.2) we attributed to disturbance of its statistic, so the rate at which a transform flips a defense’s decision is not a sufficient statistic for what the transform does: *which* clients are reweighted matters, and ρ does not measure that. We now measure the quantity that does, Δ_c (the supplement §S10): the transforms carrying the two anchors that miss are targeted far beyond any ladder rung (−1.21, −0.72 against the ladder’s −0.34) at ρ that is 3.7 and 13× smaller. That names the axis of the mismatch without supplying a replacement predictor: Δ_c ’s magnitude does not order suppression either. Second, the instrument: mean-1 normalization fixes the aggregate’s scale but not any client’s relative weight, so with the dose assigned independently of adversary status a large ρ mostly dilutes whichever client carries the poison (measured: the ladder’s Δ_c drifts monotonically negative with the dose, $0.000 \rightarrow -0.140 \rightarrow -0.240 \rightarrow -0.336$), and three arms end up measuring Theorem 8’s attenuation regime rather than C2 disturbance. The adversary-targeted dose that separates the two, recorded as not run before the freeze (caveat (iii), the supplement §S7), is run in §D.2, and in retrospect Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to 54.6) and Mode A the Δ_c -**swept** one (−1.14 to +2.47): with every adversary’s weight pinned, Krum’s statistic is disturbed in up to 80% of rounds while the admission of adversarial input does not move and ASR stays inside the pre-registered practical-equivalence margin, so *which* clients are reweighted, not how often a decision flips, is indeed the operative variable, and the ladder confound is *explained* (reputation’s rise shrinks from +0.762 to +0.178 once the adversarial share is held fixed). That isolates C2’s causal content within a single defense; it does not rescue the pooled positive direction ($R^2 = 0.239$ over 26 cells, cross-arm ordering refuted by the replication).

I EVIDENCE STATUS: WHAT IS PROVED, WHAT IS SUPPORTED, WHAT IS NOT

Proved: the invariance classes of Proposition 10; sufficient conditions for mechanism preservation on the weighting→aggregator class (Theorem 8, Lemma 2, Corollary 4); the identification boundary for outcome-gated screening (Proposition 6, with Lemma 1 its FL instance and Corollary 1 the within-defense design it licenses); the recall bound on inherited-suppression screens (Proposition 7, whose cap reaches 0 on a menu whose suppression is emergent throughout, Corollary 2); and, formalizing the interventions, statistic-level non-identifiability (Proposition 11) and the independent controllability of ρ and Δ_c (Proposition 12). **The two boundary results are general** (they are statements about the shape of a screen, not about FL), so what carries beyond this paper is the design they license, not the screen. **Empirically supported:** that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed, with adversarial admission the channel the evidence favours, *on the adjudicating Krum arm under Mode S* and, by pre-registered replication, on a downstream defense in a *different* invariance class, on a *second dataset and a differently configured CNN*, and on a *second architecture family* with architec-

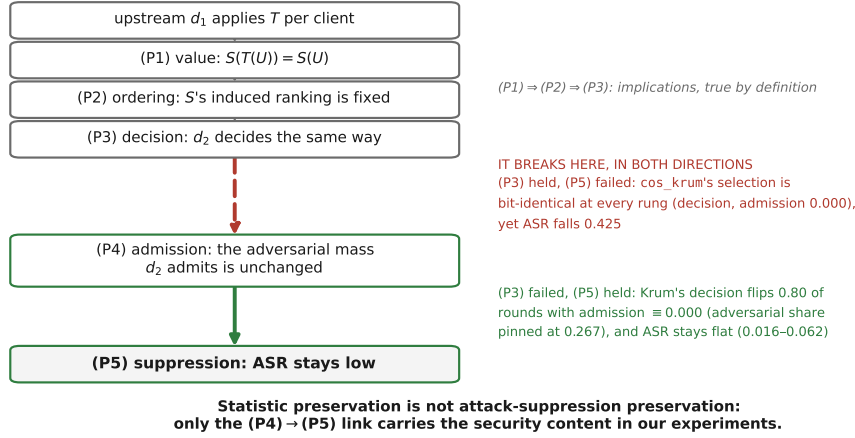


Figure A6: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **below (P3) the chain breaks, in both directions**, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 2’s attenuation channel by construction (§D.2). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder ($\rho_s = +0.351$, one-sided $p = 0.091$, 16 cells; §D), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering** ($\Delta = +0.098$ at $n=5$; the pooled two-channel model stays unconfirmed, $R^2 = 0.239$ over 26 cells), so **admission is more causally proximal than (P1)–(P3) here, and not a validated predictor**. Every number in the figure and in this caption is recomputed from the scoring scripts’ own JSON and printed by `story_chain_figure.py`.

ture varying alone (§D.2); that the screen behaves as a conservative prioritizer across 42 pairs; and that in the one region where Prop. 10(a) settles preservation *exactly*, a closed 9-cell census whose two unspent cells were registered blind contains a single low-ASR composition and it is the one the criterion certifies, at precision 1/1 and recall 1/1 for a tenth of the runs against statistic preservation’s 11% (§E.5, one cell at $n=3$, gated reading). Those two 100% are contingent on the frozen seed count twice over, and disclosure (ν) of §E.5 states the second contingency: the C1 input that certifies the pair reads 0.300 over seeds 42–44 and climbs with the seed count, to 0.490 at $n=5$ (still below the threshold, by 0.010) and to 0.584 at $n=8$, where C1 fails and the criterion certifies nothing in that census. **Not established:** the positive predictive direction of the criterion, since the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and the replication refutes its cross-arm admission ordering, so admission is a direction and not a quantitative predictor; any ability to discover emergent synergy; and generalization beyond star-topology FL, or beyond CIFAR-10 for anything but the two replicated arms, EMNIST-byclass and CIFAR-100, neither of which carries a composition suite; on the EMNIST arm no single defense measured suppresses the committed pixel backdoor alone, so C1’s per-attack precondition fails over the menu measured there and the strategy comparison is out of scope there rather than merely untested (L1), which also states what the seven do *not* cover). **The open problem**, which the sections above refer to by pointing here: every reading of (P4) in this paper comes from an instrument that knows which clients are adversarial, and no deployed defense knows that, so whether the adversarial contribution a downstream mechanism admits can be estimated *without* adversary identity is open. Nothing here bounds it, and the design we license identifies that channel rather than screening on it.

J THE PRESERVATION CHAIN IN ONE FIGURE

The lower arc, and why the last two links must be broken by intervening rather than comparing. T can reach ASR by attenuating whichever client carries the poison (Lem. 2), **without passing through d_2 ’s statistic at all**. Mode S cuts that arc by construction, pinning every adversary at $c=1$. **The three numbers annotating Figure 1(a), which Figure A6 redraws without panels, are three scopes, not one experiment:** Krum’s flip rate; the admission support *pooled* over Table 1’s four arms, so it names no arm and Krum’s own zero is a floor; and `cos_krum/pixel` at its first gate-passing rung (§6.2).

K NOTATION

symbol	meaning
c_i	upstream coefficient applied to client i 's update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i i \in A] - \mathbb{E}[c_i i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
r^*	the attack's coordinate extremeness, $\min_k \min_a \min_b u_{a,k} / u_{b,k} $ (Theorem 8(1))
S	the downstream statistic d_2 reads, as in $S(T(U))$ (§C.4)
m	RFA separation margin, $\Delta'_{\text{sep}} / 2(R'_B + \delta')$
Λ_a	aggregate adversarial influence, instantiated per aggregator: Weiszfeld/reputation weight share, selected-coordinate share, or the selection indicator
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

L EQUIVALENCE TESTING AT THE EXISTING SEED COUNTS: WHERE THE FROZEN MARGIN PASSES AND WHERE IT ONLY APPEARS TO

Why this is here and what it is not. The pre-registered rule scores a *point* estimate against a margin: $|\text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)| < 0.15$ and every $\text{rung} < 0.5$. A point estimate inside a margin is consistent with a negligible effect *and* with an effect the sample cannot locate, and those are different claims. Table A22 scores the same frozen margin as a two-one-sided- t (TOST) equivalence test on the *paired* per-seed differences, declaring equivalence only when the 90% interval lies inside $(-0.15, +0.15)$. It is **post-hoc**, reads only frozen artifacts, **adds no runs and revises no published verdict**; the emitter is `experiments/analyze_tost_existing.py`. Pairing is not a modelling choice: every rung of an arm runs at the same seeds, so each difference holds one data partition fixed and the seed-to-seed spread that dominates this suite drops out of the contrast.

What it shows, including against us. The **flagship passes**, and passes at $n=5$: $p=0.0014$, the pairing holding sd to 0.042, so *the equivalence claim on the adjudicating arm does not depend on a larger seed count*. So do the score-only control ($p=0.0071$) and the EMNIST-byclass replication ($p=0.0004$, $n=3$, $\text{sd}=0.007$, and $p < 0.0001$ at the pooled $n=5$ of §D.2, where the interval widens across zero but stays well inside the margin). **The ResNet18 replication passes as well, and it is the closest call in the table**: $p=0.040$ at $n=3$ on two endpoint rungs, its 90% interval reaching -0.137 against the -0.15 boundary. **Its 95% interval**, $[-0.183, +0.101]$, **is wider than the margin**, so that arm's equivalence is established at the frozen ± 0.15 and at no margin below 0.137; §D.2 discloses this beside its pre-registered point-estimate verdict, and §S12's ladder statement is scoped to the comparability arms for exactly this reason. The two arms whose suppression genuinely moves correctly **fail** it, which is the positive control working: reputation (+0.178) and `cos_krum` (-0.425). **And one arm passes the frozen rule while failing the test**: `coord.median`'s $+0.098$ ($n=5$) sits inside ± 0.15 , but its 90% interval reaches $+0.160$ and TOST gives $p=0.073$. We therefore report that arm's equivalence as **not established** rather than as inherited from the point estimate. **This is the one place the paper's frozen scoring rule was more permissive than the evidence warrants, and we say so rather than leave the width implicit**. `cos_krum` additionally sits above the 0.5 ceiling the frozen rule conjoins with the margin (0.584 at the identity), which the body already states as its never suppressing.

Two arms are equivalent and detected at once, and the table marks them. The EMNIST-byclass row at the frozen $n=3$ and the CIFAR-100 row of §D.3 both have 90% intervals inside the margin *and* 95% intervals excluding zero: a real effect, smaller than the margin. Nothing is inconsistent here, but a bare "equivalent" on such a row would be read as "no effect", so both carry $\dagger$. **The CIFAR-100 row is the sharper case**: §D.3 reports it as a *rise* of $+0.071$ against the confounded design's fall of -0.213 , and that reading stands. What this row adds is its size, under half the frozen margin, which is why the sign reversal there is reported as a reversal of *direction* and never as a large controlled effect. **This arm is scored here and excluded from the binding-arm search of §S12**, on the separate ground that a margin-sensitivity analysis may only rest on arms about which the paper makes an equivalence claim, and it makes none about this one.

Where ± 0.15 came from, reconstructed and not pre-registered. The margin was frozen at 5130`cec` with *no written rationale*, which we state plainly rather than supply a derivation after the fact. What the pre-freeze record does support: the effect the design was built against was the pre-registered ladder's own $+0.762$ rise at $\rho=54.6$, five times the margin, and the margin never stands alone, the frozen rule conjoining it with *every rung* < 0.5 . So the margin is deliberately generous and **the conjunction is what stops a generous margin from certifying a high-ASR arm**; on the flagship cell every rung's *mean* in fact sits below 0.07, the quantity the frozen rule scores.

Table A22: **TOST against the frozen ± 0.15 margin at the seed counts already on disk**, on the paired per-seed difference (top rung minus identity). Equivalence requires the 90% interval inside the margin, which at $\alpha=0.05$ is $\max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$. [†] **equivalent and detected**: the 95% interval also excludes zero, so the verdict means “no effect larger than 0.15”, never “no effect”. Those two readings are not exclusive and the marker exists so a checkmark cannot be read as the stronger one; on the CIFAR-100 row the paper reports a *measured rise* (§D.3) and this row does not retract it. **One row’s 95% interval is itself wider than the margin**: ResNet18’s $[-0.183, +0.101]$. TOST at $\alpha=0.05$ reads the 90% interval, so that row passes the test defined here while failing the 95%-interval criterion §S12’s margin ladder uses; both readings are disclosed at §D.2, and the two conventions are not interchangeable. Post-hoc; no runs added. Emitted by `experiments/analyze_tost_existing.py`.

Arm	n	paired Δ	95% CI	90% CI (TOST)	verdict
Krum / scaling (flagship)	5	-0.026	$[-0.078, +0.026]$	$[-0.066, +0.014]$	equivalent
Krum / scaling (score-only)	5	-0.023	$[-0.108, +0.061]$	$[-0.088, +0.042]$	equivalent
Krum / scaling (EMNIST-byclass)	3	-0.017	$[-0.033, -0.001]$	$[-0.028, -0.006]$	equivalent [†]
the same arm, pooled	5	-0.013	$[-0.034, +0.009]$	$[-0.029, +0.004]$	equivalent
Krum / scaling (ResNet18)	3	-0.041	$[-0.183, +0.101]$	$[-0.137, +0.055]$	equivalent
Reputation / scaling	5	+0.178	$[-0.112, +0.469]$	$[-0.045, +0.402]$	not est.
cos_krum / pixel	8	-0.425	$[-0.732, -0.118]$	$[-0.671, -0.179]$	not est.
coord_median / pixel	5	+0.098	$[+0.018, +0.178]$	$[+0.036, +0.160]$	not est.
coord_median / pixel (CIFAR-100)	5	+0.071	$[+0.031, +0.110]$	$[+0.040, +0.101]$	equivalent [†]

Table A23: **The frozen ± 0.15 is not load-bearing; every equivalence reading in the comparability table survives the whole ladder.** ✓ if the arm’s *controlled* 95% interval lies inside $[-m, +m]$; m^* is the smallest margin at which it does. An arm is scored only where the paper makes an equivalence reading, that is where its interval contains zero, and the others are marked absent (—) because their verdict is a detected effect and reads no margin at all. The one row marked × at every rung, reputation / scaling, is scored because its interval contains zero, but that interval is too wide ($[-0.112, +0.469]$) for any margin here and the paper makes no equivalence claim about that arm, so the headline above is not contradicted by it. **These are the controlled intervals of the comparability design** (§D.3), *not* Table A22’s paired 90% TOST intervals: the two differ by design and by seed count, so Krum / scaling is $n=20$ here against $n=5$ there, and coord_median / pixel is $n=20$ here (both legs, after the pre-registered endpoint top-up, seeds 47–61) against the frozen $n=5$ there. Two absent arms have m^* inside the ladder’s own span (0.155 and 0.110), named here so the absence cannot be mistaken for a hidden flip. **One arm of the paper is not in this artifact at all**: the resnet18 replication (§D.2) has only the controlled leg and so no comparability cell, and its own m^* is 0.183, *wider* than the frozen margin. The emitter derives and prints that number beside this table, so the ladder cannot be read as covering it. Emitted by `experiments/analyze_margin_sensitivity.py` part (D); the fuller sweep, including the ± 0.05 AGREE tolerance over all of $[0, 1]$, is supplement §S12.

Arm	$m=0.05$	0.10	0.15	0.20	m^*
krum / scaling	✓	✓	✓	✓	0.032
cos_krum / pixel	—	—	—	—	0.732
reputation / scaling	×	×	×	×	0.469
coord_median / pixel	—	—	—	—	0.155
coord_median / scaling	—	—	—	—	0.280
krum / scaling, EMNIST	✓	✓	✓	✓	0.034
coord_median / pixel, CIFAR-100	—	—	—	—	0.110

What the ladder shows, and the one arm it marks ×. Three arms carry an equivalence reading and none of their verdicts changes anywhere on the ladder, so the reading the body rests on would be identical at ± 0.05 . The exception is reputation / scaling, which fails at every rung: its controlled interval contains zero but is wide, $[-0.112, +0.469]$, so no margin in this range licenses an equivalence claim there, and the paper makes none. *The sign reversal turns on neither threshold*: it is defined by opposite signs with both intervals excluding zero.

M COMPOSITION AND ADAPTIVE-ATTACK DETAILS

M.1 COMPOSITION BASELINE DETAILS

Simultaneous composition (NormClip $\tau=5$ then reputation every round), $N=10$, $K=5$, $f=0.2$, 50 rounds, cifar_cnn, 5 seeds. Per-seed results:

Seed	model_scaling		backdoor_pixel	
	Accuracy	ASR	Accuracy	ASR
42	0.783	0.766	0.774	0.837
43	0.788	0.783	0.774	0.792
44	0.794	0.835	0.784	0.828
45	0.762	0.889	0.771	0.931
46	0.780	0.763	0.766	0.823
Mean $\pm$ std	0.781 ± 0.011	0.807 ± 0.048	0.774 ± 0.006	0.842 ± 0.047

Composition partially suppresses scaling (ASR 0.807 vs. pure NormClip ~ 0.95 , and 0.961 under the NE3 temporal-mixing policy defined in §M.2) but does not approach pure reputation (0.017). Norm clipping preserves the attack direction while reducing magnitude; the clipped update then passes reputation’s consensus-distance filter. Pixel attacks are similarly persistent under composition (ASR 0.842). Both attack families remain effective despite simultaneous application of both defenses.

M.2 DBA AT $f=0.4$: PER-SEED DETAILS

Three run labels are carried over from an earlier framing and used here only to identify runs, not to make any equilibrium claim: *NE3* is the mixed server policy that framing computed, which randomizes over defenses across rounds; *reputation eq* is the pure-reputation policy; and *BR* is the adversary’s best response to the policy it faces. NE3 BR on CIFAR-10 with DBA in place of model scaling at $f=0.4$ (raising adversarial co-participation to $K \cdot f = 2.0$ from the $f=0.2$ value of 1.0). $N=10$, $K=5$, 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

seed	DBA realized ASR	backdoor_pixel realized ASR
42	0.222	0.972
43	0.103	0.945
44	0.176	0.913
45	0.603	0.993
46	0.077	0.934
mean $\pm$ std	0.236 ± 0.190	0.952 ± 0.028

The DBA per-seed spread is $7.8\times$ between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold (≥ 0.20) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at $f=0.4$ (0.952 ± 0.028), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether ≥ 2 of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at $f=0.4$ is suggestive but not a robust second-attack demonstration.

M.3 DBA AT $f=0.6$: PER-SEED DETAILS AND BIMODALITY REPRODUCTION

Follow-up to Appendix M.2, designed to test whether the DBA bimodality at $f=0.4$ is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher f . At $f=0.6$, expected adversarial co-participation per round is $K \cdot f = 3.0$, making ≥ 2 adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds, $N=10$, $K=5$, cifar_cnn. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

seed	DBA ASR (NE3 mix)	DBA ASR (reputation eq)
42	0.360	0.339
43	0.295	0.278
44	0.386	0.415
45	0.734	0.705
46	0.154	0.130
mean $\pm$ std	0.386 $\pm$ 0.192	0.373 $\pm$ 0.190

Observations. (i) Mean DBA ASR improves over $f=0.4$ (0.236) to ~ 0.38 on both policies, but does not reach the ≥ 0.85 persistence band scaling achieves. (ii) Per-seed bimodality persists at $f=0.6$: seed 45 alone reaches > 0.7 in both policies; the remaining four seeds cluster in $[0.13, 0.42]$. (iii) Crucially, *the same seed is the strong-persistence seed under both policies* (NE3 mix: 0.734; reputation eq: 0.705), and *the same four seeds are the weak-persistence seeds under both*: the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli, but at a much lower mean (admissibility prevalence $\approx 1/5$ for DBA at $f=0.6$ vs. $\approx 4/5$ for scaling at $f=0.2$).

Pre-specified verdict: INTERMEDIATE. Overall mean 0.380 falls between PARTIAL (> 0.5) and NEGATIVE (< 0.3). Reading: DBA at $f=0.6$ does *not* replicate the persistence collapse with the cleanness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising f . The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.

N SCALE, ARCHITECTURE, AND EXTERNAL VALIDITY

N.1 SCALE VALIDATION ($N = 100$) AND DEFENSE-INFERENCE CHECK

Scale check at $N = 100$ (4×5 game, 5 seeds). Residual material from an earlier framing, recorded rather than dropped. Nothing in the screen, the identification result or the dose-response evidence depends on it. Write VoPD (value of policy diversity) for the gain the server’s equilibrium strategy achieves over its best *pure* defense, so VoPD = 0 exactly when the equilibrium is pure. **This is not the quantity an earlier draft called “realized VoPD”, and that draft described its own quantity with the bound direction inverted, which is worth recording rather than quietly dropping.** There the reported difference was oracle-minus-committed, and it was said to *lower*-bound the gap an adversary with no information about the server’s draw could realize. It does the opposite: the committed policies are a *subset* of that adversary’s available policies, so its best policy is at least as good as the best committed one, and subtracting the larger quantity makes oracle-minus-committed an *upper* bound on the gap. That quantity is withdrawn with its framing, nothing in this paper uses it, and the VoPD defined above is a different object, non-negative by construction and sharing only the name. The equilibrium is pure on seeds 43 to 46 (model scaling vs. RFA, VoPD = 0) and mixed on seed 42 (pixel + scaling vs. FedAvg + RFA, VoPD = 0.012); per-seed values in Table A24.

Table A24: Per-seed NE, CIFAR-10 $N = 100$ (4×5 game). Seeds 43–46 identical.

Seed	Type	Adversary	Server	VoPD
42	Mixed	Pixel + Scaling	FedAvg + RFA	0.012
43–46	Pure	Model scaling	RFA	0.000

N.2 RESNET18 ARCHITECTURE-REPLICATION MINI-SWEEP: PER-SEED DETAILS

Per-seed details for the ResNet18 architecture-replication mini-sweep behind the architecture-conditional collapse boundary in §H. `resnet18` with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to 32×32 (kernel 3, stride 1, maxpool $\rightarrow$ Identity); 11.2M parameters. Identical FL parameters to `cifar.cnn`: $N=10$, $K=5$, $f=0.2$, 50 rounds, $\tau = 5$. Committed-scaling BR, 3 mix points $\times$ 5 seeds = 15 runs at ~ 100 min/run on MPS. **Reading the table.** *mix (NC/rep)* is the server’s per-round mixture over the two upstream defenses, `norm_clip` (NC) and `reputation` (rep), so the 90/10 row draws `norm_clip` in 90% of rounds and `reputation` in the other 10%; every cell is trigger ASR under the committed scaling attack; the two *mean* columns are over the same five seeds, the second on `cifar.cnn` from `results/cifar10_mix_ratio_sweep/summary.json`, whose sweep carries five mix points of which the three run here are shown. Both spreads are population standard deviations over the five seeds, recomputed here from the per-seed values and not the Student- t intervals the rest of the paper reports.

mix (NC/rep)	s42	s43	s44	s45	s46	ResNet18 mean	cifar_cnn mean
90/10	0.926	0.929	0.965	0.974	0.962	0.951 ± 0.020	0.945 ± 0.021
50/50	0.924	0.971	0.813	0.920	0.903	0.906 ± 0.052	0.815 ± 0.104
10/90	0.917	0.733	0.783	0.975	0.915	0.865 ± 0.091	0.097 ± 0.090

NC-dominant endpoint replicates within 0.01; mid-mix is $\sim 9\%$ higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict: MEDIUM** (monotonic ordering preserved, endpoint criterion fails). The cifar_cnn collapse boundary at $\sim 70\text{--}90\%$ reputation is therefore a property of the smaller-architecture regime. This sweep bounds the architecture-conditional claim of §H and establishes no boundary of its own; we draw no conclusion here about which mixture of the two defenses survives, since mixture policies are outside this paper’s scope (§C).

O IS RANDOMIZATION ITSELF COSTLY? A COMPUTE-MATCHED, PRE-REGISTERED TEST

What this is, and what it is not. This is a new experiment, pre-registered in `experiments/pre_registration_randomized_composition.md` and committed at `e3fb038` before the first run. It is *not* a reinstatement of the randomized-menu comparison §G withdraws: that artifact stays withdrawn, because it is unpaired and compute-confounded, and nothing below is scored against it. The question here is narrower and answerable: holding defense computation fixed, does randomizing *which* composition is applied each round cost suppression relative to committing to the best member of the same set? The direction was fixed in advance as $H_1 : \Delta > 0$, randomization *hurts*, because that is the claim the withdrawn comparison made and the claim we were asked to substantiate.

Three defects in the predecessor, and what each fix costs. (a) *The arms were not seed-matched.* The old runner tested `randomize_order` and `rng.random() < 0.5`, which short-circuits, so the fixed arm never drew that number and the randomized arm consumed one extra draw per round from the same generator that selects participants. The arms therefore ran on divergent participant streams and no paired test was licensed. Fixed with a separate `policy_rng`, leaving `rng` for participants only, which is why every arm here was re-run rather than partly reused. (b) *A defense-count confound.* Every CM→FG round reduces to FoolsGold alone, so the old gap conflated randomization with dropping a stage. Every member of every randomized set here keeps FoolsGold as a genuine per-client upstream, so no round degenerates. (c) *A mixed- n trap.* Every leg of every contrast below is recomputed per seed from `results/randomized_composition/summary.json` at seeds 42–46; the analyzer prints both legs’ n on every row and marks any mismatch.

arm	condition	def./rd	scaling ASR	pixel ASR	acc. (sc./px.)
1	fixed_fg_cm	2	0.1068 ± 0.0306	0.0988 ± 0.0201	0.682/0.664
1	fixed_fg_rfa	2	0.0510 ± 0.0096	0.0452 ± 0.0118	0.499/0.495
1	fixed_fg_tm	2	0.3843 ± 0.2594	0.2495 ± 0.1026	0.694/0.705
2	rand_comp_strong	2	0.1522 ± 0.1091	0.1115 ± 0.0440	0.607/0.598
2	rand_comp_all3	2	0.2317 ± 0.2780	0.0879 ± 0.0301	0.623/0.635
3 [†]	temporal_mix	1	0.5712 ± 0.3737	0.6600 ± 0.1543	0.307/0.765
4 [†]	pure_coord_median	1	0.5186 ± 0.1413	0.4425 ± 0.1290	0.764/0.767

All seven conditions, $n = 5$ seeds 42–46, population sd. [†] **Compute matching is claimed for arms 1–2 only:** arms 3 and 4 apply one defense per round and use roughly half the defense computation, so they are reference points and not comparators. Arm 2’s primary set is the *smaller* one, {FG→CM, FG→RFA}, whose members each suppress both committed attacks alone; FG→TM is a criterion-FAIL pair whose canonical scaling arm is 0.508 ± 0.269 , above threshold, so a set containing it could degrade merely because a non-suppressing member was drawn. That 0.508 is the frozen $n=3$ reading (seeds 42–44) and is *not* the table’s 0.3843, which is the same arm at $n=5$; the two agree exactly on the three shared seeds by control 3, and the difference is seeds 45–46 alone. Both sets were fixed by rule in the pre-registration, not by any outcome measured here.

Four positive controls, because this project has twice shipped a hook that silently never fired. (1) *The hook fired:* every member of every randomized set appears in the stored per-round policy log across all 15 streams. (2) *The arms differ:* neither randomized arm is bit-identical to any fixed arm at any seed, the closest approach being max per-seed $|d| = 0.2207$ (rand_comp_strong/scaling against fixed_fg_cm) and 0.0489 (rand_comp_all3/pixel). (3) *The harness did not change the run:* all eight per-seed identity checks of the fixed arms against the canonical artifacts return $\max |d| = 0.000e+00$ on their shared seeds, against all_compositions (seeds 42–44) and fg_cm_survivor’s composed (42–46). (4) *The end-to-end path reproduces a published standalone:* pure_coord_median returns 0.5186 and 0.4425 against

wave2_held_out's fedavg_then_coord_median at $\max |d| = 0.000\text{e}+00$, the same $n = 5$ seeds. Controls 3 and 4 are what license comparing any arm here to a number already in print.

Control 1 fails as literally pre-registered, and we report it rather than restate it. The frozen assertion was a per-stream two-sided 99% binomial interval with no family-wise correction. Three of the 15 streams carry an excursion (four member-level flags): `rand_comp_strong` seed 46 draws 36/14 against [16, 34] (tail $P = 0.00130$), and `rand_comp_all3` draws 26 at seed 45 and 27 at seed 46 against [8, 25] ($P = 0.00492$, $P = 0.00208$). Family-wise, $P(\text{at least one excursion}) = 1 - 0.99^{15} = 0.140$, so excursions at this rate are expected. We do not treat this as invalidating, and say why: the failure mode the control targets is a *dead* hook, which yields a member with count 0 or a 50/0 log, and every stream here draws every member. The pre-registered assertion is nonetheless the stricter one and it did not hold on all streams; that is the honest statement. One further correction of the pre-registration's own arithmetic: it wrote `rand_comp_all3`'s interval as approximately 8–26, the exact interval is [8, 25], and the exact one is used above.

The primary contrast, paired per seed. `rand_comp_strong` against the best member of its own set, where "best" is fixed by rule as the lower full-artifact mean on that same attack, which selects `fixed_fg_rfa` on both. On *model scaling* the per-seed gaps are +0.0352, +0.2899, +0.1564, +0.0102, +0.0143, giving $\bar{\Delta} = +0.1012$, 5/5 concordant signs, one-sided Wilcoxon $p = 0.0312$ and paired $t = 1.866$ ($p = 0.0677$). On the *pixel backdoor* they are +0.0816, +0.0428, +0.1581, +0.0133, +0.0356, giving $\bar{\Delta} = +0.0663$, 5/5 signs, Wilcoxon $p = 0.0312$ and $t = 2.603$ ($p = 0.0299$). At $n = 5$ the minimum attainable Wilcoxon p is $1/32 = 0.031$, stated in advance, so only 5/5 signs can reach $p < 0.05$. Post hoc: the randomized arm exceeds *both* members, not just the better one, in 3 of 5 seeds on each attack, so the gap is not interpolation between the two fixed arms. The secondary contrast, `rand_comp_all3` against the best of all three arm-1 conditions, runs the same way: $\bar{\Delta} = +0.1806$ with 4/5 signs on scaling ($p = 0.0625$) and +0.0427 with 4/5 on pixel ($p = 0.0625$).

The pre-registered outcome: all three admissible outcomes obtained, on three different objects. Materiality required *both* $\bar{\Delta} \geq 0.10$ and at least 4/5 positive signs. (i) *Material, in the pre-registered direction, on model scaling:* $\bar{\Delta} = +0.1012$ with 5/5 signs, so we state, scoped to this menu and this attack, that randomizing over non-degenerate compositions costs suppression at matched defense computation. (ii) *Not material on the pixel backdoor:* $\bar{\Delta} = +0.0663$ clears the sign leg but not the magnitude leg, and the honest phrasing at $n = 5$ is that no material difference was detected and the smallest effect this design resolves is ≈ 0.10 . (iii) *The accuracy floor is breached by one arm:* `temporal_mix` under model scaling means 0.307 clean accuracy against a frozen floor of 0.35, so that cell is reported as uninterpretable for ASR rather than compared, and every contrast involving it inherits the exclusion. This is why arm 3 is a reference point in the table and carries no verdict.

Three caveats, and what survives them. First, $\bar{\Delta}$ clears the 0.10 threshold by 0.0012 and is carried by two of five seeds (+0.2899 and +0.1564); on the three others the gap is ≤ 0.036 . A threshold met by 1.2% of itself is met, and it is also the weakest way a threshold can be met, so we do not describe the scaling result as large. Second, the two tests disagree in opposite directions across the two attacks: Wilcoxon reaches $p = 0.0312$ on both while the t -test reaches $p = 0.0299$ on pixel, where materiality *fails*, and only $p = 0.0677$ on scaling, where it holds. The stable part of the result is the sign structure, 10/10 across both attacks, and not either magnitude. Third, control 1 fails as literally pre-registered, as recorded above. What survives all three is the weakest claim consistent with them and the one the design was built to license: at matched defense computation and with the degeneracy confound removed, randomizing over a set of compositions did not *help* on either attack and was consistently worse on both, materially so on one. Nothing here licenses a claim about randomized menus in general, about randomized *internals* such as noise or subsampling (L3), or about any menu but this one.

And the attack is committed in both arms, which is the scope condition that bites hardest on what this test can be read as answering. Randomization's theoretical case is against an adversary that *adapts* to the composition: a menu the adversary cannot predict is a menu it cannot optimize against. Both attacks here are fixed in advance, identical in every arm, and blind to the draw. So this arm bounds the *cost* of randomizing under a committed threat and measures none of the benefit randomization is proposed for, and a favourable outcome would not have shown that benefit either. No adaptive-adversary test of a randomized menu is run here or anywhere else in this paper: the criterion-aware adversary of §F adapts to a composition that is *fixed* and known to it, which is the opposite of the setting randomization is meant to deny an adversary, and building an adversary that adapts to a distribution over menus is a separate experiment we do not run.